7

Linear Filtering Theory

1. INTRODUCTION

In this chapter, we specialize the nonlinear theory developed in Chapter 6 to linear filtering problems. As was already noted in Section 5.2, in linear problems the conditional density function is Gaussian and is, therefore, completely characterized by its mean vector and covariance matrix. This fact makes linear filtering problems particularly tractable. As we have seen in some special cases (Examples 6.1, 6.2, and 6.4), the equation for the conditional covariance matrix is independent of the observations. Thus, it is also the unconditional covariance matrix and may be precomputed in advance. We shall see that this is true, in general, in linear problems. As a result, the filtering algorithm essentially consists of the equation for the conditional mean. Appropriately, the conditional mean is sometimes referred to as a *sufficient statistic.* We recall that, according to the Corollary of Theorem 5.2, the conditional mean minimizes the expected loss for a large class of loss functions. In any case, it is, of course, the minimum variance estimate (Theorem 5.3).

By specializing the nonlinear results to the linear case, we are following the probabilistic approach outlined in Section 5.2. We will use some of the statistical methods (Section 5.3) to derive these same results in various examples. In addition, some examples will be devoted to the

generalizations described in Section 5.4 and to the derivation of a linear smoother.

The bulk of the remainder of this chapter is devoted to the study of some important properties of linear filters. The concepts of observability, information, and controllability are defined and exploited in the study of the asymptotic behavior of linear filters. Useful bounds and stability properties are developed. Of great importance in applications is the sensitivity of the filtering algorithm to dynamical model and statistical model errors. Considerable attention is devoted to this subject. Finally, linear limited memory filter algorithms are developed. These, as we shall see in Chapter 8, have important applications in problems of optimal filter divergence.

2. CONTINUOUS-DISCRETE FILTER

Our linear dynamical system is described by the linear vector (Itô) stochastic differential equation

$$dx_t = F(t)\, x_t\, dt + G(t)\, d\beta_t\,, \qquad t \geqslant t_0\,, \tag{7.1}$$

where x_t is the n-vector state, F and G are, respectively, $n \times n$ and $n \times r$ nonrandom, continuous matrix time-functions, and $\{\beta_t\,, t \geqslant t_0\}$ is an r-vector Brownian motion process with $\mathscr{E}\{d\beta_t\, d\beta_t{}^{\mathrm{T}}\} = Q(t)\, dt$. Discrete, linear observations are taken at time instants t_k :

$$y_k = M(t_k)\, x_{t_k} + v_k\,; \qquad k = 1, 2,\ldots\,; \qquad t_{k+1} > t_k \geqslant t_0\,, \tag{7.2}$$

where y_k is the m-vector observation, M is an $m \times n$ nonrandom, bounded matrix function, and $\{v_k\,, k = 1, 2,\ldots\}$ is an m-vector, white Gaussian sequence, $v_k \sim N(0, R_k)$, $R_k > 0$. The distribution of x_{t_0} is Gaussian, $x_{t_0} \sim N(\hat{x}_{t_0}\,, P_{t_0})$, and $x_{t_0}\,$, $\{\beta_t\}$, and $\{v_k\}$ are assumed independent.

For the linear system (7.1), Kolmogorov's forward equation (6.5) becomes

$$\partial p/\partial t = -p\,\mathrm{tr}(F) - p_x{}^{\mathrm{T}} F x + \tfrac{1}{2}\,\mathrm{tr}(GQG^{\mathrm{T}} p_{xx}). \tag{7.3}$$

The conditional density in Eq. (6.12) is

$$p(y_k \mid x) = (1/(2\pi)^{m/2} \mid R_k \mid^{1/2}) \exp\{-\tfrac{1}{2}(y_k - M(t_k)\, x)^{\mathrm{T}}\, R_k^{-1}(y_k - M(t_k)\, x)\}, \tag{7.4}$$

and Theorem 6.1 is specialized to the linear case.

We obtain the equations for the conditional mean and covariance matrix, between observations, directly from Eq. (6.25) of Theorem 6.2. Since

$$\hat{f}^t = F(t)\,\hat{x}_t{}^t,$$

$$\mathscr{E}^t\{x_t f^{\mathrm{T}}\} - \hat{x}_t{}^t \hat{f}^{t\mathrm{T}} = \widehat{(x_t x_t^{\mathrm{T}})^t} - \hat{x}_t{}^t \hat{x}_t^{t\mathrm{T}})\,F^{\mathrm{T}}(t) = P_t{}^t F^{\mathrm{T}}(t),$$

$$\mathscr{E}^t\{GQG^{\mathrm{T}}\} = GQG^{\mathrm{T}},$$

we have

$$\begin{aligned} d\hat{x}_t{}^t/dt &= F(t)\,\hat{x}_t{}^t, \\ dP_t{}^t/dt &= F(t)\,P_t{}^t + P_t{}^t F^{\mathrm{T}}(t) + G(t)\,Q(t)\,G^{\mathrm{T}}(t). \end{aligned} \tag{7.5}$$

To compute the change in $\hat{x}$ and P at an observation at t_k, we use Eq. (6.14) of Theorem 6.1.[1] Carrying out the integration in the denominator of (6.14), we have

$$p(x, t_k \mid Y_{t_k}) = \frac{p(y_k \mid x)\,p(x, t_k \mid Y_{t_k^-})}{p(y_k \mid Y_{t_k^-})}.^{2} \tag{7.6}$$

All the densities appearing in (7.6) are Gaussian. By definition,[3]

$$p(x, t_k \mid Y_{t_k^-}) \sim N(\hat{x}_{t_k}^{t_k^-}, P_{t_k}^{t_k^-}).$$

$p(y_k \mid x)$ has already been computed in (7.4). Now

$$\mathscr{E}\{y_k \mid Y_{t_k^-}\} = M(t_k)\,\hat{x}_{t_k}^{t_k^-},$$

in view of (7.2), and

$$\begin{aligned} &\mathscr{E}\{(y_k - \mathscr{E}\{y_k \mid Y_{t_k^-}\})(y_k - \mathscr{E}\{y_k \mid Y_{t_k^-}\})^{\mathrm{T}} \mid Y_{t_k^-}\} \\ &\quad = M(t_k)\,P_{t_k}^{t_k^-}\,M^{\mathrm{T}}(t_k) + R_k, \end{aligned}$$

so that

$$p(y_k \mid Y_{t_k^-}) \sim N(M(t_k)\,\hat{x}_{t_k}^{t_k^-},\ M(t_k)\,P_{t_k}^{t_k^-} M^{\mathrm{T}}(t_k) + R_k).$$

Combining these results, and dropping the t_k subscripts and superscripts for the moment, we have

$$p(x, t_k \mid Y_{t_k}) = \frac{|\,MP^- M^{\mathrm{T}} + R\,|^{1/2}}{(2\pi)^{n/2}\,|\,R\,|^{1/2}\,|\,P^-\,|^{1/2}} \exp[-\tfrac{1}{2}\{\cdot\}], \tag{7.7}$$

[1] The following derivation was given by Ho and Lee [27] and Lee [37].

[2] Recall that $Y_\tau = \{y_l : t_l \leqslant \tau\}$ (6.3); $Y_{t_k} = \{y_1, \ldots, y_k\}$; $Y_{t_k^-} = Y_{t_{k-1}}$.

[3] Recall that $\hat{\varphi}^\tau(x_t) = [\varphi(x_t)]^{\wedge\tau} = \mathscr{E}^\tau\{\varphi(x_t)\} = \mathscr{E}\{\varphi(x_t) \mid Y_\tau\}$ (6.16).

where

$$\{\cdot\} = (y - Mx)^{\mathrm{T}} R^{-1}(y - Mx) + (x - \hat{x}^-)^{\mathrm{T}} P^{-^{-1}}(x - \hat{x}^-) - (y - M\hat{x}^-)^{\mathrm{T}} (MP^-M^{\mathrm{T}} + R)^{-1} (y - M\hat{x}^-). \tag{7.8}$$

Now, by definition,

$$p(x, t_k \mid Y_{t_k}) \sim N(\hat{x}_{t_k}^{t_k^+}, P_{t_k}^{t_k^+}),$$

so that the term in braces in (7.8) must be

$$\{\cdot\} = (x - \hat{x}^+)^{\mathrm{T}} P^{+^{-1}}(x - \hat{x}^+). \tag{7.9}$$

To put (7.8) in the form of (7.9), we rewrite it as

$$\begin{aligned}\{\cdot\} = {} & x^{\mathrm{T}}(M^{\mathrm{T}}R^{-1}M + P^{-^{-1}})\, x - 2(y^{\mathrm{T}}R^{-1}M + \hat{x}^{-\mathrm{T}}P^{-^{-1}})(M^{\mathrm{T}}R^{-1}M + P^{-^{-1}})^{-1} \\ & \times (M^{\mathrm{T}}R^{-1}M + P^{-^{-1}})\, x + y^{\mathrm{T}}R^{-1}y + \hat{x}^{-\mathrm{T}}P^{-^{-1}}\hat{x}^- \\ & - (y - M\hat{x}^-)^{\mathrm{T}} (MP^-M^{\mathrm{T}} + R)^{-1} (y - M\hat{x}^-)\end{aligned}$$

and complete the squares by adding and subtracting

$$(y^{\mathrm{T}}R^{-1}M + \hat{x}^{-\mathrm{T}}P^{-^{-1}})(M^{\mathrm{T}}R^{-1}M + P^{-^{-1}})^{-1} (M^{\mathrm{T}}R^{-1}y + P^{-^{-1}}\hat{x}^-).$$

We thus obtain

$$\begin{aligned}\{\cdot\} = {} & [x - (M^{\mathrm{T}}R^{-1}M + P^{-^{-1}})^{-1} (M^{\mathrm{T}}R^{-1}y + P^{-^{-1}}\hat{x}^-)]^{\mathrm{T}} [M^{\mathrm{T}}R^{-1}M + P^{-^{-1}}] \\ & \times [x - (M^{\mathrm{T}}R^{-1}M + P^{-^{-1}})^{-1} (M^{\mathrm{T}}R^{-1}y + P^{-^{-1}}\hat{x}^-)] + r,\end{aligned} \tag{7.10}$$

where, with the aid of the matrix equalities developed in Appendix 7B, r can be shown to be zero. Comparing (7.10) with (7.9), we conclude that

$$\begin{aligned}\hat{x}_{t_k}^{t_k^+} &= (M^{\mathrm{T}}(t_k)\, R_k^{-1} M(t_k) + P_{t_k}^{t_k^{-^{-1}}})^{-1} (M^{\mathrm{T}}(t_k)\, R_k^{-1} y_k + P_{t_k}^{t_k^{-^{-1}}} \hat{x}_{t_k}^{t_k^-}), \\ P_{t_k}^{t_k^{+^{-1}}} &= P_{t_k}^{t_k^{-^{-1}}} + M^{\mathrm{T}}(t_k)\, R_k^{-1} M(t_k).\end{aligned} \tag{7.11}$$

Equations (7.11) are the desired relations. We see [(7.5), (7.11)] that the conditional covariance matrix is independent of observations and is, therefore, the unconditional covariance matrix:

$$P_t{}^t \triangleq \mathscr{E}\{(x_t - \hat{x}_t{}^t)(x_t - \hat{x}_t{}^t)^{\mathrm{T}} \mid Y_t\} = \mathscr{E}\{(x_t - \hat{x}_t{}^t)(x_t - \hat{x}_t{}^t)^{\mathrm{T}}\}. \tag{7.12}$$

Consequently, we shall sometimes write $P_t{}^t$ as $P(t \mid t)$ and even as $P(t)$ when no confusion can arise.

The computation of Eqs. (7.11) involves the inversion of $n \times n$ matrices. We next reduce (7.11) to a form that requires the inversion of an $m \times m$ matrix (m is the dimension of the observation vector). Often, in applications, $m < n$, and, in case R is diagonal, we may always take $m = 1$. This is done by processing the components of y_k one by one, and supposing that there is no change in $\hat{x}$ or P due to the dynamics (7.5) in the meantime. That this is equivalent to processing the whole observation vector at once is conceptually clear and can be proved algebraically (very tedious!).

Using (7B.5) of Appendix 7B, we rewrite the second of Eqs. (7.11) as

$$P_{t_k}^{t_k^+} = P_{t_k}^{t_k^-} - P_{t_k}^{t_k^-} M^{\mathrm{T}}(t_k)[M(t_k)\, P_{t_k}^{t_k^-} M^{\mathrm{T}}(t_k) + R_k]^{-1}\, M(t_k)\, P_{t_k}^{t_k^-}. \tag{7.13}$$

With the aid of (7B.5) and (7B.6) of Appendix 7B, the first of Eqs. (7.11) becomes

$$\hat{x}_{t_k}^{t_k^+} = \hat{x}_{t_k}^{t_k^-} + P_{t_k}^{t_k^-} M^{\mathrm{T}}(t_k)[M(t_k)\, P_{t_k}^{t_k^-} M^{\mathrm{T}}(t_k) + R_k]^{-1}\, (y_k - M(t_k)\, \hat{x}_{t_k}^{t_k^-}). \tag{7.14}$$

If we define the *Kalman gain* [32, 36]

$$K(t_k) \triangleq P_{t_k}^{t_k^-} M^{\mathrm{T}}(t_k)[M(t_k)\, P_{t_k}^{t_k^-} M^{\mathrm{T}}(t_k) + R_k]^{-1}, \tag{7.15}$$

we have

$$\begin{aligned} \hat{x}_{t_k}^{t_k^+} &= \hat{x}_{t_k}^{t_k^-} + K(t_k)(y_k - M(t_k)\, \hat{x}_{t_k}^{t_k^-}), \\ P_{t_k}^{t_k^+} &= P_{t_k}^{t_k^-} - K(t_k)\, M(t_k)\, P_{t_k}^{t_k^-} \\ &= [I - K(t_k)\, M(t_k)]\, P_{t_k}^{t_k^-}[I - K(t_k)\, M(t_k)]^{\mathrm{T}} + K(t_k)\, R_k K^{\mathrm{T}}(t_k). \end{aligned} \tag{7.16}$$

We summarize these results in the following theorem.

Theorem 7.1. *The optimal (minimum variance) filter for the continuous-discrete system* (7.1), (7.2) *consists of equations of evolution for the conditional mean* $\hat{x}_t{}^t$ *and covariance matrix* $P_t{}^t$. *Between observations, these satisfy the differential equations*

$$\begin{aligned} d\hat{x}_t{}^t/dt &= F(t)\, \hat{x}_t{}^t, \\ dP_t{}^t/dt &= F(t)\, P_t{}^t + P_t{}^t F^{\mathrm{T}}(t) + G(t) Q(t)\, G^{\mathrm{T}}(t), \qquad t_k \leqslant t < t_{k+1}. \end{aligned} \tag{7.17}$$

At an observation at t_k, *they satisfy the difference equations*

$$\begin{aligned} \hat{x}_{t_k}^{t_k^+} &= \hat{x}_{t_k}^{t_k^-} + K(t_k)(y_k - M(t_k)\, \hat{x}_{t_k}^{t_k^-}), \\ P_{t_k}^{t_k^+} &= P_{t_k}^{t_k^-} - K(t_k)\, M(t_k)\, P_{t_k}^{t_k^-}, \end{aligned} \tag{7.18}$$

where the Kalman gain K is given in (7.15). *Prediction for* $t > \tau$ $(\hat{x}_t^{\tau}, P_t^{\tau})$, *based on* Y_τ (6.3), *is clearly accomplished via* (7.17), *with initial condition* $(\hat{x}_\tau^{\tau}, P_\tau^{\tau})$.

Theorem 7.1 gives the celebrated Kalman–Bucy filter [32, 33, 36] for the continuous-discrete problem. Compare this with Examples 6.1 and 6.2. The equations for the estimate are linear, with a forcing term that is linear in the observation. The forcing term is proportional to the *residual*

$$y_k - M(t_k)\,\hat{x}_{t_k}^{t_k^-} = y_k - \mathscr{E}\{y_k \mid Y_{t_k^-}\}.$$

The initial conditions $(\hat{x}_{t_0}, P_{t_0})$ are part of the problem statement and reflect the *prior* knowledge of x_t before the filtering begins. As will be seen from other derivations of these same results, we need only that $P_{t_0} \geqslant 0$. In connection with the question of prior information, see Appendix 7A and also Sections 5 and 10. The parameters $(\hat{x}_t^{\,t}, P_t^{\,t})$ comprise the *state* of the linear filter. They determine the density $p(x, t \mid Y_t)$. However, as has already been noted, $P_t^{\,t}$ and the filter gain $K(t_k)$ may be *precomputed*, since they do not depend on the noise samples, but only on the noise statistics which are part of the problem statement. Thus, once the observation schedule has been settled, the real-time implementation of the Kalman–Bucy filter requires only the computation of the first of Eqs. (7.17) and the first of Eqs. (7.18). This happy situation is, of course, unique to the linear case.

Now Eq. (7.1) can be integrated over intervals $[t_k, t_{k+1}]$:

$$x_{t_{k+1}} = \Phi(t_{k+1}, t_k)\,x_{t_k} + \int_{t_k}^{t_{k+1}} \Phi(t_{k+1}, \tau)\,G(\tau)\,d\beta_\tau, \qquad k = 0, 1, \ldots, \tag{7.19}$$

where the $n \times n$ matrix Φ is the *fundamental matrix* of the homogeneous part of (7.1), or the *state transition matrix* of (7.1). That is,

$$d\Phi(t, \tau)/dt = F(t)\,\Phi(t, \tau), \qquad \Phi(\tau, \tau) = I \quad \text{all} \quad \tau, \tag{7.20}$$

$$\Phi(t, \tau)\,\Phi(\tau, \xi) = \Phi(t, \xi) \qquad \text{all} \quad t, \tau, \xi. \tag{7.21}$$

From (7.21), it follows that

$$\Phi^{-1}(t, \tau) = \Phi(\tau, t) \qquad \text{all} \quad t, \tau. \tag{7.22}$$

We write (7.19) as

$$x_{t_{k+1}} = \Phi(t_{k+1}, t_k)\,x_{t_k} + w_{k+1}, \tag{7.23}$$

where

$$w_{k+1} = \int_{t_k}^{t_{k+1}} \Phi(t_{k+1}, \tau)\,G(\tau)\,d\beta_\tau. \tag{7.24}$$

It is easy to see that $\{w_k\}$ is a zero-mean, white Gaussian sequence with (see Theorem 4.1)

$$\mathscr{E}\{w_{k+1}w_{k+1}^{\mathrm{T}}\} = \int_{t_k}^{t_{k+1}} \Phi(t_{k+1}, \tau)\, G(\tau)\, Q(\tau)\, G^{\mathrm{T}}(\tau)\, \Phi^{\mathrm{T}}(t_{k+1}, \tau)\, d\tau. \tag{7.25}$$

Thus the continuous-discrete filter can be imbedded in a discrete filter, and we need not study the properties of the continuous-discrete filter separately.

It is easy to verify that (7.17) of Theorem 7.1 are equivalent to

$$\begin{aligned} \hat{x}_t^{\,t} &= \Phi(t, t_k)\, \hat{x}_{t_k}^t, \\ P_t^{\,t} &= \Phi(t, t_k)\, P_{t_k}^t \Phi^{\mathrm{T}}(t, t_k) + \mathscr{E}\{w_{k+1}w_{k+1}^{\mathrm{T}}\}, \qquad t_k \leqslant t < t_{k+1}. \end{aligned} \tag{7.26}$$

3. DISCRETE FILTER

The discrete linear system is described by the vector difference equation

$$x_{k+1} = \Phi(t_{k+1}, t_k)\, x_k + \Gamma(t_k)\, w_{k+1}, \qquad k = 0, 1,..., \tag{7.27}$$

where x_k is the n-vector state at t_k, Φ is the $n \times n$, nonsingular state transition matrix, Γ is $n \times r$, and $\{w_k, k = 1,...\}$ is an r-vector, white Gaussian sequence, $w_k \sim N(0, Q_k)$. The discrete, linear observations are given in (7.2):

$$y_k = M(t_k)\, x_k + v_k. \tag{7.2}$$

x_0, $\{w_k\}$, and $\{v_k\}$ are assumed independent.

Difference equations for $\hat{x}$ and P at an observation are already given in Theorem 7.1. We compute difference equations between observations directly from (7.27). This procedure avoids the computation of the transition probability density $p(x_{k+1} \mid x_k)$ and the Chapman–Kolmogorov equation (6.62) of Theorem 6.4. We know that all the densities are Gaussian. Now

$$\hat{x}_{k+1}^k = \mathscr{E}\{x_{k+1} \mid Y_k\} = \Phi(t_{k+1}, t_k)\, \hat{x}_k^{\,k},$$

in view of (7.27). Also,

$$\begin{aligned} P_{k+1}^k &= \mathscr{E}\{(x_{k+1} - \hat{x}_{k+1}^k)(x_{k+1} - \hat{x}_{k+1}^k)^{\mathrm{T}} \mid Y_k\} \\ &= \mathscr{E}\{[\Phi(x_k - \hat{x}_k^{\,k}) + \Gamma w_{k+1}][\Phi(x_k - \hat{x}_k^{\,k}) + \Gamma w_{k+1}]^{\mathrm{T}} \mid Y_k\} \\ &= \Phi P_k^{\,k} \Phi^{\mathrm{T}} + \Gamma Q_{k+1} \Gamma^{\mathrm{T}}. \end{aligned}$$

Compare this with (7.26).

The discrete filter is summarized in the following theorem.

Theorem 7.2. *The optimal (minimum variance) filter for the discrete system* (7.27), (7.2) *consists of difference equations for the conditional mean and covariance matrix. Between observations,*

$$\begin{aligned} \hat{x}_{k+1}^{k} &= \Phi(t_{k+1}, t_k)\, \hat{x}_k^{\,k}, \\ P_{k+1}^{k} &= \Phi(t_{k+1}, t_k)\, P_k^{\,k} \Phi^{\mathrm{T}}(t_{k+1}, t_k) + \Gamma(t_k) Q_{k+1} \Gamma^{\mathrm{T}}(t_k). \end{aligned} \tag{7.28}$$

At observations,

$$\begin{aligned} \hat{x}_k^{\,k} &= \hat{x}_k^{k-1} + K(t_k)(y_k - M(t_k)\, \hat{x}_k^{k-1}), \\ P_k^{\,k} &= P_k^{k-1} - K(t_k)\, M(t_k)\, P_k^{k-1}, \end{aligned} \tag{7.29}$$

where

$$K(t_k) = P_k^{k-1} M^{\mathrm{T}}(t_k)[M(t_k)\, P_k^{k-1} M^{\mathrm{T}}(t_k) + R_k]^{-1} \tag{7.15}$$

is the Kalman gain. Prediction for $t_l > t_k$ $(\hat{x}_l^{\,k}, P_l^{\,k})$ *is accomplished via* (7.28) *with initial condition* $(\hat{x}_k^{\,k}, P_k^{\,k})$.

This is the Kalman–Bucy filter [32, 36] for the discrete problem. The discussion of Theorem 7.1 applies in this case as well. We may note that, if $P_0 \geqslant 0$, then $P_k^{\,k} \geqslant 0$ for all k, since P is a covariance matrix.

Example 7.1. (*Orthogonal Projections*). In his original derivation of the discrete filter, Kalman [32] used the concept of orthogonal projection. We develop this concept here and, using it, rederive the discrete filter.

As was pointed out in footnote 7 of Chapter 3, the space of random variables with finite second moments, with scalar product

$$\langle x, y \rangle = \mathscr{E}\{xy\}$$

and norm

$$\| x \| = \mathscr{E}^{1/2}\{x^2\},$$

is a (complete) normed space (an infinite-dimensional Hilbert space). Now, by definition, the minimum variance estimate minimizes the error norm

$$\mathscr{E}\{(x - \hat{x})^2\}.$$

The following lemma gives a condition that is equivalent to the minimization of the norm.

Orthogonal Projection Lemma. *Let X be a normed space, $x \in X$, and let Y be a subspace of X. Then*

$$\min_{\alpha \in Y} \| x - \alpha \|^2 = \| x - \hat{x} \|^2$$

if, and only if,

$$\langle x - \hat{x}, \alpha \rangle = 0 \qquad \text{for all} \quad \alpha \in Y.$$

The space Y is the "approximation space." That is, we want to find that $\alpha \in Y$ which minimizes the error norm for every $x \in X$. The orthogonal projection lemma states that the error is orthogonal to the approximation space Y. That is, if we write

$$x = \tilde{x} + \hat{x},$$

where $\tilde{x} = x - \hat{x}$ is the error and $\hat{x}$ is the best approximation (estimate), then $\hat{x}$ is the orthogonal projection of x into Y.

PROOF: We first prove the "if" part. Suppose $\langle x - \hat{x}, Y \rangle = 0$. Take any $\alpha \in Y$, $\alpha \neq 0$. Then

$$\begin{aligned} \langle x - \hat{x} + \alpha, x - \hat{x} + \alpha \rangle &= \langle x - \hat{x}, x - \hat{x} \rangle + 2\langle x - \hat{x}, \alpha \rangle + \langle \alpha, \alpha \rangle \\ &= \langle x - \hat{x}, x - \hat{x} \rangle + \langle \alpha, \alpha \rangle \\ &> \langle x - \hat{x}, x - \hat{x} \rangle. \end{aligned}$$

Now we shall prove the "only if" part. Suppose there exists an α such that

$$\langle x - \hat{x}, \alpha \rangle = \beta \neq 0.$$

Then, for any scalar λ,

$$\langle x - \hat{x} + \lambda\alpha, x - \hat{x} + \lambda\alpha \rangle = \| x - \hat{x} \|^2 + 2\lambda\beta + \lambda^2 \| \alpha \|^2.$$

Then, for $\lambda = -\beta / \| \alpha \|^2$,

$$\| x - \hat{x} + \lambda\alpha \|^2 = \| x - \hat{x} \|^2 - \frac{2\beta^2}{\| \alpha \|^2} + \frac{\beta^2}{\| \alpha \|^2} < \| x - \hat{x} \|^2. \quad \blacksquare$$

In our filtering problem, the approximation space is $Y_k = \{\text{space spanned by } y_1, \ldots, y_k\}$. This is a km-dimensional space. The space X is infinite-dimensional. Let the set of m-vectors $\{u_1, \ldots, u_k\}$ be an orthonormal basis for Y_k, that is,

$$\mathcal{E}\{u_i u_i^{\mathrm{T}}\} = I, \qquad \mathcal{E}\{u_i u_j^{\mathrm{T}}\} = 0, \qquad i \neq j.$$

It is interesting that the minimization of

$$\mathcal{E}\{(x_k - \hat{x}_k^k)^T (x_k - \hat{x}_k^k)\} \tag{7.30}$$

is accomplished componentwise. That is, in minimizing (7.30), we shall minimize each component of the expected error.

Now, from the orthogonal projection lemma, we have

$$\mathcal{E}\{(x_k - \hat{x}_k^k)\, u_i^T\} = 0 \qquad \text{for all} \quad i,$$

or

$$\mathcal{E}\{x_k u_i^T\} = \mathcal{E}\{\hat{x}_k^k u_i^T\}.$$

Multiplying by u_i and summing,

$$\sum_{i=1}^{k} \mathcal{E}\{x_k u_i^T\}\, u_i = \sum_{i=1}^{k} \mathcal{E}\{\hat{x}_k^k u_i^T\}\, u_i\,. \tag{7.31}$$

According to the lemma, $\hat{x}_k^k$ is the orthogonal projection of x_k into Y. Therefore,

$$\hat{x}_k^k = \sum_{i=1}^{k} A_i u_i \tag{7.32}$$

for some set of constant matrices A_i. Using (7.32), it is easy to show that the right-hand side of (7.31) is simply $\hat{x}_k^k$; thus,

$$\hat{x}_k^k = \sum_{i=1}^{k} \mathcal{E}\{x_k u_i^T\}\, u_i$$

or

$$\hat{x}_k^k = \sum_{i=1}^{k-1} \mathcal{E}\{x_k u_i^T\}\, u_i + \mathcal{E}\{x_k u_k^T\}\, u_k\,,$$

and, using (7.27),

$$\hat{x}_k^k = \sum_{i=1}^{k-1} \mathcal{E}\{[\Phi(k, k-1)\, x_{k-1} + \Gamma w_k]\, u_i^T\}\, u_i + \mathcal{E}\{x_k u_k^T\}\, u_k\,.$$

Now w_k is independent of Y_{k-1}, and so we get

$$\hat{x}_k^k = \Phi(k, k-1)\, \hat{x}_{k-1}^{k-1} + \mathcal{E}\{x_k u_k^T\}\, u_k\,. \tag{7.33}$$

Now u_k is orthogonal to Y_{k-1}. We next show that

$$[y_k - M(t_k)\, \Phi(k, k-1)\, \hat{x}_{k-1}^{k-1}]$$

is also orthogonal to Y_{k-1}, and, since it clearly lies in Y_k, we must have

$$\mathscr{E}\{x_k u_k^{\mathrm{T}}\}\, u_k = K_k[y_k - M(t_k)\,\Phi(k, k-1)\,\hat{x}_{k-1}^{k-1}]$$

for some constant matrix K_k. Clearly,

$$\mathscr{E}\{(x_{k-1} - \hat{x}_{k-1}^{k-1})\, y_i^{\mathrm{T}}\} = 0, \qquad i = 1,\ldots, k-1,$$

and, multiplying this by Φ and using (7.27),

$$\mathscr{E}\{(x_k - \Phi\hat{x}_{k-1}^{k-1})\, y_i^{\mathrm{T}}\} = 0, \qquad i = 1,\ldots, k-1.$$

Multiplying this by M and using (7.2),

$$\mathscr{E}\{(y_k - M\Phi\hat{x}_{k-1}^{k-1})\, y_i^{\mathrm{T}}\} = 0, \qquad i = 1,\ldots, k-1.$$

Therefore,

$$\hat{x}_k^{\,k} = \Phi(k, k-1)\,\hat{x}_{k-1}^{k-1} + K_k(y_k - M(t_k)\,\Phi(k, k-1)\,\hat{x}_{k-1}^{k-1}). \tag{7.34}$$

It remains to determine the gain K_k.

The estimation error is given by

$$\begin{aligned}\tilde{x}_k^{\,k} = x_k - \hat{x}_k^{\,k} &= \Phi(k, k-1)\,\tilde{x}_{k-1}^{k-1} + \Gamma(k-1)\,w_k - K_k M(k)\,\Phi(k, k-1)\,\tilde{x}_{k-1}^{k-1} \\ &\quad - K_k[M(k)\,\Gamma(k-1)\,w_k + v_k],\end{aligned} \tag{7.35}$$

where we have used (7.27) and (7.2); and

$$y_k = M(k)\,\Phi(k, k-1)(\tilde{x}_{k-1}^{k-1} + \hat{x}_{k-1}^{k-1}) + M(k)\,\Gamma(k-1)\,w_k + v_k\,.$$

Therefore, by the orthogonal projection lemma,

$$\begin{aligned}0 = \mathscr{E}\{\tilde{x}_k^{\,k} y_k^{\mathrm{T}}\} &= \Phi(k, k-1)\,P_{k-1}^{k-1}\Phi^{\mathrm{T}}(k, k-1)\,M^{\mathrm{T}}(k) \\ &\quad + \Gamma(k-1)\,Q_k\Gamma^{\mathrm{T}}(k-1)\,M^{\mathrm{T}}(k) \\ &\quad - K_k[M(k)\,\Phi(k, k-1)\,P_{k-1}^{k-1}\Phi^{\mathrm{T}}(k, k-1)\,M^{\mathrm{T}}(k) \\ &\quad + M(k)\,\Gamma(k-1)\,Q_k\Gamma^{\mathrm{T}}(k-1)\,M^{\mathrm{T}}(k) + R_k],\end{aligned} \tag{7.36}$$

where we have defined

$$P_{k-1}^{k-1} \triangleq \mathscr{E}\{\tilde{x}_{k-1}^{k-1}\tilde{x}_{k-1}^{k-1\,\mathrm{T}}\}.$$

Defining in addition

$$P_k^{k-1} \triangleq \Phi(k, k-1)\,P_{k-1}^{k-1}\Phi^{\mathrm{T}}(k, k-1) + \Gamma(k-1)\,Q_k\Gamma^{\mathrm{T}}(k-1),$$

we get from (7.36)

$$K_k = P_k^{k-1}M^{\mathrm{T}}(k)[M(k)\,P_k^{k-1}M^{\mathrm{T}}(k) + R_k]^{-1}.$$

Using (7.35), it is easy to compute the difference equation for

$$P_k^{\,k} = \mathscr{E}\{\tilde{x}_k^{\,k}\tilde{x}_k^{k\mathrm{T}}\}$$

and thus to see that we have derived the Kalman–Bucy filter in Theorem 7.2. ▲

In the next several examples, we shall rederive the Kalman–Bucy filter using some of the statistical methods outlined in Chapter 5.

Example 7.2 (Recursive Least Squares). This method has been used by many authors (e.g., Swerling [53], Gainer [19], Fagin [16], Mowery [40], and Bryson and Ho [10]) to derive the discrete Kalman–Bucy filter. We give a version of the method here. For simplicity, we assume that the dynamical system is noise-free [$w_k \equiv 0$ in (7.27)].

In this case, the least squares problem (5.22), (5.23) at time t_k is to minimize, with respect to $\{x_0, \ldots, x_k\}$,

$$J_k' = \frac{1}{2}(x_0 - \bar{x}_0)^{\mathrm{T}} P_0^{-1}(x_0 - \bar{x}_0) + \frac{1}{2}\sum_{i=1}^{k}(y_i - M(i)\,x_i)^{\mathrm{T}} R_i^{-1}(y_i - M(i)\,x_i),$$

subject to the constraints

$$x_{i+1} = \Phi(i+1, i)\,x_i\,, \qquad i = 0, \ldots, k-1.$$

In view of the constraints, we may alternatively minimize

$$J_k = \frac{1}{2}(x_k - \bar{x}_k)^{\mathrm{T}}\,\Phi^{\mathrm{T}}(0, k)\,P_0^{-1}\Phi(0, k)(x_k - \bar{x}_k)$$

$$+ \frac{1}{2}\sum_{i=1}^{k}(y_i - M(i)\,\Phi(i, k)\,x_k)^{\mathrm{T}}\,R_i^{-1}(y_i - M(i)\,\Phi(i, k)\,x_k) \quad (7.37)$$

with respect to x_k. Note that $\Phi(i, k) = \Phi^{-1}(k, i)$. Since the *prior* estimate of x_0 is $\bar{x}_0$, that of x_k is $\bar{x}_k = \Phi(k, 0)\bar{x}_0$.

To minimize J_k in (7.37), we set its gradient with respect to x_k equal to zero:

$$\Phi^{\mathrm{T}}(0, k)\,P_0^{-1}\Phi(0, k)(x_k - \bar{x}_k) - \sum_{i=1}^{k}\Phi^{\mathrm{T}}(i, k)\,M^{\mathrm{T}}(i)\,R_i^{-1}(y_i - M(i)\,\Phi(i, k)\,x_k) = 0. \quad (7.38)$$

Now defining

$$\mathscr{I}_{k,1} \triangleq \sum_{i=1}^{k}\Phi^{\mathrm{T}}(i, k)\,M^{\mathrm{T}}(i)\,R_i^{-1}M(i)\,\Phi(i, k) \quad (7.39)$$

(this is the *information matrix* defined in Section 5; see also Appendix 7A) and

$$H_k \triangleq \sum_{i=1}^{k} \Phi^{\mathrm{T}}(i, k)\, M^{\mathrm{T}}(i)\, R_i^{-1} y_i\,, \tag{7.40}$$

the solution $\bar{x}_k^{\,k}$ of (7.38) is written as

$$\bar{x}_k^{\,k} = [\Phi^{\mathrm{T}}(0, k)\, P_0^{-1} \Phi(0, k) + \mathscr{I}_{k,1}]^{-1}\, [H_k + \Phi^{\mathrm{T}}(0, k)\, P_0^{-1} \Phi(0, k)\, \bar{x}_k]. \tag{7.41}$$

Note that the indicated inverse in (7.41) always exists since, by assumption, $P_0 > 0$, and $R_k > 0$ implies $\mathscr{I}_{k,1} \geqslant 0$ for all k.

Now clearly the minimization of J_{k+1} yields

$$\begin{aligned} \bar{x}_{k+1}^{k+1} = {} & [\Phi^{\mathrm{T}}(0, k+1)\, P_0^{-1} \Phi(0, k+1) + \mathscr{I}_{k+1,1}]^{-1} \\ & \times [H_{k+1} + \Phi^{\mathrm{T}}(0, k+1)\, P_0^{-1} \Phi(0, k+1)\, \bar{x}_{k+1}]. \end{aligned} \tag{7.42}$$

It is easy to derive the following difference equations for $\mathscr{I}_{k,1}$ and H_k :

$$\begin{aligned} \mathscr{I}_{k+1,1} &= \Phi^{\mathrm{T}}(k, k+1)\, \mathscr{I}_{k,1} \Phi(k, k+1) + M^{\mathrm{T}}(k+1)\, R_{k+1}^{-1} M(k+1), \\ H_{k+1} &= \Phi^{\mathrm{T}}(k, k+1)\, H_k + M^{\mathrm{T}}(k+1)\, R_{k+1}^{-1} y_{k+1}\,, \end{aligned} \tag{7.43}$$

and, using these and the definition

$$P_k^{k-1} \triangleq \Phi^{\mathrm{T}}(0, k)\, P_0^{-1} \Phi(0, k) + \mathscr{I}_{k,1}\,, \tag{7.44}$$

to rewrite (7.42) as

$$\begin{aligned} \bar{x}_{k+1}^{k+1} = {} & [\Phi^{\mathrm{T}}(k, k+1)\, P_k^{k-1} \Phi(k, k+1) + M^{\mathrm{T}}(k+1)\, R_{k+1}^{-1} M(k+1)]^{-1} \\ & \times [\Phi^{\mathrm{T}}(k, k+1)(H_k + \Phi^{\mathrm{T}}(0, k)\, P_0^{-1} \Phi(0, k)\, \bar{x}_k) + M^{\mathrm{T}}(k+1)\, R_{k+1}^{-1} y_{k+1}]. \end{aligned} \tag{7.45}$$

Now, eliminating

$$(H_k + \Phi^{\mathrm{T}}(0, k)\, P_0^{-1} \Phi(0, k)\, \bar{x}_k)$$

via (7.41) and defining

$$P_{k+1}^{k-1} \triangleq \Phi^{\mathrm{T}}(k, k+1)\, P_k^{k-1} \Phi(k, k+1), \tag{7.46}$$

$$\bar{x}_{k+1}^{k} \triangleq \Phi(k+1, k)\, \bar{x}_k^{\,k}, \tag{7.47}$$

Eq. (7.45) takes the form

$$\bar{x}_{k+1}^{k+1} = [P_{k+1}^{k-1} + M^{\mathrm{T}}(k+1)\, R_{k+1}^{-1} M(k+1)]^{-1}\, [P_{k+1}^{k-1} \bar{x}_{k+1}^{k} + M^{\mathrm{T}}(k+1) R_{k+1}^{-1} y_{k+1}]. \tag{7.48}$$

It is easy to show by induction that the difference equations

$$P_{i+1}^{i-1} = \Phi^{\mathrm{T}}(i, i+1)\, P_i^{i-1} \Phi(i, i+1), \tag{7.49}$$

$$P_{i+1}^{i+1-1} = P_{i+1}^{i-1} + M^{\mathrm{T}}(i+1)\, R_{i+1}^{-1} M(i+1); \qquad i = 0, 1, \ldots; \qquad P_0^{0-1} = P_0^{-1}, \tag{7.50}$$

applied in the order listed, have P_k^{k-1} as defined in (7.44), as their solution.

Comparing (7.47), (7.49) with (7.28), and (7.48), (7.50) with (7.11), we see that we have derived the discrete Kalman–Bucy filter. At least, our recursive least squares filter has the same form as the Kalman–Bucy filter. We have not attached any statistical significance to the weighting matrices P_0 and R_k and need not do so. We simply have a recursive algorithm, which requires no data storage, for generating the least squares estimate. Once we make the identifications

$$x_0 \sim N(\bar{x}_0, P_0), \qquad v_k \sim N(0, R_k),$$

then we have the Kalman–Bucy filter. Then it is easy to compute from (7.41) directly that

$$\mathscr{E}\{(x_k - \bar{x}_k^k)(x_k - \bar{x}_k^k)^{\mathrm{T}}\} = P_k^k,$$

where P_k^k is given in (7.44). ▲

Example 7.3 (*Maximum Likelihood*). Ho [25], Schmidt [47], Rauch *et al.* [45], and others used a maximum likelihood approach to derive the discrete Kalman–Bucy filter (see also Smith *et al.* [50]). We derive the maximum likelihood (Bayesian) estimator (see Section 5.3) in this example.

Since all the densities in our linear problem are Gaussian, their means correspond to their respective modes. As a consequence, the mode (mean) and its error covariance matrix satisfy (7.28) between observations. The recursion for the mode (mean) at an observation is obtained by maximizing, with respect to x, the conditional density

$$p(x, t_k \mid Y_{t_k}),$$

which is given in Eq. (7.7). This is equivalent to minimizing the term in braces in (7.8). Setting the gradient of Eq. (7.8) with respect to x equal to zero, we get

$$-M^{\mathrm{T}}(k)\, R_k^{-1}(y_k - M(k)\, x_k) + P_k^{k-1-1}(x_k - \hat{x}_k^{k-1}) = 0. \tag{7.51}$$

The solution of (7.51), denote it by $\hat{x}_k^k$, is

$$\hat{x}_k^k = (M^{\mathrm{T}}(k)\, R_k^{-1} M(k) + P_k^{k-1^{-1}})^{-1}\, (M^{\mathrm{T}}(k)\, R_k^{-1} y_k + P_k^{k-1^{-1}} \hat{x}_k^{k-1}),$$

which is just the first of Eqs. (7.11). Using the matrix identities in Appendix 7B, we get, as before,

$$\hat{x}_k^k = \hat{x}_k^{k-1} + P_k^{k-1} M^{\mathrm{T}}(k)[M(k)\, P_k^{k-1} M^{\mathrm{T}}(k) + R_k]^{-1}\, (y_k - M(k)\, \hat{x}_k^{k-1}), \tag{7.14}$$

and the estimation error

$$\tilde{x}_k^k \triangleq x_k - \hat{x}_k^k = [I - K(k)\, M(k)]\, \tilde{x}_k^{k-1} - K(k)\, v_k\,, \tag{7.52}$$

where we have used (7.14), (7.2), and (7.15). Therefore, the estimation error covariance matrix is computed to be

$$\begin{aligned} P_k^k = \mathscr{E}\{\tilde{x}_k^k \tilde{x}_k^{k\mathrm{T}}\} &= [I - K(k)\, M(k)]\, P_k^{k-1} [I - K(k)\, M(k)]^{\mathrm{T}} + K(k)\, R_k K^{\mathrm{T}}(k) \\ &= P_k^{k-1} - K(k)\, M(k)\, P_k^{k-1}. \end{aligned} \tag{7.53}$$

Comparing (7.14) and (7.53) with (7.29), we see that we have derived the Kalman–Bucy filter. ▲

Example 7.4 (*Linear Minimum Variance Estimator*). Suppose the estimate and its error covariance evolve according to (7.28) between observations. At an observation, assume a *linear* estimator of the form

$$\hat{x}_k^k = \hat{x}_k^{k-1} + K(k)[y_k - M(k)\, \hat{x}_k^{k-1}], \tag{7.54}$$

where the gain $K(k)$ is to be determined so as to minimize the error variance

$$\mathscr{E}\{(x_k - \hat{x}_k^k)^{\mathrm{T}}\, (x_k - \hat{x}_k^k)\} = \operatorname{tr} \mathscr{E}\{(x_k - \hat{x}_k^k)(x_k - \hat{x}_k^k)^{\mathrm{T}}\} \triangleq \operatorname{tr} P_k^k. \tag{7.55}$$

This approach to the linear filtering problem was taken by Battin [2, 3], Schmidt [47], and Sorenson [51].

With considerable hindsight, let

$$K(k) = P_k^{k-1} M^{\mathrm{T}}(k)[M(k)\, P_k^{k-1} M^{\mathrm{T}}(k) + R_k]^{-1} + \hat{K}(k), \tag{7.56}$$

and find that $\hat{K}(k)$ which minimizes (7.55). Using (7.54)–(7.56), we get

$$\begin{aligned} \operatorname{tr} P_k^k = \operatorname{tr}[&P_k^{k-1} - P_k^{k-1} M^{\mathrm{T}} (M P_k^{k-1} M^{\mathrm{T}} + R_k)^{-1}\, M P_k^{k-1} \\ &+ \hat{K}(M P_k^{k-1} M^{\mathrm{T}} + R_k)\, \hat{K}^{\mathrm{T}}], \end{aligned} \tag{7.57}$$

where

$$P_k^{k-1} \triangleq \mathcal{E}\{(x_k - \hat{x}_k^{k-1})(x_k - \hat{x}_k^{k-1})^T\}.$$

Now

$$MP_k^{k-1}M^T + R_k > 0,$$

so that

$$\operatorname{tr} \hat{K}(MP_k^{k-1}M^T + R_k)\,\hat{K}^T \geqslant 0$$

for all $\hat{K}$. Therefore, the minimizing $\hat{K} = 0$, and $K(k)$ is the Kalman gain. From (7.57), we have the recursion for P. It is easy to rederive a known result. ▲

The several examples that follow consider some extensions to our filtering problem, as outlined in Section 5.4.

Example 7.5 (*Correlated System and Measurement Noise*: 1). Suppose the system noise (w_k) and measurement noise (v_k) are correlated, that is,

$$\mathcal{E}\{w_k v_l^T\} = C_k\,\delta_{kl}\,. \tag{7.58}$$

The discrete Kalman–Bucy filter for this case was derived by Kalman [36]. [Note that his problem is slightly different from ours in that the noise in our system (7.27) is $\Gamma(k)w_{k+1}$, whereas Kalman has $\Gamma(k)w_k$.]

We shall use the orthogonal projection lemma of Example 7.1 in our derivation. The reader can verify that the analysis in Example 7.1 up to, but not including, Eq. (7.36) applies in the present case. Equation (7.36) now becomes

$$\begin{aligned} 0 = {} & P_k^{k-1}M^T(k) + \Gamma(k-1)\,C_k - K_k{}^c[M(k)\,P_k^{k-1}M^T(k) + M(k)\,\Gamma(k-1)\,C_k \\ & + C_k{}^T\Gamma^T(k-1)\,M^T(k) + R_k], \end{aligned} \tag{7.59}$$

where, as before,

$$P_k^{k-1} = \Phi(k, k-1)\,P_{k-1}^{k-1}\Phi^T(k, k-1) + \Gamma(k-1)\,Q_k\Gamma^T(k-1). \tag{7.60}$$

Therefore, the Kalman gain for this problem is

$$\begin{aligned} K_k{}^c = {} & [P_k^{k-1}M^T(k) + \Gamma(k-1)\,C_k][M(k)\,P_k^{k-1}M^T(k) + M(k)\,\Gamma(k-1)\,C_k \\ & + C_k{}^T\Gamma^T(k-1)\,M^T(k) + R_k]^{-1}. \end{aligned} \tag{7.61}$$

Using (7.35) and (7.61), it is easy to compute the recursion for P:

$$P_k^k = P_k^{k-1} - K_k{}^c[M(k)\,P_k^{k-1} + C_k{}^T\Gamma^T(k-1)]. \tag{7.62}$$

Thus, between observations, the estimate and covariance matrix still evolve according to (7.28). At an observation, the estimate satisfies the first of Eqs. (7.29), but with K_k replaced by $K_k{}^c$. The covariance matrix satisfies (7.62). ▲

Example 7.6 (*Correlated System and Measurement Noise*: 2). Suppose that instead of (7.27), (7.2), our system is

$$\begin{aligned} x_{k+1} &= \Phi(k+1, k)\, x_k + \Gamma(k)\, w_k\,, \\ y_k &= M(k)\, x_k + v_k\,, \end{aligned} \tag{7.63}$$

with

$$\mathscr{E}\left\{\begin{bmatrix} w_k \\ v_k \end{bmatrix} [w_l{}^{\mathrm{T}}, v_l{}^{\mathrm{T}}]\right\} = \begin{bmatrix} Q_k & C_k \\ C_k{}^{\mathrm{T}} & R_k \end{bmatrix} \delta_{kl}\,. \tag{7.64}$$

This corresponds to the problem treated by Kalman [36] and others. We derive the optimal filter for this problem using the orthogonal projection lemma again, but in a slightly different form.

Instead of minimizing (7.30), we minimize

$$\mathscr{E}\{(x_{k+1} - \hat{x}_{k+1}^k)^{\mathrm{T}}\, (x_{k+1} - \hat{x}_{k+1}^k)\}.$$

This will enable us to compute a linear recursion for $\hat{x}_{k+1}^k$; that is, we will compute $\hat{x}_{k+1}^k$ as a linear function of $\hat{x}_k^{k-1}$. (In Example 7.1, we computed the recursion for $\hat{x}_k{}^k$.) The Kalman gain for this problem is more conveniently computed within this structure.

By the orthogonal projection lemma, we have

$$\mathscr{E}\{(x_{k+1} - \hat{x}_{k+1}^k)\, u_i{}^{\mathrm{T}}\} = 0 \qquad \text{for all} \quad i \leqslant k.$$

Computations analogous to those in Example 7.1 lead to

$$\hat{x}_{k+1}^k = \Phi(k+1, k)\, \hat{x}_k^{k-1} + \mathscr{E}\{x_{k+1} u_k{}^{\mathrm{T}}\}\, u_k\,,$$

which is the analog of (7.33). It is easy to show that

$$y_k - M(k)\, \hat{x}_k^{k-1}$$

is orthogonal to Y_{k-1} , and, since it is contained in Y_k , that

$$\mathscr{E}\{x_{k+1} u_k{}^{\mathrm{T}}\}\, u_k = K_k{}^d[y_k - M(k)\, \hat{x}_k^{k-1}].$$

Thus, the filter for this problem has the structure

$$\hat{x}_{k+1}^k = \Phi(k+1, k)\, \hat{x}_k^{k-1} + K_k{}^d[y_k - M(k)\, \hat{x}_k^{k-1}]. \tag{7.65}$$

It remains to determine the Kalman gain $K_k{}^d$.

From (7.63) and (7.65),

$$\tilde{x}_{k+1}^{k} \triangleq x_{k+1} - \hat{x}_{k+1}^{k} = \Phi(k+1,k)\,\tilde{x}_{k}^{k-1} + \Gamma(k)\,w_k - K_k{}^{d}[M(k)\,\tilde{x}_{k}^{k-1} + v_k], \tag{7.66}$$

and

$$y_k = M(k)(\tilde{x}_{k}^{k-1} + \hat{x}_{k}^{k-1}) + v_k\,.$$

Therefore, by the orthogonal projection lemma,

$$\begin{aligned} 0 &= \mathscr{E}\{\tilde{x}_{k+1}^{k} y_k{}^{\mathrm{T}}\} \\ &= \Phi(k+1,k)\,P_k^{k-1}M^{\mathrm{T}}(k) + \Gamma(k)\,C_k - K_k{}^{d}[M(k)\,P_k^{k-1}M^{\mathrm{T}}(k) + R_k], \end{aligned}$$

so that the Kalman gain is

$$K_k{}^{d} = [\Phi(k+1,k)\,P_k^{k-1}M^{\mathrm{T}}(k) + \Gamma(k)\,C_k][M(k)\,P_k^{k-1}M^{\mathrm{T}}(k) + R_k]^{-1}. \tag{7.67}$$

Using (7.66) and (7.67), it is easy to compute the recursion for P:

$$\begin{aligned} P_{k+1}^{k} \triangleq \mathscr{E}\{\tilde{x}_{k+1}^{k}\tilde{x}_{k+1}^{k\mathrm{T}}\} &= \Phi(k+1,k)\,P_k^{k-1}\Phi^{\mathrm{T}}(k+1,k) + \Gamma(k)Q_k\Gamma^{\mathrm{T}}(k) \\ &\quad - K_k{}^{d}[M(k)\,P_k^{k-1}\Phi^{\mathrm{T}}(k+1,k) + C_k{}^{\mathrm{T}}\Gamma^{\mathrm{T}}(k)].^{4} \end{aligned} \tag{7.68}$$

By setting $\Phi = I$, $\Gamma = 0$ in (7.65) and (7.68) (so that $x_{k+1} = x_k$), we have

$$\begin{aligned} \hat{x}_k{}^{k} &= \hat{x}_{k}^{k-1} + K_k(y_k - M(k)\,\hat{x}_{k}^{k-1}), \\ P_k{}^{k} &= P_k^{k-1} - K_kM(k)\,P_k^{k-1}, \end{aligned} \tag{7.69}$$

where

$$K_k = P_k^{k-1}M^{\mathrm{T}}(k)[M(k)\,P_k^{k-1}M^{\mathrm{T}}(k) + R_k]^{-1}. \tag{7.15}$$

Then, using (7.69) with (7.65) and (7.68),

$$\begin{aligned} \hat{x}_{k+1}^{k} &= \Phi(k+1,k)\,\hat{x}_k{}^{k} + \Gamma(k)\,C_k[M(k)\,P_k^{k-1}M^{\mathrm{T}}(k) + R_k]^{-1}\,(y_k - M(k)\,\hat{x}_{k}^{k-1}), \\ P_{k+1}^{k} &= \Phi(k+1,k)\,P_k{}^{k}\Phi^{\mathrm{T}}(k+1,k) + \Gamma(k)Q_k\Gamma^{\mathrm{T}}(k) \\ &\quad - \Gamma(k)\,C_k[M(k)\,P_k^{k-1}M^{\mathrm{T}}(k) + R_k]^{-1}\,C_k{}^{\mathrm{T}}\Gamma^{\mathrm{T}}(k) \\ &\quad - \Phi(k+1,k)\,K_kC_k{}^{\mathrm{T}}\Gamma^{\mathrm{T}}(k) - \Gamma(k)\,C_kK_k{}^{\mathrm{T}}\Phi^{\mathrm{T}}(k+1,k). \end{aligned} \tag{7.70}$$

Prediction for $k > l+1$, based on Y_l, is clearly accomplished via

$$\begin{aligned} \hat{x}_{k+1}^{l} &= \Phi(k+1,k)\,\hat{x}_k{}^{l}, \\ P_{k+1}^{l} &= \Phi(k+1,k)\,P_k{}^{l}\Phi^{\mathrm{T}}(k+1,k) + \Gamma(k)Q_k\Gamma^{\mathrm{T}}(k), \qquad k = l+1, l+2,..., \end{aligned}$$

[4] Equation (III$_d$), p. 303 in Kalman [36], which is analogous to this equation, is in error.

where $\hat{x}_{l+1}^{l}$, P_{l+1}^{l} is the filtering solution. This is easily verified directly from (7.63) (take the expectation with Y_l fixed). ▲

Example 7.7 (*Sequentially Correlated* (*Colored*) *Measurement Noise*). Consider the system (7.63), where $\{w_k\}$ is white, Gaussian, $w_k \sim N(0, Q_k)$, but $\{v_k\}$ is sequentially correlated, the output of the linear system (see Example 4.12)

$$v_{k+1} = \Psi(k+1, k)\, v_k + u_k, \qquad v_0 \sim N(0, V), \qquad \mathcal{E}\{x_0 v_0^{\mathrm{T}}\} = 0. \tag{7.71}$$

Here $\{u_k\}$ is also white, Gaussian, $u_k \sim N(0, S_k)$, independent of $\{w_k\}$.

We could proceed as follows. Define the *augmented* state vector

$$\bar{x}_k^{\mathrm{T}} = [x_k^{\mathrm{T}}, \quad v_k^{\mathrm{T}}]$$

and the matrices

$$\bar{\Phi} = \begin{bmatrix} \Phi, & 0 \\ 0, & \Psi \end{bmatrix}, \qquad \bar{\Gamma} = \begin{bmatrix} \Gamma, & 0 \\ 0, & I \end{bmatrix}, \qquad \bar{w}_k^{\mathrm{T}} = [w_k^{\mathrm{T}}, \quad u_k^{\mathrm{T}}],$$

$$\bar{M} = [M, \quad I], \qquad \bar{Q}_k = \begin{bmatrix} Q_k, & 0 \\ 0, & S_k \end{bmatrix}.$$

Then the augmented dynamical system is

$$\bar{x}_{k+1} = \bar{\Phi}(k+1, k)\, \bar{x}_k + \bar{\Gamma}(k)\, \bar{w}_k, \qquad y_k = \bar{M}(k)\, \bar{x}_k, \tag{7.72}$$

with *perfect* measurements uncontaminated by noise. Assuming $\bar{M}(k)\, \bar{P}_k^{k-1} \bar{M}^{\mathrm{T}}(k)$ is positive definite,[5] the Kalman–Bucy filter of Theorem 7.2 can be used in system (7.72). Recall that its derivation in Example 7.1 does not require a nonsingular measurement noise covariance matrix. In fact, the derivation stands in the absence of measurement noise ($R_k \equiv 0$). Thus the difference equations for the augmented state estimation error covariance matrix are

$$\begin{aligned} \bar{P}_k^{\,k} &= \bar{P}_k^{k-1} - \bar{P}_k^{k-1} \bar{M}^{\mathrm{T}}(k) [\bar{M}(k)\, \bar{P}_k^{k-1} \bar{M}^{\mathrm{T}}(k)]^{-1}\, \bar{M}(k)\, \bar{P}_k^{k-1}, \\ \bar{P}_{k+1}^{k} &= \bar{\Phi}(k+1, k)\, \bar{P}_k^{\,k} \bar{\Phi}^{\mathrm{T}}(k+1, k) + \bar{\Gamma}(k)\, \bar{Q}_k \bar{\Gamma}^{\mathrm{T}}(k). \end{aligned} \tag{7.73}$$

Now since the linear combinations of state variables $\bar{M}(k)\bar{x}_k$ are known precisely at time t_k, $\bar{P}_k^{\,k}$ must be singular. In fact, from (7.73),

$$\bar{M}(k)\, \bar{P}_k^{\,k} \bar{M}^{\mathrm{T}}(k) = 0.$$

[5] In fact, this matrix need not be positive definite, if, following Kalman [36], we use the pseudo-inverse rather than the inverse in (7.73).

As a consequence, $\bar{P}_{k+1}^{k}$ can be ill-conditioned, leading to difficulties in the computation of $\bar{P}_{k+1}^{k+1}$. Furthermore, the dimension of the augmented state filter is $n + m$, which also compounds the computational problems.

Following Bryson and Hendrickson [9], we introduce a measurement differencing scheme that results in measurements with additive white noise and leads to an equivalent filter of dimension n, in which computational ill-conditioning is eliminated, or at least reduced. With the aid of (7.71), the measurement can be written as

$$y_k = M(k)\, x_k + \Psi(k, k-1)\, v_{k-1} + u_{k-1}\,.$$

It is easily seen that v_{k-1} can be eliminated from y_k by subtracting

$$\Psi(k, k-1)\, y_{k-1}\,.$$

Thus,

$$\begin{aligned} y_k - \Psi(k, k-1)\, y_{k-1} &= M(k)\, x_k + u_{k-1} - \Psi(k, k-1)\, M(k-1)\, x_{k-1} \\ &= [M(k)\, \Phi(k, k-1) - \Psi(k, k-1)\, M(k-1)]\, x_{k-1} \\ &\quad + M(k)\, \Gamma(k-1)\, w_{k-1} + u_{k-1}\,. \end{aligned}$$

Let

$$H(k-1) \triangleq M(k)\, \Phi(k, k-1) - \Psi(k, k-1)\, M(k-1), \tag{7.74}$$

and define the measurements

$$\begin{aligned} &\xi_1 \triangleq y_1 = M(1)\, x_1 + v_1\,, \\ &\xi_k \triangleq y_k - \Psi(k, k-1)\, y_{k-1} = H(k-1)\, x_{k-1} + M(k)\, \Gamma(k-1)\, w_{k-1} + u_{k-1}\,, \\ &\qquad\qquad k \geqslant 2. \end{aligned} \tag{7.75}$$

The measurements ξ_k, $k \geqslant 2$, now contain additive white noise which, we note, is correlated with the system noise (w). The general approach is this. We process ξ_1 with the augmented-state filter described earlier. Subsequent observations ξ_k will be processed via the filter developed in Example 7.6, where the system is

$$x_k = \Phi(k, k-1)\, x_{k-1} + \Gamma(k-1)\, w_{k-1}\,.$$

Thus, aside from the starting procedure (processing ξ_1), the filter will be of dimension n. Our measurement noise covariance matrix will be

$$R_{k-1} = M(k)\, \Gamma(k-1)\, Q_{k-1} \Gamma^{\mathrm{T}}(k-1)\, M^{\mathrm{T}}(k) + S_{k-1}\,,$$

which, we assume, is positive definite.

Assume, for simplicity, that $t_0 = t_1$. That is, the first observation is made at the initial time t_0. [If this is not the case, then a prediction via (7.28) propagates the initial conditions to t_1.] Then applying the augmented-state filter to ξ_1 results in the following estimate of x_1 (*not* $\bar{x}_1$) and the associated covariance matrix:

$$\begin{aligned} \hat{x}_1^1 &= \hat{x}_0 + P_0 M^{\mathrm{T}}(1)[M(1)\, P_0 M^{\mathrm{T}}(1) + V]^{-1}(y_1 - M(1)\,\hat{x}_0), \\ P_1^1 &= P_0 - P_0 M^{\mathrm{T}}(1)[M(1)\, P_0 M^{\mathrm{T}}(1) + V]^{-1} M(1)\, P_0 . \end{aligned} \tag{7.76}$$

Now we use the n-dimensional filter of Example 7.6 on measurements ξ_k, $k \geqslant 2$, with initial conditions $\hat{x}_1^1$ and P_1^1 from (7.76).

The system is

$$\begin{aligned} &x_{k+1} = \Phi(k+1, k)\, x_k + \Gamma(k)\, w_k , \\ &\xi_{k+1} = y_{k+1} - \Psi(k+1, k)\, y_k = H(k)\, x_k + M(k+1)\, \Gamma(k)\, w_k + u_k , \quad k \geqslant 1, \\ &H(k) = M(k+1)\, \Phi(k+1, k) - \Psi(k+1, k)\, M(k). \end{aligned} \tag{7.77}$$

Note that ξ_{k+1}, although it depends on x_k and is to be formally treated as a measurement at t_k, depends on y_{k+1}. As a consequence, in applying the filter equations (7.65) and (7.68) to (7.77), $\hat{x}_{k+1}^k$ and P_{k+1}^k in those filter equations are, in fact, $\hat{x}_{k+1}^{k+1}$ and P_{k+1}^{k+1}, respectively. That is to say, they are conditioned on Y_{k+1}, *not* on Y_k. Keeping this in mind, and noting that y_k in (7.65) is now ξ_{k+1}, and M is now H, we have

$$\begin{aligned} \hat{x}_{k+1}^{k+1} &= \Phi(k+1, k)\, \hat{x}_k^{\,k} + [\Phi(k+1, k)\, P_k^{\,k} H^{\mathrm{T}}(k) + \Gamma(k)\, C_k] \\ &\quad \times [H(k)\, P_k^{\,k} H^{\mathrm{T}}(k) + R_k]^{-1} [\xi_{k+1} - H(k)\, \hat{x}_k^{\,k}], \\ P_{k+1}^{k+1} &= \Phi(k+1, k)\, P_k^{\,k} \Phi^{\mathrm{T}}(k+1, k) + \Gamma(k) Q_k \Gamma^{\mathrm{T}}(k) \\ &\quad - [\Phi(k+1, k)\, P_k^{\,k} H^{\mathrm{T}}(k) + \Gamma(k)\, C_k] \\ &\quad \times [H(k)\, P_k^{\,k} H^{\mathrm{T}}(k) + R_k]^{-1} [H(k)\, P_k^{\,k} \Phi^{\mathrm{T}}(k+1, k) + C_k^{\mathrm{T}} \Gamma^{\mathrm{T}}(k)], \end{aligned} \tag{7.78}$$

where

$$\begin{aligned} C_k &= Q_k \Gamma^{\mathrm{T}}(k)\, M^{\mathrm{T}}(k+1), \\ R_k &= M(k+1)\, \Gamma(k) Q_k \Gamma^{\mathrm{T}}(k)\, M^{\mathrm{T}}(k+1) + S_k . \end{aligned} \tag{7.79}$$

Note that the reduced filter (7.78) requires the storage of one measurement, since ξ_k is the difference of two consecutive measurements.

The reduced filter does not generate estimates of v_k, but these

estimates are, in general, not desired. It is easy to compute them, however. From the measurement equation (7.63), it readily follows that

$$\begin{aligned} \hat{v}_k{}^k &= y_k - M(k)\,\hat{x}_k{}^k, \\ V_k{}^k &\triangleq \mathscr{E}\{(v_k - \hat{v}_k{}^k)(v_k - \hat{v}_k{}^k)^{\mathrm{T}}\} = M(k)\,P_k{}^k M^{\mathrm{T}}(k). \end{aligned} \tag{7.80}$$

From the system equations (7.77) and (7.71), it is easy to compute the prediction equations

$$\begin{aligned} \hat{x}_{k+1}^l &= \Phi(k+1,k)\,\hat{x}_k{}^l, \\ v_{k+1}^l &= \Psi(k+1,k)\,\hat{v}_k{}^l, \\ P_{k+1}^l &= \Phi(k+1,k)\,P_k{}^l\Phi^{\mathrm{T}}(k+1,k) + \Gamma(k)Q_k\Gamma^{\mathrm{T}}(k), \\ V_{k+1}^l &= \Psi(k+1,k)\,V_k{}^l\Psi^{\mathrm{T}}(k+1,k) + S_k\,, \qquad k = l, l+1,..., \end{aligned} \tag{7.81}$$

where $\hat{x}_l{}^l$, $v_l{}^l$, $P_l{}^l$, $V_l{}^l$ are the filtering solutions.

Now the reduced filter (7.78) was derived formally, assuming that optimal filtering of observations ξ_k is equivalent to optimal filtering of observations y_k. This latter fact remains to be proved. Actually, it is easy to prove using the orthogonal projection lemma (see Rishel [46]). The fact that (7.78) is the (unique) minimum variance filter will be proved if we show that

$$\mathscr{E}\{(x_k - \hat{x}_k{}^k)\,y_l{}^{\mathrm{T}}\} = 0, \qquad \text{all} \quad l \leqslant k. \tag{7.82}$$

Now it was proved in Example 7.6 that

$$\mathscr{E}\{(x_k - \hat{x}_k{}^k)\,\xi_l{}^{\mathrm{T}}\} = 0, \qquad \text{all} \quad l \leqslant k. \tag{7.83}$$

But $\xi_1 = y_1$, so (7.82) is true for $l = 1$. Suppose (7.82) is true for l. Then

$$\mathscr{E}\{(x_k - \hat{x}_k{}^k)\,y_{l+1}^{\mathrm{T}}\} = \mathscr{E}\{(x_k - \hat{x}_k{}^k)\,\xi_{l+1}^{\mathrm{T}}\} + \mathscr{E}\{(x_k - \hat{x}_k{}^k)\,y_l{}^{\mathrm{T}}\Psi^{\mathrm{T}}\} = 0,$$

using (7.77), (7.83), and the inductive hypothesis. This proves (7.82). Since

$$v_k - \hat{v}_k{}^k = M(k)(x_k - \hat{x}_k{}^k),$$

(7.82) also implies that

$$\mathscr{E}\{(v_k - \hat{v}_k{}^k)\,y_l{}^{\mathrm{T}}\} = 0, \qquad \text{all} \quad l \leqslant k. \quad \blacktriangle$$

Example 7.8 (Linear Smoother). We develop here one form of the linear smoother using the Bayesian maximum likelihood approach of

Rauch *et al.* [45]. Other derivations and (equivalent) forms of the linear smoother may be found in the references cited in Section 5.1.

Our system is (7.27), (7.2) with $\{w_k\}$ and $\{v_k\}$ independent. We assume for simplicity (see Section 3.9) that $\Gamma(k)$ and $Q(k)$ are $n \times n$ and nonsingular. The solution in the general case, without this assumption, is the same, but the derivation requires the cumbersome machinery developed in Section 3.9 to get an expression for the transition density. We leave that case for the reader. Hint: the Dirac delta function is formally maximized when its argument is zero. Set its argument equal to zero, and use that relation as a constraint.

Assuming the filtering solution $(\hat{x}_k^k, P_k^k)$ is available, we seek a recursion for $\hat{x}_k^l$, $k < l$. Now the conditional means $\hat{x}_k^l$ and $\hat{x}_{k+1}^l$ are those values of x_k and x_{k+1}, respectively, which maximize the (Gaussian) density

$$p(x_k, x_{k+1} \mid Y_l).$$

Now

$$\begin{aligned} p(x_k, x_{k+1} \mid Y_l) &= \frac{p(x_k, x_{k+1}, Y_l)}{p(Y_l)} = \frac{p(x_k, x_{k+1}, Y_k, y_{k+1}, \ldots, y_l)}{p(Y_l)} \\ &= \frac{p(Y_k)}{p(Y_l)}\, p(x_k, x_{k+1}, y_{k+1}, \ldots, y_l \mid Y_k) \\ &= \frac{p(Y_k)}{p(Y_l)}\, p(y_{k+1}, \ldots, y_l \mid x_k, x_{k+1}, Y_k)\, p(x_k, x_{k+1} \mid Y_k). \end{aligned}$$

But

$$p(y_{k+1}, \ldots, y_l \mid x_k, x_{k+1}, Y_k) = p(y_{k+1}, \ldots, y_l \mid x_{k+1}),$$

and

$$p(x_k, x_{k+1} \mid Y_k) = p(x_{k+1} \mid x_k, Y_k)\, p(x_k \mid Y_k) = p(x_{k+1} \mid x_k)\, p(x_k \mid Y_k),$$

in view of the properties of our system, so that

$$p(x_k, x_{k+1} \mid Y_l) = c(x_{k+1})\, p(x_{k+1} \mid x_k)\, p(x_k \mid Y_k), \tag{7.84}$$

where c is independent of x_k.

The densities in (7.84) are Gaussian, specifically,

$$\begin{aligned} p(x_{k+1} \mid x_k) &\sim N[\Phi(k+1, k)\, x_k, \Gamma(k) Q_{k+1} \Gamma^{\mathrm{T}}(k)], \\ p(x_k \mid Y_k) &\sim N(\hat{x}_k^k, P_k^k). \end{aligned}$$

Therefore, to maximize (7.84) with respect to x_k, x_{k+1}, we minimize

$$\begin{aligned} &\tfrac{1}{2}[x_{k+1} - \Phi(k+1, k)\, x_k]^{\mathrm{T}} [\Gamma(k) Q_{k+1} \Gamma^{\mathrm{T}}(k)]^{-1} [x_{k+1} - \Phi(k+1, k)\, x_k] \\ &\quad + \tfrac{1}{2}(x_k - \hat{x}_k^k)^{\mathrm{T}} P_k^{k-1} (x_k - \hat{x}_k^k) + d(x_{k+1}), \end{aligned}$$

where d is independent of x_k. Suppose $\hat{x}_{k+1}^{l}$, the minimizing x_{k+1}, is available. Then $\hat{x}_k{}^{l}$ minimizes

$$J = \tfrac{1}{2}[\hat{x}_{k+1}^{l} - \Phi x_k]^{\mathrm{T}} [\Gamma Q_{k+1} \Gamma^{\mathrm{T}}]^{-1} [\hat{x}_{k+1}^{l} - \Phi x_k] + \tfrac{1}{2}(x_k - \hat{x}_k{}^{k})^{\mathrm{T}} P_k^{k-1} (x_k - \hat{x}_k{}^{k})$$

with respect to x_k. Setting the gradient of J with respect to x_k equal to zero, we get

$$\hat{x}_k{}^{l} = [P_k^{k-1} + \Phi^{\mathrm{T}}(k+1, k)[\Gamma(k) Q_{k+1} \Gamma^{\mathrm{T}}(k)]^{-1} \Phi(k+1, k)]^{-1} \times [P_k^{k-1} \hat{x}_k{}^{k} + \Phi^{\mathrm{T}}(k+1, k)[\Gamma(k) Q_{k+1} \Gamma^{\mathrm{T}}(k)]^{-1} \hat{x}_{k+1}^{l}]. \tag{7.85}$$

Applying Eqs. (7B.5) and (7B.6) of Appendix 7B, we have

$$\hat{x}_k{}^{l} = \hat{x}_k{}^{k} + S_k(\hat{x}_{k+1}^{l} - \Phi(k+1, k)\, \hat{x}_k{}^{k}), \tag{7.86}$$

where

$$S_k = P_k{}^{k} \Phi^{\mathrm{T}}(k+1, k)[\Phi(k+1, k)\, P_k{}^{k} \Phi^{\mathrm{T}}(k+1, k) + \Gamma(k) Q_{k+1} \Gamma^{\mathrm{T}}(k)]^{-1} = P_k{}^{k} \Phi^{\mathrm{T}}(k+1, k)\, P_{k+1}^{k-1}. \tag{7.87}$$

Equation (7.86) is the smoothing algorithm. The computation goes backwards (in the index k) with $\hat{x}_l{}^{l}$, the filtering solution, as initial condition. The filter solutions $(\hat{x}_k{}^{k}, P_k{}^{k})$ are required in the computations.

We now develop a recursion for the smoothing error covariance matrix. From (7.86),

$$x_k - \hat{x}_k{}^{l} = x_k - \hat{x}_k{}^{k} - S_k\, (\hat{x}_{k+1}^{l} - \Phi \hat{x}_k{}^{k}),$$

or

$$\tilde{x}_k{}^{l} + S_k \hat{x}_{k+1}^{l} = \tilde{x}_k{}^{k} + S_k \Phi \hat{x}_k{}^{k}.$$

Squaring both sides and taking the expectation, we get

$$P_k{}^{l} + S_k \mathscr{E}\{\hat{x}_{k+1}^{l} \hat{x}_{k+1}^{l^{\mathrm{T}}}\}\, S_k{}^{\mathrm{T}} = P_k{}^{k} + S_k \Phi \mathscr{E}\{\hat{x}_k{}^{k} \hat{x}_k^{k^{\mathrm{T}}}\}\, \Phi^{\mathrm{T}} S_k{}^{\mathrm{T}}, \tag{7.88}$$

where we have used the fact that

$$\mathscr{E}\{\tilde{x}_k{}^{l} \hat{x}_{k+1}^{l^{\mathrm{T}}}\} = \mathscr{E}\{\tilde{x}_k{}^{k} \hat{x}_k^{k^{\mathrm{T}}}\} = 0. \tag{7.89}$$

It is easy to see that (7.89) holds. For example,

$$\mathscr{E}\{\tilde{x}_k{}^{l} \hat{x}_{k+1}^{l^{\mathrm{T}}}\} = \mathscr{E}\{\tilde{x}_k{}^{l} \hat{x}_k^{l^{\mathrm{T}}}\}\, \Phi^{\mathrm{T}} = \mathscr{E}\{\mathscr{E}\{\tilde{x}_k{}^{l} \hat{x}_k^{l^{\mathrm{T}}} \mid Y_l\}\}\, \Phi^{\mathrm{T}} = \mathscr{E}\{\mathscr{E}\{\tilde{x}_k{}^{l} \mid Y_l\}\, \hat{x}_k^{l^{\mathrm{T}}}\}\, \Phi^{\mathrm{T}} = \mathscr{E}\{(\hat{x}_k{}^{l} - \hat{x}_k{}^{l})\, \hat{x}_k^{l^{\mathrm{T}}}\}\, \Phi^{\mathrm{T}} = 0.$$

Now

$$\mathscr{E}\{x_{k+1}x_{k+1}^{\mathrm{T}}\} = \mathscr{E}\{\hat{x}_{k+1}^{l}\hat{x}_{k+1}^{l\mathrm{T}}\} + P_{k+1}^{l} \tag{7.90}$$

in view of (7.89). Similarly,

$$\mathscr{E}\{x_k x_k^{\mathrm{T}}\} = \mathscr{E}\{\hat{x}_k^{k}\hat{x}_k^{k\mathrm{T}}\} + P_k^{k}. \tag{7.91}$$

Also,

$$\mathscr{E}\{x_{k+1}x_{k+1}^{\mathrm{T}}\} = \Phi\mathscr{E}\{x_k x_k^{\mathrm{T}}\}\,\Phi^{\mathrm{T}} + \Gamma Q_{k+1}\Gamma^{\mathrm{T}}. \tag{7.92}$$

Using (7.90)–(7.92) in (7.88), we get

$$P_k^{l} = P_k^{k} + S_k[P_{k+1}^{l} - \Phi(k+1,k)\,P_k^{k}\Phi^{\mathrm{T}}(k+1,k) - \Gamma(k)\,Q_{k+1}\Gamma^{\mathrm{T}}(k)]\,S_k^{\mathrm{T}} \tag{7.93}$$

or

$$P_k^{l} = P_k^{k} + S_k(P_{k+1}^{l} - P_{k+1}^{k})\,S_k^{\mathrm{T}}, \tag{7.94}$$

which is the desired recursion. It is also computed backwards, with P_l^{l} as initial condition.

Suppose our system is noise-free ($\Gamma = 0$). Then, from (7.86) and (7.94), it is easy to see that smoothing is simply *backward* prediction:

$$\hat{x}_k^{l} = \Phi^{-1}(k+1,k)\,\hat{x}_{k+1}^{l},$$

$$P_k^{l} = \Phi^{-1}(k+1,k)\,P_{k+1}^{l}\Phi^{-\mathrm{T}}(k+1,k).$$

See Bryson and Henrikson [9] for the linear smoothing algorithm in the case of sequentially correlated measurement noise. ▲

4. CONTINUOUS FILTER

Our continuous linear system is described by the vector (Itô) stochastic differential equation (7.1), which we repeat here,

$$dx_t = F(t)\,x_t\,dt + G(t)\,d\beta_t, \qquad t \geqslant t_0, \tag{7.95}$$

and the vector (Itô) observation equation,

$$dz_t = M(t)\,x_t\,dt + d\eta_t, \qquad t \geqslant t_0. \tag{7.96}$$

Equation (7.95) is described in Section 2. The observed process $\{z_t, t \geqslant t_0\}$ is an m-vector process, $M(t)$ is an $m \times n$, nonrandom, continuous matrix time-function, and $\{\eta_t, t \geqslant t_0\}$ is an m-vector Brownian motion process with $\mathscr{E}\{d\eta_t\,d\eta_t^{\mathrm{T}}\} = R(t)\,dt$, $R(t) > 0$. x_{t_0},

$\{\beta_t\}$, and $\{\eta_t\}$ are assumed independent. The formal analogy between (7.95), (7.96), and

$$dx_t/dt = F(t)\,x_t + G(t)\,w_t\,, \tag{7.95'}$$

$$y_t = M(t)\,x_t + v_t\,, \tag{7.96'}$$

where $\{w_t\}$ and $\{v_t\}$ are white Gaussian noise processes, $w_t \sim N(0, Q(t))$, $v_t \sim N(0, R(t))$, was noted in Section 5.1. (See also Chapter 4, especially Section 4.8.)

For this linear filtering problem, Kushner's equation (6.79) of Theorem 6.5 becomes

$$\begin{aligned} dp = {} & -p\,\mathrm{tr}(F)\,dt - p_x{}^{\mathrm{T}}Fx\,dt + \tfrac{1}{2}\,\mathrm{tr}(GQG^{\mathrm{T}}p_{xx})\,dt \\ & + (x - \hat{x})^{\mathrm{T}}\,M^{\mathrm{T}}R^{-1}(dz_t - M\hat{x}\,dt)\,p. \end{aligned} \tag{7.97}$$

We obtain the equations for the conditional mean and covariance matrix by specializing Theorem 6.6 to the linear case. Since $f = Fx$, $h = Mx$,

$$(\widehat{xh}^{\mathrm{T}} - \hat{x}\hat{h}^{\mathrm{T}}) = (\widehat{xx}^{\mathrm{T}} - \hat{x}\hat{x}^{\mathrm{T}})\,M^{\mathrm{T}},$$

and similarly for the terms involving xf^{T}. Since, in this case, the conditional density is Gaussian, its third central moments are zero,

$$[(x_i - \hat{x}_i)(x_j - \hat{x}_j)(x_k - \hat{x}_k)]\hat{} = 0,$$

and, as a consequence, the last term in (6.101) is zero. We therefore have

Theorem 7.3. *The optimal (minimum variance) filter for the continuous system* (7.95), (7.96) *consists of the following differential equations for the conditional mean and covariance matrix*:

$$d\hat{x}_t{}^t = F(t)\,\hat{x}_t{}^t\,dt + P_t{}^tM^{\mathrm{T}}(t)\,R^{-1}(t)(dz_t - M(t)\,\hat{x}_t{}^t\,dt), \tag{7.98}$$

$$dP_t{}^t/dt = F(t)\,P_t{}^t + P_t{}^tF^{\mathrm{T}}(t) + G(t)\,Q(t)\,G^{\mathrm{T}}(t) - P_t{}^tM^{\mathrm{T}}(t)\,R^{-1}(t)\,M(t)\,P_t{}^t,$$

$$t \geqslant t_0\,. \tag{7.99}$$

Prediction is accomplished as in the continuous-discrete problem (Theorem 7.1). [*Set* $R^{-1} \equiv 0$ *in* (7.98) *and* (7.99).]

This is the now well-known continuous Kalman–Bucy filter [33, 36]. The *variance equation* (7.99) is the well-known matrix *Riccati* equation. It is independent of the observed process and, as a result, the Kalman gain

$$K(t) = P_t{}^tM^{\mathrm{T}}(t)\,R^{-1}(t) \tag{7.100}$$

may be precomputed. The discussion of Theorem 7.1 applies to Theorem 7.3 as well. Since the Kalman gain is a nonrandom time-function, the coefficient of dz_t in the filter equation (7.98) is nonrandom. As a result, Eq. (7.98) may be manipulated by formal rules (see Section 4.6). We may write (7.98) as

$$d\hat{x}_t{}^t/dt = F(t)\,\hat{x}_t{}^t + P_t{}^t M^{\mathrm{T}}(t)\,R^{-1}(t)(y_t - M(t)\,\hat{x}_t{}^t). \tag{7.98}$$

Compare Theorem 7.3 with Example 6.4.

The mechanization of the continuous Kalman–Bucy filter requires the solution of the variance equation (7.99). We must verify that the variance equation has a unique solution existing for all $t \geqslant t_0$.

Now since the matrices F, G, M, Q, and R are assumed continuous, and, being quadratic in P, the right-hand side of (7.99) is locally Lipschitzian, standard existence and uniqueness proofs (see Coddington and Levinson [11]) show that the variance equation has a unique solution locally. We now show that the unique solution of (7.99) can be continued for all $t \geqslant t_0$, provided P_{t_0} is positive semidefinite (a covariance matrix). It suffices to show that $P_t{}^t$ is bounded for all $t \geqslant t_0$ [11]. By Theorem 2.13,

$$P_t{}^t = P_t^{t_0} - B,$$

where $B \geqslant 0$ and $P_t^{t_0}$ is the unconditioned covariance matrix of the solution of (7.95) given only the initial condition. That is,

$$P_t{}^t \leqslant P_t^{t_0}. \tag{7.101}$$

But it is easy to compute from (7.95) [see (7.19)–(7.25)] that

$$P_t^{t_0} = \Phi(t, t_0)\,P_{t_0}\Phi^{\mathrm{T}}(t, t_0) + \int_{t_0}^{t} \Phi(t, \tau)\,G(\tau)\,Q(\tau)\,G^{\mathrm{T}}(\tau)\,\Phi^{\mathrm{T}}(t, \tau)\,d\tau, \tag{7.102}$$

which is bounded for all $t \geqslant t_0$. This, together with (7.101), shows that $P_t{}^t$ is bounded for all $t \geqslant t_0$. As a result, the variance equation has a unique solution existing for all $t \geqslant t_0$.

It is useful to note, and can easily be verified, that the solution of the (nonlinear) variance equation can be obtained as a solution of the linear system:

$$\begin{aligned} dX/dt &= -F^{\mathrm{T}}(t)\,X + M^{\mathrm{T}}(t)\,R^{-1}(t)\,M(t)\,Y, & X(t_0) &= I, \\ dY/dt &= G(t)Q(t)\,G^{\mathrm{T}}(t)\,X + F(t)\,Y, & Y(t_0) &= P_{t_0}. \end{aligned} \tag{7.103}$$

For then

$$Y(t) = P_t{}^t X(t) \qquad \text{for all} \quad t \geqslant t_0, \tag{7.104}$$

so that

$$P_t^t = Y(t)\, X^{-1}(t). \tag{7.105}$$

This result was derived by Kalman [36].

The next few examples give alternate derivations of the continuous Kalman–Bucy filter.

Example 7.9 (Conditional Characteristic Function). We use here the method of Example 4.15 (characteristic function), together with Itô's stochastic calculus, to derive the Kalman–Bucy filter from Kushner's equation (7.97). The conditional characteristic function (2.122) is defined as

$$\varphi(u, t, z_t) = \int e^{iu^{\mathrm{T}}x} p(x, t \mid Y_t)\, dx.$$

Therefore,

$$d\varphi = \int e^{iu^{\mathrm{T}}x}\, (dp)\, dx.$$

Substituting from Kushner's equation and carrying out the integrations by parts, we get

$$d\varphi = u^{\mathrm{T}}F\varphi_u\, dt - \tfrac{1}{2}\varphi u^{\mathrm{T}}GQG^{\mathrm{T}}u\, dt - (i\varphi_u{}^{\mathrm{T}} + \varphi \hat{x}_t^{t\mathrm{T}})\, M^{\mathrm{T}}(t)\, R^{-1}(t)\, (dz_t - M(t)\, \hat{x}_t{}^t\, dt). \tag{7.106}$$

This is a stochastic partial differential equation for the conditional characteristic function.

Now since p is Gaussian, we know that the conditional characteristic function is given by (2.130):

$$\varphi(u, t, z_t) = \exp(iu^{\mathrm{T}}\hat{x}_t{}^t - \tfrac{1}{2}u^{\mathrm{T}}P_t{}^t u), \tag{7.107}$$

so that

$$\varphi_u = \varphi(i\hat{x}_t{}^t - P_t{}^t u).$$

Substituting this in (7.106), we get

$$\begin{aligned} d\varphi = {} & \varphi iu^{\mathrm{T}}[F\hat{x}_t{}^t\, dt + P_t{}^t M^{\mathrm{T}} R^{-1}\, (dz_t - M\hat{x}_t{}^t\, dt)] \\ & - \tfrac{1}{2}\varphi u^{\mathrm{T}}(FP_t{}^t + P_t{}^t F^{\mathrm{T}} + GQG^{\mathrm{T}})\, u\, dt. \end{aligned} \tag{7.108}$$

We next compute $d\varphi$ from (7.107). Since φ is a function of $\hat{x}_t{}^t$ and $P_t{}^t$, which may be solutions of stochastic differential equations, we must use Itô's Lemma 4.2. Applying Itô's lemma, we first compute the imaginary part of $d\varphi$ from (7.107). Note that this part will not involve

partials with respect to the elements of P_t^t, nor will it involve $\varphi_{\hat{x}\hat{x}}$, since these are real. Thus we have

$$\operatorname{Im}(d\varphi/\varphi) = iu^{\mathrm{T}}\, d\hat{x}_t^t.$$

Comparing this with (7.108), we immediately have

$$d\hat{x}_t^t = F\hat{x}_t^t\, dt + P_t^t M^{\mathrm{T}} R^{-1}\,(dz_t - M\hat{x}_t^t\, dt), \tag{7.98}$$

which is (7.98).

Now we compute $\operatorname{Re}(d\varphi)$ from (7.107). Using Itô's lemma,

$$\begin{aligned}\operatorname{Re}(d\varphi/\varphi) = {} & -\tfrac{1}{2}\operatorname{tr} P_t^t M^{\mathrm{T}} R^{-1} R R^{-1} M P_t^t u u^{\mathrm{T}}\, dt - \tfrac{1}{2}\operatorname{tr} u u^{\mathrm{T}}\, dP_t^t \\ & + (\text{terms quartic in the elements of } u \text{ resulting from second} \\ & \quad \text{partials of } \varphi \text{ with respect to the elements of } P_t^t),\end{aligned} \tag{7.109}$$

where we have used the fact that $\hat{x}_t^t$ satisfies (7.98). Now there are no terms quartic in the elements of u in (7.108). Therefore, such terms must have zero coefficients in (7.109). Thus (7.109) becomes

$$\operatorname{Re}(d\varphi/\varphi) = -\tfrac{1}{2}u^{\mathrm{T}} P_t^t M^{\mathrm{T}} R^{-1} M P_t^t u\, dt - \tfrac{1}{2}u^{\mathrm{T}}\, dP_t^t u,$$

and, comparing this with (7.108), we get (7.99). ▲

Example 7.10 (Limit of Discrete Filter). We obtain the continuous filter (Theorem 7.3) from the discrete filter (Theorem 7.2) by the formal limiting process that was already employed in Example 4.18. For $\Delta t = t_{k+1} - t_k$ small, Eq. (7.95′) is integrated to produce the difference equation

$$x_{t_{k+1}} - x_{t_k} = F(t_k)\, x_{t_k}\, \Delta t + \Delta t\, G(t_k)\, w_{t_{k+1}} + o(\Delta t)$$

or

$$x_{t_{k+1}} = [I + \Delta t\, F(t_k)]\, x_{t_k} + \Delta t\, G(t_k)\, w_{t_{k+1}}, \tag{7.110}$$

where

$$\mathscr{E}\{w_{t_k} w_{t_k}^{\mathrm{T}}\} = Q(t_k)/\Delta t,$$

according to Example 3.20. The observations (7.96′) are

$$y_{t_k} = M(t_k)\, x_{t_k} + v_{t_k}, \tag{7.111}$$

with

$$\mathscr{E}\{v_{t_k} v_{t_k}^{\mathrm{T}}\} = R(t_k)/\Delta t.$$

Comparing (7.110), (7.111) with (7.27), (7.2),

$$\Phi(t_{k+1}, t_k) = I + \Delta t\, F(t_k), \qquad \Gamma(t_k) = \Delta t\, G(t_k). \tag{7.112}$$

Using these relations in Theorem 7.2, we get

$$\begin{aligned}
(\hat{x}_{t_{k+1}}^{t_{k+1}} - \hat{x}_{t_k}^{t_k})/\Delta t &= F(t_k)\,\hat{x}_{t_k}^{t_k} + (P_{t_k}^{t_k} + O(\Delta t))\,M^{\mathrm{T}}(t_k) \\
&\quad \times [\Delta t\, M(P_{t_k}^{t_k} + O(\Delta t))\,M^{\mathrm{T}} + R(t_k)]^{-1} \\
&\quad \times (y_{t_k} - M(t_k)\,\hat{x}_{t_k}^{t_k} - O(\Delta t)) + o(\Delta t)/\Delta t, \\
(P_{t_{k+1}}^{t_{k+1}} - P_{t_k}^{t_k})/\Delta t &= F(t_k)\,P_{t_k}^{t_k} + P_{t_k}^{t_k}F^{\mathrm{T}}(t_k) + G(t_k)\,Q(t_{k+1})\,G^{\mathrm{T}}(t_k) \\
&\quad - (P_{t_k}^{t_k} + O(\Delta t))\,M^{\mathrm{T}}(t_k)\,[\Delta t\, M(P_{t_k}^{t_k} + O(\Delta t))\,M^{\mathrm{T}} + R(t_k)]^{-1} \\
&\quad \times [M(t_k)(P_{t_k}^{t_k} + O(\Delta t))] + o(\Delta t)/\Delta t.
\end{aligned}$$

In the $\Delta t = 0$ limit, we get Theorem 7.3. ▲

Example 7.11 (*Calculus of Variations*). We solve the continuous least squares problem (5.24) using the calculus of variations [20]. Minimize

$$\begin{aligned}
J_t' &= \frac{1}{2}(x_{t_0} - \bar{x}_{t_0})^{\mathrm{T}}\,P_{t_0}^{-1}(x_{t_0} - \bar{x}_{t_0}) \\
&\quad + \frac{1}{2}\int_{t_0}^{t} (y_\tau - M(\tau)\,x_\tau)^{\mathrm{T}}\,R^{-1}(\tau)(y_\tau - M(\tau)\,x_\tau)\,d\tau \\
&\quad + \frac{1}{2}\int_{t_0}^{t} w_\tau{}^{\mathrm{T}} Q^{-1}(\tau)\,w_\tau\,d\tau
\end{aligned} \tag{7.113}$$

with respect to x_τ and w_τ, $t_0 \leqslant \tau \leqslant t$, subject to the constraint

$$dx_\tau/d\tau = F(\tau)\,x_\tau + G(\tau)\,w_\tau, \qquad t_0 \leqslant \tau \leqslant t. \tag{7.114}$$

This variational approach to estimation was first taken by Bryson and Frazier [7]. The solution $\hat{x}_\tau{}^t$ is the smoothing solution.

In view of the constraint (7.114), x_{t_0} and w_τ, $t_0 \leqslant \tau \leqslant t$, determine x_τ, $t_0 \leqslant \tau \leqslant t$. Thus, we minimize J_t' with respect to x_{t_0} and w_τ, $t_0 \leqslant \tau \leqslant t$. $\hat{x}_\tau{}^t$ will then be determined by (7.114) once the minimizing x_{t_0} and w_τ are established. Using the standard calculus of variations

technique, we adjoin the constraint (7.114) to J_t' via a vector Lagrange multiplier function $\lambda(\tau)$:

$$J_t = \frac{1}{2}(x_{t_0} - \bar{x}_{t_0})^{\mathrm{T}} P_{t_0}^{-1}(x_{t_0} - \bar{x}_{t_0}) + \frac{1}{2}\int_{t_0}^{t} [(y_\tau - Mx_\tau)^{\mathrm{T}} R^{-1}(y_\tau - Mx_\tau) + w_\tau^{\mathrm{T}} Q^{-1} w_\tau + 2\lambda^{\mathrm{T}}(\dot{x}_\tau - Fx_\tau - Gw_\tau)]\, d\tau.$$

We compute the first variation of J_t:

$$\delta J_t = [(x_{t_0} - \bar{x}_{t_0})^{\mathrm{T}} P_{t_0}^{-1} - \lambda^{\mathrm{T}}(t_0)]\, \delta x_{t_0} + \lambda^{\mathrm{T}}(t)\, \delta x_t + \int_{t_0}^{t} \{-[(y_\tau - Mx_\tau)^{\mathrm{T}} R^{-1}M + \dot{\lambda}^{\mathrm{T}} + \lambda^{\mathrm{T}}F]\, \delta x_\tau + [w_\tau^{\mathrm{T}} Q^{-1} - \lambda^{\mathrm{T}}G]\, \delta w_\tau\}\, d\tau.$$

Necessary conditions for $\delta J_t = 0$ are

$$(x_{t_0} - \bar{x}_{t_0})^{\mathrm{T}} P_{t_0}^{-1} - \lambda^{\mathrm{T}}(t_0) = 0, \tag{7.115}$$

$$\lambda(t) = 0, \tag{7.116}$$

$$d\lambda/d\tau = -F^{\mathrm{T}}\lambda - M^{\mathrm{T}}R^{-1}(y_\tau - Mx_\tau), \tag{7.117}$$

$$w_\tau^{\mathrm{T}} Q^{-1} - \lambda^{\mathrm{T}}G = 0. \tag{7.118}$$

Eliminating w_τ from (7.117) and (7.114) via (7.118), we have the two-point boundary-value problem:

$$d\lambda(\tau)/d\tau = -F^{\mathrm{T}}\lambda(\tau) + M^{\mathrm{T}}R^{-1}Mx_\tau - M^{\mathrm{T}}R^{-1}y_\tau, \tag{7.119}$$

$$dx_\tau/d\tau = GQG^{\mathrm{T}}\lambda(\tau) + Fx_\tau, \tag{7.120}$$

$$x_{t_0} = \bar{x}_{t_0} + P_{t_0}\lambda(t_0), \tag{7.121}$$

$$\lambda(t) = 0. \tag{7.122}$$

Compare (7.119), (7.120) with (7.103).

This problem is linear and easy to solve [7]. Let $y(\tau)$ and $\xi(\tau)$ be particular solutions of (7.120) and (7.119), respectively, with initial conditions

$$y(t_0) = \bar{x}_{t_0}, \qquad \xi(t_0) = 0. \tag{7.123}$$

Let $Y(\tau)$ and $X(\tau)$ be as defined in (7.103). Then it is easy to show that the smoothing solution is

$$\hat{x}_\tau^{\,t} = y(\tau) - Y(\tau)\, X^{-1}(t)\, \xi(t) \tag{7.124}$$

and

$$\lambda(\tau) = \xi(\tau) - X(\tau)\, X^{-1}(t)\, \xi(t).$$

Now we are interested in the filtering solution

$$\hat{x}_t{}^t = y(t) - Y(t)\, X^{-1}(t)\, \xi(t) = y(t) - P_t{}^t \xi(t) \tag{7.125}$$

in view of (7.105). We already saw that $P_t{}^t$ satisfies the variance equation. To obtain the differential equation for the estimate, we differentiate (7.125) with respect to t:

$$d\hat{x}_t{}^t/dt = dy(t)/dt - (dP_t{}^t/dt)\, \xi(t) - P_t{}^t\, d\xi(t)/dt.$$

Using the variance equation, (7.119), and (7.120), it is easy to show that

$$d\hat{x}_t{}^t/dt = F\hat{x}_t{}^t + P_t{}^t M^{\mathrm{T}} R^{-1}(y_t - M\hat{x}_t{}^t). \quad \blacktriangle$$

Example 7.12 (*Dynamic Programming*). The solution of the continuous least squares problem of Example 7.11 via dynamic programming [4] was first given by Cox [13]. The problem of minimizing J_t' (7.113) subject to constraint (7.114) is *imbedded* in a family of problems parameterized by values of x_t by defining the *cost* function

$$V(x_t\,, t) = \min_{\{w_\tau, t_0 \leqslant \tau \leqslant t\}} J_t'. \tag{7.126}$$

Keeping in mind the probabilistic interpretation of the least squares problem (Sect. 5.3), $-V(x, t)$ can be interpreted as a measure of the likelihood of the most likely trajectory passing through $x_t = x$. The filtering solution $\hat{x}_t{}^t$ is clearly that value of x_t which minimizes V.

By the principle of optimality [4], we have the following functional equation for V:

$$V(x_t\,, t) = \min_{\{w_\tau, t-\delta t \leqslant \tau \leqslant t\}} \Big\{ V(x_t - \delta x,\, t - \delta t) + \frac{1}{2} \int_{t-\delta t}^{t} [(y_\tau - Mx_\tau)^{\mathrm{T}} R^{-1}(y_\tau - Mx_\tau) + w_\tau{}^{\mathrm{T}} Q^{-1} w_\tau]\, d\tau \Big\}. \tag{7.127}$$

Now for δt small, the integral in (7.127) is

$$[(y_t - Mx_t)^{\mathrm{T}} R^{-1}(y_t - Mx_t) + w_t{}^{\mathrm{T}} Q^{-1} w_t]\, \delta t + o(\delta t)$$

and

$$V(x_t - \delta x,\, t - \delta t) = V(x_t\,, t) \quad V_t(x_t\,, t)\, \delta t - V_x{}^{\mathrm{T}}(x_t\,, t)\, \delta x + o(\delta t),$$

so that (7.127) becomes

$$V_t = \min_{\{w_\tau, t-\delta t \leqslant \tau \leqslant t\}} \{-V_x^{\mathrm{T}}(\delta x/\delta t) + \tfrac{1}{2}(y_t - Mx_t)^{\mathrm{T}} R^{-1}(y_t - Mx_t) + \tfrac{1}{2} w_t^{\mathrm{T}} Q^{-1} w_t + o(\delta t)/\delta t\}.$$

Substituting for $\delta x/\delta t$ from (7.114) and passing to the $\delta t = 0$ limit, we formally get the Bellman partial differential equation

$$V_t = \min_{w_t}\{-V_x^{\mathrm{T}}(Fx_t + Gw_t) + \tfrac{1}{2}(y_t - Mx_t)^{\mathrm{T}} R^{-1}(y_t - Mx_t) + \tfrac{1}{2} w_t^{\mathrm{T}} Q^{-1} w_t\}. \tag{7.128}$$

The minimizing w_t is easy to compute by setting the derivative of the term in braces equal to zero:

$$w_t = QG^{\mathrm{T}} V_x . \tag{7.129}$$

Comparing (7.129) with (7.118), we see that V_x plays the role of the Lagrange multiplier. Thus we finally have

$$V_t = -V_x^{\mathrm{T}} F x_t - \tfrac{1}{2} V_x^{\mathrm{T}} GQG^{\mathrm{T}} V_x + \tfrac{1}{2}(y_t - Mx_t)^{\mathrm{T}} R^{-1}(y_t - Mx_t). \tag{7.130}$$

The boundary condition is clearly

$$V(x_{t_0}, t_0) = \tfrac{1}{2}(x_{t_0} - \bar{x}_{t_0})^{\mathrm{T}} P_{t_0}^{-1}(x_{t_0} - \bar{x}_{t_0}).$$

The solution of the estimation problem is reduced to the solution of (7.130). Once $V(x(\tau), \tau)$ is found, then, in view of (7.129), backward integration of (7.114) produces $\hat{x}_\tau{}^t$.

To get a formal filtering solution, assume V of the form

$$V(x_t, t) = \tfrac{1}{2}(x_t - \hat{x}_t{}^t)^{\mathrm{T}} P_t^{t-1}(x_t - \hat{x}_t{}^t) + a(t). \tag{7.131}$$

$\hat{x}_t{}^t$ clearly minimizes V with respect to x_t. From (7.131),

$$V_x = P_t^{t-1}(x_t - \hat{x}_t{}^t),$$

$$V_t = \dot{a} - (x_t - \hat{x}_t{}^t)^{\mathrm{T}} P_t^{t-1}\, d\hat{x}_t{}^t/dt + \tfrac{1}{2}(x_t - \hat{x}_t{}^t)^{\mathrm{T}} [d(P_t^{t-1})/dt]\,(x_t - \hat{x}_t{}^t).$$

Substituting these expressions into (7.130) and equating like powers of x_t, we get

$$\begin{aligned} d(P_t^{t-1})/dt &= -P_t^{t-1}F - F^{\mathrm{T}}P_t^{t-1} - P_t^{t-1}GQG^{\mathrm{T}}P_t^{t-1} + M^{\mathrm{T}}R^{-1}M, \\ d\hat{x}_t{}^t/dt &= F\hat{x}_t{}^t + P_t{}^t M^{\mathrm{T}} R^{-1}(y_t - M\hat{x}_t{}^t). \end{aligned} \tag{7.132}$$

Since

$$PP^{-1} = I,$$

$$dP/dt = -P[d(P^{-1})/dt]\,P.$$

Using this fact, (7.132) gives the variance equation. ▲

Example 7.13 (*Invariant Imbedding*). We return to the variational two-point boundary-value problem (7.119)–(7.122) of Example 7.11, and obtain the filtering solution by invariant imbedding (see Bellman *et al.* [5] and Detchmendy and Sridhar [14]).

Suppose we solve the problem (7.119)–(7.122) and obtain the missing terminal condition on x_t, call it $\hat{x}_t{}^t$. Then, for a new time $t' > t$, the two-point boundary-value problem has to be re-solved to produce $\lambda(t') = 0$ and obtain $\hat{x}_t^{t'}$. If the solution of the original problem is continued from t to t', then, in general, $\lambda(t') = c \neq 0$. Clearly, x_t is a function of t and c:

$$x_t = r(c, t) \tag{7.133}$$

and

$$\hat{x}_t{}^t = r(0, t). \tag{7.134}$$

We have imbedded the original problem in a family of problems parameterized by c. $r(0, t)$ is the solution of our filtering problem.

Now, on the one hand,

$$r(c + \delta c, t + \delta t) = r(c, t) + (GQG^{\mathrm{T}}c + Fr)\,\delta t + o(\delta t) \tag{7.135}$$

in view of (7.120), and on the other, using a Taylor series expansion,

$$r(c + \delta c, t + \delta t) = r(c, t) + r_c\,\delta c + r_t\,\delta t + o(\delta t), \tag{7.136}$$

where, from (7.119),

$$\delta c = [-F^{\mathrm{T}}c + M^{\mathrm{T}}R^{-1}Mr - M^{\mathrm{T}}R^{-1}y_t]\,\delta t\,.$$

Equating (7.135) and (7.136), dividing by δt, and passing to the $\delta t = 0$ limit, we formally get

$$r_t + r_c(-F^{\mathrm{T}}c + M^{\mathrm{T}}R^{-1}Mr - M^{\mathrm{T}}R^{-1}y_t) = GQG^{\mathrm{T}}c + Fr. \tag{7.137}$$

This is a partial differential equation for r.

Assume a solution of the form

$$r(c, t) = \hat{x}_t{}^t + P(t)\,c. \tag{7.138}$$

This is motivated by (7.134). Then, from (7.138),

$$r_c = P(t), \qquad r_t = \dot{\hat{x}}_t^t + \dot{P}(t)\, c. \tag{7.139}$$

Substituting (7.138) and (7.139) into (7.137), and regarding the result as an identity in c (equate coefficients of c^0 and c to zero), produces the continuous Kalman–Bucy filter. ▲

In the next three examples, we obtain the continuous filter in the case of correlated system and measurement noise, colored measurement noise, and the continuous linear smoother by using the limiting procedure of Example 7.10 in Examples 7.5–7.8.

Example 7.14 (Correlated System and Measurement Noise). Suppose in the system (7.95), (7.96) that

$$\mathscr{E}\{d\beta_t\, d\eta_t^{\mathrm{T}}\} = C(t)\, dt.$$

Then, using the limiting procedure of Example 7.10 in the filter of Example 7.5, where

$$\mathscr{E}\{w_{t_k} v_{t_k}^{\mathrm{T}}\} = C(t_k)/\Delta t,$$

we easily obtain

$$\begin{aligned} d\hat{x}_t^t/dt &= F(t)\,\hat{x}_t^t + [P_t^t M^{\mathrm{T}}(t) + G(t)\,C(t)]\,R^{-1}(t)(y_t - M(t)\,\hat{x}_t^t), \\ dP_t^t/dt &= F(t)\,P_t^t + P_t^t F^{\mathrm{T}} + G(t)\,Q(t)\,G^{\mathrm{T}}(t) \\ &\quad - [P_t^t M^{\mathrm{T}}(t) + G(t)\,C(t)]\,R^{-1}(t)[M(t)\,P_t^t + C^{\mathrm{T}}(t)\,G^{\mathrm{T}}(t)]. \end{aligned} \tag{7.140}$$

The reader can verify for himself that the same result is obtained from the discrete filter of Example 7.6. The filter (7.140) was first derived by Kalman [36], and also by Cox [13]. ▲

Example 7.15 (Colored Measurement Noise). Suppose v_t in (7.96′) is generated by

$$dv_t/dt = L(t)\,v_t + u_t\,, \qquad v_{t_0} \sim N(0, V), \qquad \mathscr{E}\{x_{t_0} v_{t_0}^{\mathrm{T}}\} = 0. \tag{7.141}$$

$\{u_t\}$ is white, Gaussian, $u_t \sim N[0, S(t)]$, independent of $\{w_t\}$. We apply the limiting procedure of Example 7.10 to the discrete filter of Example 7.7. We have

$$v_{t_{k+1}} - v_{t_k} = L(t_k)\,v_{t_k}\,\Delta t + \Delta t\, u_{t_k} + o(\Delta t),$$

where

$$\mathscr{E}\{u_{t_k} u_{t_k}^{\mathrm{T}}\} = S(t_k)/\Delta t,$$

so that

$$\Psi(t_{k+1}, t_k) = I + \Delta t\, L(t_k).$$

Note that u_k in (7.71) is replaced by $\Delta t\, u_{t_k}$.
Then, $t_k \leqslant t \leqslant t_{k+1}$,

$$\begin{aligned}
\xi_{k+1} &= [dy_t/dt - L(t)\, y_t]\, \Delta t + o(\Delta t), \\
H(k) &= \bar{H}(t)\, \Delta t + o(\Delta t), \\
\bar{H}(t) &\triangleq dM/dt + [M(t)F(t) - L(t)\, M(t)], \qquad (7.142) \\
C_k &= Q(t)\, G^{\mathrm{T}}(t)\, M^{\mathrm{T}}(t) + O(\Delta t), \\
R_k &= \bar{R}(t)\, \Delta t + o(\Delta t), \\
\bar{R}(t) &\triangleq M(t)\, G(t) Q(t)\, G^{\mathrm{T}}(t)\, M^{\mathrm{T}}(t) + S(t). \qquad (7.143)
\end{aligned}$$

Therefore, from (7.78),

$$\begin{aligned}
(\hat{x}_{t_{k+1}}^{t_{k+1}} - \hat{x}_{t_k}^{t_k})/\Delta t = F\hat{x}_{t_k}^{t_k} &+ [P_{t_k}^{t_k}\bar{H}^{\mathrm{T}} + GQG^{\mathrm{T}}M^{\mathrm{T}} + O(\Delta t)] \\
&\times [\bar{R}\, \Delta t + o(\Delta t)]^{-1}[(dy/dt)\, \Delta t - Ly\, \Delta t - \bar{H}\hat{x}_{t_k}^{t_k}\, \Delta t + o(\delta t)],
\end{aligned}$$

where we have omitted the time argument t. In the $\Delta t = 0$ limit, we get

$$\begin{aligned}
d\hat{x}_t^t/dt = F(t)\, \hat{x}_t^t &+ [P_t^t \bar{H}^{\mathrm{T}}(t) + G(t) Q(t)\, G^{\mathrm{T}}(t)\, M^{\mathrm{T}}(t)] \\
&\times \bar{R}^{-1}(t)[dy_t/dt - L(t)\, y_t - \bar{H}(t)\, \hat{x}_t^t]. \qquad (7.144)
\end{aligned}$$

The filter equation (7.144) involves the differentiation of observations. This can be removed for practical applications in the following way. Define

$$B(t) \triangleq [P_t^t \bar{H}^{\mathrm{T}}(t) + G(t) Q(t)\, G^{\mathrm{T}}(t)\, M^{\mathrm{T}}(t)]\, \bar{R}^{-1}(t). \qquad (7.145)$$

Since

$$\frac{d}{dt}(\hat{x}_t^t - B(t)\, y_t) = d\hat{x}_t^t/dt - B(t)\, dy_t/dt - \dot{B}(t)\, y_t,$$

(7.144) becomes

$$\frac{d}{dt}(\hat{x}_t^t - B(t)\, y_t) = F(t)\, \hat{x}_t^t + B(t)[-L(t)\, y_t - \bar{H}(t)\, \hat{x}_t^t] - \dot{B}(t)\, y_t. \qquad (7.146)$$

Similarly, from (7.78) we get the variance equation

$$dP_t^t/dt = F(t)\,P_t^t + P_t^t F^{\mathrm{T}}(t) + G(t)\,Q(t)\,G^{\mathrm{T}}(t) - B(t)\,\bar{R}(t)\,B^{\mathrm{T}}(t). \quad (7.147)$$

The initial conditions for (7.146), (7.147) follow from (7.76):

$$\begin{aligned} \hat{x}_{t_0}^{t_0} &= \hat{x}_{t_0} + P_{t_0}M^{\mathrm{T}}(t_0)[M(t_0)\,P_{t_0}M^{\mathrm{T}}(t_0) + V]^{-1}\,(y_{t_0} - M(t_0)\,\hat{x}_{t_0}), \\ P_{t_0}^{t_0} &= P_{t_0} - P_{t_0}M^{\mathrm{T}}(t_0)[M(t_0)\,P_{t_0}M^{\mathrm{T}}(t_0) + V]^{-1}\,M(t_0)\,P_{t_0}. \end{aligned} \quad (7.148)$$

Thus, there is a discontinuity in the estimate and the covariance matrix at the initial time.

This filter was first derived by Bryson and Johansen [8], and later by Mehra and Bryson [39], who used variational techniques. The latter reference also derives the smoother for the colored measurement noise case. ▲

Example 7.16 (Continuous Linear Smoother). Using again the limiting procedure of Example 7.10, applied to the discrete smoother of Example 7.8, we derive the continuous smoother for system (7.95), (7.96). Let $t_k \leqslant t \leqslant t_{k+1} \leqslant t_l$, t_l fixed. From (7.86),

$$\begin{aligned} \hat{x}_{t_{k+1}}^{t_l} - \hat{x}_{t_k}^{t_l} &= \Phi(t_{k+1}, t_k)\,\hat{x}_{t_k}^{t_k} - \hat{x}_{t_k}^{t_l} + S_k^{-1}(\hat{x}_{t_k}^{t_l} - \hat{x}_{t_k}^{t_k}) \\ &= (\hat{x}_{t_k}^{t_k} - \hat{x}_{t_k}^{t_l}) + \Delta t\,F(t_k)\,\hat{x}_{t_k}^{t_k} + P_{t_k}^{t_k}[I + \Delta t\,F^{\mathrm{T}}(t_k)]^{-1}\,P_{t_k}^{t_k\,-1}(\hat{x}_{t_k}^{t_l} - \hat{x}_{t_k}^{t_k}) \\ &\quad + \Delta t[F(t_k)\,P_{t_k}^{t_k} + P_{t_k}^{t_k}F^{\mathrm{T}}(t_k) + G(t_k)\,Q(t_k)\,G^{\mathrm{T}}(t_k)] \\ &\quad \times [I + O(\Delta t)]^{-1}\,P_{t_k}^{t_k\,-1}(\hat{x}_{t_k}^{t_l} - \hat{x}_{t_k}^{t_k}) + o(\Delta t). \end{aligned}$$

But

$$[I + \Delta t\,F^{\mathrm{T}}(t_k)]^{-1} = I - \Delta t\,F^{\mathrm{T}}(t_k)[I + \Delta t\,F^{\mathrm{T}}(t_k)]^{-1},$$

and so, in the $\Delta t = 0$ limit, we get

$$d\hat{x}_t^{t_l}/dt = F(t)\,\hat{x}_t^{t_l} + G(t)\,Q(t)\,G^{\mathrm{T}}(t)\,P_t^{t\,-1}(\hat{x}_t^{t_l} - \hat{x}_t^{t}). \quad (7.149)$$

Similarly,

$$\begin{aligned} dP_t^{t_l}/dt &= [F(t) + G(t)\,Q(t)\,G^{\mathrm{T}}(t)\,P_t^{t\,-1}]\,P_t^{t_l} + P_t^{t_l}[F^{\mathrm{T}}(t) + P_t^{t\,-1}G(t)\,Q(t)\,G^{\mathrm{T}}(t)] \\ &\quad - G(t)\,Q(t)\,G^{\mathrm{T}}(t). \end{aligned} \quad (7.150)$$

Equations (7.149) and (7.150) are to be integrated backwards from t_l with initial conditions

$$\hat{x}_{t_l}^{t_l}, \qquad P_{t_l}^{t_l}.$$

This smoothing algorithm was first derived by Rauch, Tung, and Striebel [45] (see also Bryson and Frazier [7]). ▲

5. OBSERVABILITY AND INFORMATION

Assuming that a dynamical system (dynamics and measurement equation) has been defined, the preceding sections have dealt with derivations of the optimal filter for that system. Now it is important to ask what, if anything, can be gained from filtering the data. How much information about the state of the system is contained in the data? Can the state be determined from the data? Intuitively, it would seem that answers to such questions are related to the system model itself, and indeed this is so. The importance of such questions is obvious. If little is to be gained from filtering, then we should consider remodeling the system. This might involve taking additional or alternate measurements, or redesigning the dynamics of the system.

How well the state is known is measured by the estimation error covariance matrix $P_t{}^t$. But $P_t{}^t$ depends on its initial condition P_{t_0} (the initial data) and does not reflect the uncertainty in the estimate by virtue of filtering the data alone. Consider the discrete least squares problem of Example 7.2. Set $P_0^{-1} = 0$, which means that no weight is attached to the prior estimate $\bar{x}_0$; there is no prior information. Then from (7.41), it is clear that, in order to determine x_k, the *information matrix*

$$\mathscr{I}(k, 1) \triangleq \sum_{i=1}^{k} \Phi^{\mathrm{T}}(i, k)\, M^{\mathrm{T}}(i)\, R_i^{-1} M(i)\, \Phi(i, k) \tag{7.151}$$

must be positive definite. If $\mathscr{I}(k, 1)$ is singular, then certain linear combinations of the elements of x_k cannot be determined; there is no information about them in the data $\{y_1, \ldots, y_k\}$. Notice that the information matrix depends on Φ and M, the system model, and *not* on the data themselves. Also note that the dynamical system in Example 7.2 is noise-free. From (7.44), we see that uncertainty is inversely proportional to information.

This discussion motivates the following definitions, which are due to Kalman [29, 34]. The discrete dynamical system (7.27), (7.2) is said to be *completely observable* if, and only if,

$$\mathscr{I}(k, 0) > 0 \qquad \text{for some} \quad k > 0.^{6} \tag{7.152}$$

[6] We assume that the first observation is taken at t_0 rather than at t_1.

It is *uniformly completely observable* if there exist a positive integer N and positive constants α, β such that

$$0 < \alpha I \leqslant \mathscr{I}(k, k-N) \leqslant \beta I, \tag{7.153}$$

for *all* $k \geqslant N$.[7] Finally, we say that x_k *is completely observable with respect to* $\{y_l, y_{l+1}, \ldots, y_k\}$ if, and only if,

$$\mathscr{I}(k, l) > 0 \qquad (k \geqslant l). \tag{7.154}$$

We next note some properties of the information matrix, already developed in Example 7.2, for a noise-free system ($w_k \equiv 0$). We saw that the information matrix satisfies the following difference equation:

$$\mathscr{I}(k+1, 0) = \Phi^{\mathrm{T}}(k, k+1)\, \mathscr{I}(k, 0)\, \Phi(k, k+1) + M^{\mathrm{T}}(k+1)\, R_{k+1}^{-1} M(k+1), \tag{7.155}$$

and is related to the covariance matrix by

$$P_k^{k-1} = \Phi^{\mathrm{T}}(0, k)\, P_0^{-1} \Phi(0, k) + \mathscr{I}(k, 0). \tag{7.156}$$

A concept dual to that of observability is the concept of controllability, also introduced by Kalman [29, 34, 35]. Consider the effect of the forcing term in (7.27). At t_k, the difference between x_k and $\Phi(k, 0)x_0$ is

$$\Delta_k = \sum_{i=0}^{k-1} \Phi(k, i+1)\, \Gamma(i)\, w_{i+1}$$

and

$$\mathscr{E}\{\Delta_k\} \equiv 0, \qquad \mathscr{E}\{\Delta_k\, \Delta_k^{\mathrm{T}}\} = \sum_{i=0}^{k-1} \Phi(k, i+1)\, \Gamma(i) Q_{i+1} \Gamma^{\mathrm{T}}(i)\, \Phi^{\mathrm{T}}(k, i+1).$$

We define the *controllability matrix*

$$\mathscr{C}(k, 0) \triangleq \sum_{i=0}^{k-1} \Phi(k, i+1)\, \Gamma(i) Q_{i+1} \Gamma^{\mathrm{T}}(i)\, \Phi^{\mathrm{T}}(k, i+1). \tag{7.157}$$

The discrete dynamical system (7.27), (7.2) is said to be *completely controllable* if, and only if,

$$\mathscr{C}(k, 0) > 0 \qquad \text{for some} \quad k > 0. \tag{7.158}$$

[7] For two symmetric matrices A and B, $A > B$ ($A \geqslant B$) means that $A - B > 0$ ($A - B \geqslant 0$) is positive definite (positive semidefinite).

It is *uniformly completely controllable* if there exist a positive integer N and positive constants α, β such that

$$0 < \alpha I \leqslant \mathscr{C}(k, k - N) \leqslant \beta I \tag{7.159}$$

for *all* $k \geqslant N$.

Uniform complete observability and controllability imply stochastic observability and controllability as defined by Aoki [1]. We prefer the present definitions because they involve intrinsic properties of the system.

Completely analogous definitions are made for the continuous dynamical system (7.95), (7.96). In the limit of continuous observations [replace R_i by $R(t_i)/\Delta t$, Q_i by $Q(t_i)/\Delta t$, and $\Gamma(i)$ by $\Delta t\, G(t_i)$; see Example 7.10], we have the *information matrix*

$$\mathscr{I}(t, t_0) \triangleq \int_{t_0}^{t} \Phi^{\mathrm{T}}(\tau, t)\, M^{\mathrm{T}}(\tau)\, R^{-1}(\tau)\, M(\tau)\, \Phi(\tau, t)\, d\tau \tag{7.160}$$

and the *controllability matrix*

$$\mathscr{C}(t, t_0) \triangleq \int_{t_0}^{t} \Phi(t, \tau)\, G(\tau)\, Q(\tau)\, G^{\mathrm{T}}(\tau)\, \Phi^{\mathrm{T}}(t, \tau)\, d\tau. \tag{7.161}$$

The continuous dynamical system (7.95), (7.96) is said to be *completely observable* (*controllable*) if, and only if,

$$\mathscr{I}(t, t_0) > 0 \qquad (\mathscr{C}(t, t_0) > 0) \qquad \text{for some} \quad t > t_0\,. \tag{7.162}$$

It is *uniformly completely observable* (*controllable*) if there exist positive constants σ, α, β such that

$$0 < \alpha I \leqslant \mathscr{I}(t, t - \sigma) \leqslant \beta I \qquad (0 < \alpha I \leqslant \mathscr{C}(t, t - \sigma) \leqslant \beta I)$$

$$\text{for all} \quad t \geqslant t_0 + \sigma. \tag{7.163}$$

See Kalman [29, 34, 35] for equivalent and simpler conditions for controllability and observability of constant (stationary) continuous systems.

By the limiting procedure of Example 7.10 applied to (7.155) (or differentiate 7.160), we get the differential equation for the information matrix in the continuous case:

$$d\mathscr{I}/dt = -F^{\mathrm{T}}(t)\, \mathscr{I} - \mathscr{I} F(t) + M^{\mathrm{T}}(t)\, R^{-1}(t)\, M(t). \tag{7.164}$$

$M^{\mathrm{T}}(t)\, R^{-1}(t)\, M(t)$ is appropriately called the *information rate* matrix. Comparing (7.164) with (7.132), we see that $\mathscr{I}$ is analogous to P_t^{-1}. $\mathscr{I}^{-1}$ satisfies the variance equation in the case of no system noise.

In the next several sections, we use the concepts of observability and controllability in a study of qualitative properties of linear filters.

6. BOUNDS AND STABILITY—DISCRETE FILTER

This section is devoted to the development of qualitative properties of the discrete filter (Theorem 7.2). Our main objective is to prove, under appropriate conditions, the uniform asymptotic stability of the discrete filter. In this connection, we note that optimality does not imply stability. The stability needs to be proved. In the course of proving stability, we develop upper and lower bounds on the covariance matrix P_k^k and study its asymptotic properties. Our development is based on the work of Kalman [36], Deyst and Price [15], and Sorenson [52].

Before proceeding, it will be useful to write the discrete filter in the equivalent form [see (7.11)]

$$\hat{x}_k^k = P(k \mid k)[P^{-1}(k \mid k-1)\,\Phi(k, k-1)\,\hat{x}_{k-1}^{k-1} + M^{\mathrm{T}}(k)\,R_k^{-1}y_k], \quad (7.165)$$

$$P(k \mid k) = [P^{-1}(k \mid k-1) + M^{\mathrm{T}}(k)\,R_k^{-1}M(k)]^{-1},$$

$$P(k+1 \mid k) = \Phi(k+1, k)\,P(k \mid k)\,\Phi^{\mathrm{T}}(k+1, k) + \Gamma(k)\,Q_{k+1}\Gamma^{\mathrm{T}}(k). \quad (7.166)$$

The filter equation (7.165) can also be written as

$$\hat{x}_k^k = [I - K(k)\,M(k)]\,\Phi(k, k-1)\,\hat{x}_{k-1}^{k-1} + K(k)\,y_k\,. \quad (7.167)$$

$[I - K(k)\,M(k)]\,\Phi(k, k-1) \triangleq \psi(k, k-1)$ is the state transition matrix of the filter, and, in view of (7B.1), it is nonsingular.

Lemma 7.1. *If the dynamical system* (7.27), (7.2) *is uniformly completely observable and uniformly completely controllable, and if* $P_0 \geqslant 0$, *then* $P(k \mid k)$ *is uniformly bounded from above for all* $k \geqslant N$,

$$P(k \mid k) \leqslant \mathscr{I}^{-1}(k, k-N) + \mathscr{C}(k, k-N) \leqslant \left(\frac{1+\alpha\beta}{\alpha}\right) I, \qquad k \geqslant N.$$

PROOF: Consider the estimate

$$\bar{x}_k^N = \mathscr{I}^{-1}(k, k-N) \sum_{i=k-N}^{k} \Phi^{\mathrm{T}}(i, k)\,M^{\mathrm{T}}(i)\,R_i^{-1}y_i\,, \qquad k \geqslant N.$$

In view of Eq. (7.41) of Example 7.2, this is the least squares estimate of x_k based on the most recent N observations, ignoring the system

noise w_k. This estimate is suboptimal (not minimum variance) and, as a consequence,

$$P(k \mid k) \leqslant \mathscr{E}\{(x_k - \bar{x}_k^N)(x_k - \bar{x}_k^N)^{\mathrm{T}}\}. \tag{7.168}$$

We compute the covariance matrix in (7.168). Since, from (7.27),

$$x_i = \Phi(i, k)\, x_k - \Phi(i, k) \sum_{j=1}^{k-1} \Phi(k, j+1)\, \Gamma(j)\, w_{j+1},$$

the measurement y_i can be written as

$$y_i = M(i)\, \Phi(i, k)\, x_k + v_i - M(i)\, \Phi(i, k) \sum_{j=1}^{k-1} \Phi(k, j+1)\, \Gamma(j)\, w_{j+1}$$

Therefore,

$$\begin{aligned} x_k - \bar{x}_k^N = & -\mathscr{I}^{-1}(k, k-N) \sum_{i=k-N}^{k} \Phi^{\mathrm{T}}(i, k)\, M^{\mathrm{T}}(i)\, R_i^{-1} v_i \\ & + \mathscr{I}^{-1}(k, k-N) \sum_{i=k-N}^{k} \Phi^{\mathrm{T}}(i, k)\, M^{\mathrm{T}}(i)\, R_i^{-1} M(i)\, \Phi(i, k) \\ & \times \sum_{j=i}^{k-1} \Phi(k, j+1)\, \Gamma(j)\, w_{j+1}, \end{aligned}$$

and

$$\begin{aligned} \mathscr{E}\{(x_k - \bar{x}_k^N)(x_k - \bar{x}_k^N)^{\mathrm{T}}\} &= \mathscr{I}^{-1}(k, k-N) \\ &\quad + \operatorname{cov}\Big\{\mathscr{I}^{-1}(k, k-N) \sum_{i=k-N}^{k} \Phi^{\mathrm{T}}(i, k)\, M^{\mathrm{T}}(i)\, R_i^{-1} M(i)\, \Phi(i, k) \\ &\quad \times \sum_{j=i}^{k-1} \Phi(k, j+1)\, \Gamma(j)\, w_{j+1}\Big\} \\ &\leqslant \mathscr{I}^{-1}(k, k-N) \\ &\quad + \operatorname{cov}\Big\{\mathscr{I}^{-1}(k, k-N) \sum_{i=k-N}^{k} \Phi^{\mathrm{T}}(i, k)\, M^{\mathrm{T}}(i)\, R_i^{-1} M(i)\, \Phi(i, k) \\ &\quad \times \sum_{j=k-N}^{k-1} \Phi(k, j+1)\, \Gamma(j)\, w_{j+1}\Big\} \\ &= \mathscr{I}^{-1}(k, k-N) + \mathscr{C}(k, k-N). \end{aligned}$$

This, together with (7.168), proves the lemma. ■

By considering the estimate

$$\bar{x}_k^k = \mathscr{I}^{-1}(k, 0) \sum_{i=0}^{k} \Phi^{\mathrm{T}}(i, k)\, M^{\mathrm{T}}(i)\, R_i^{-1} y_i$$

based on all the observations, and imitating the proof of the lemma, we can derive the bound

$$P(k \mid k) \leqslant \mathscr{I}^{-1}(k, 0) + \mathscr{C}(k, 0), \qquad k \geqslant N. \tag{7.169}$$

Suppose our dynamical system is *uniformly completely observable* and *noise-free* ($\mathscr{C} \equiv 0$). Then

$$P(k \mid k) \leqslant \mathscr{I}^{-1}(k, 0), \qquad k \geqslant N. \tag{7.170}$$

[This also follows directly from (7.156).] Defining

$$\tilde{\mathscr{I}}(k, 0) \triangleq \Phi^{\mathrm{T}}(k, 0)\, \mathscr{I}(k, 0)\, \Phi(k, 0) = \sum_{i=0}^{k} \Phi^{\mathrm{T}}(i, 0)\, M^{\mathrm{T}}(i)\, R_i^{-1} M(i)\, \Phi(i, 0), \tag{7.171}$$

(7.170) becomes

$$P(k \mid k) \leqslant \Phi(k, 0)\, \tilde{\mathscr{I}}^{-1}(k, 0)\, \Phi^{\mathrm{T}}(k, 0). \tag{7.172}$$

Now

$$\tilde{\mathscr{I}}(k, 0) = \tilde{\mathscr{I}}(N, 0) + \tilde{\mathscr{I}}(2N, N) + \cdots,$$

the sum of positive definite matrices. Thus, if $\| \tilde{\mathscr{I}}^{-1}(k, 0)\| \to 0$ faster than $\| \Phi(k, 0)\|^2$ increases, as $k \to \infty$, $P(k \mid k) \to 0$ $(k \to \infty)$. The state of the system can be estimated as accurately as desired. This also shows that the initial statistic P_0 is gradually forgotten. The system noise, in the general system (7.27), prevents us from estimating the state exactly.

Returning to the *noise-free* system, we see from (7.156) that, if the system is *completely observable*, then $P(k \mid k)$ is positive definite for some k. Furthermore, it is evident that, once $P(k \mid k) > 0$, it remains positive definite thereafter. [This last property is also true for the general system, in view of (7.166).] In contrast with this, we saw in Example 7.7 that, if the measurements are noise-free, $P(k \mid k)$ is *never* positive definite.

Lemma 7.2. *If the dynamical system* (7.27), (7.2) *is uniformly completely observable and uniformly completely controllable, and if* $P_0 > 0$, *then* $P(k \mid k)$ *is uniformly bounded from below for all* $k \geqslant N$,

$$P(k \mid k) \geqslant [\mathscr{I}(k, k - N) + \mathscr{C}^{-1}(k, k - N)]^{-1} \geqslant \left(\frac{\alpha}{1 + \alpha\beta}\right) I, \qquad k \geqslant N.$$

PROOF: It is evident from (7.166) that $P(k \mid k) > 0$ for all k, since $P_0 > 0$. Define

$$S(k \mid k) \triangleq P^{-1}(k \mid k), \tag{7.173}$$

$$\bar{S}(k \mid k) \triangleq S(k \mid k) - M^{\mathrm{T}}(k) R_k^{-1} M(k), \tag{7.174}$$

$$\bar{S}(k+1 \mid k) \triangleq \Phi^{-\mathrm{T}}(k+1, k)\, S(k \mid k)\, \Phi^{-1}(k+1, k). \tag{7.175}$$

Then, using (7.166),

$$\begin{aligned} \bar{S}(k \mid k) &= [\bar{S}^{-1}(k \mid k-1) + \Gamma(k-1) Q_k \Gamma^{\mathrm{T}}(k-1)]^{-1}, \\ \bar{S}(k+1 \mid k) &= \Phi^{-\mathrm{T}}(k+1, k)\, \bar{S}(k \mid k)\, \Phi^{-1}(k+1, k) \\ &\quad + \Phi^{-\mathrm{T}}(k+1, k)\, M^{\mathrm{T}}(k)\, R_k^{-1} M(k)\, \Phi^{-1}(k+1, k). \end{aligned} \tag{7.176}$$

Comparing (7.176) with (7.166), we see that $\bar{S}(k \mid k)$ is the estimation error covariance matrix of the system

$$\begin{aligned} \bar{x}_{k+1} &= \Phi^{-\mathrm{T}}(k+1, k)\, \bar{x}_k + \Phi^{-\mathrm{T}}(k+1, k)\, M^{\mathrm{T}}(k)\, \bar{w}_k, \qquad \mathscr{E}\{\bar{w}_k \bar{w}_k^{\mathrm{T}}\} = R_k^{-1}, \\ \bar{y}_k &= \Gamma^{\mathrm{T}}(k-1)\, \bar{x}_k + \bar{v}_k, \qquad \mathscr{E}\{\bar{v}_k \bar{v}_k^{\mathrm{T}}\} = Q_k^{-1}.^{8} \end{aligned} \tag{7.177}$$

It is easy to verify that (7.177) is uniformly completely observable and uniformly completely controllable, since the dynamical system (7.27), (7.2) is. Therefore, applying Lemma 7.1 to (7.177), we have that

$$\begin{aligned} \bar{S}(k \mid k) &\leqslant \left[\sum_{i=k-N}^{k} \Phi^{-1}(i, k)\, \Gamma(i-1) Q_i \Gamma^{\mathrm{T}}(i-1)\, \Phi^{-\mathrm{T}}(i, k) \right]^{-1} \\ &\quad + \sum_{i=k-N}^{k-1} \Phi^{-\mathrm{T}}(k, i+1)\, \Phi^{-\mathrm{T}}(i+1, i) \\ &\quad \times M^{\mathrm{T}}(i) R_i^{-1} M(i)\, \Phi^{-1}(i+1, i)\, \Phi^{-1}(k, i+1) \\ &= \left[\sum_{i=k-N-1}^{k-1} \Phi(k, i+1)\, \Gamma(i) Q_{i+1} \Gamma^{\mathrm{T}}(i)\, \Phi^{\mathrm{T}}(k, i+1) \right]^{-1} \\ &\quad + \sum_{i=k-N}^{k-1} \Phi^{\mathrm{T}}(i, k)\, M^{\mathrm{T}}(i) R_i^{-1} M(i)\, \Phi(i, k). \end{aligned}$$

Now, in view of (7.174),

$$S(k \mid k) \leqslant \mathscr{C}^{-1}(k, k-N) + \mathscr{S}(k, k-N),$$

[8] The assumption that $Q_k \geqslant 0$ is no loss of generality.

and, in view of (7.173),

$$P(k \mid k) \geqslant [\mathscr{C}^{-1}(k, k-N) + \mathscr{I}(k, k-N)]^{-1}. \quad \blacksquare$$

Lemma 7.3. *If the dynamical system* (7.27), (7.2) *is uniformly completely controllable and* $P_0 \geqslant 0$, *then* $P(k \mid k) > 0$ *for all* $k \geqslant N$.

PROOF: In view of (7.166), if $P(N \mid N) > 0$, then $P(k \mid k) > 0$ for all $k \geqslant N$. So it suffices to prove that $P(N \mid N)$ is nonsingular. Suppose $P(N \mid N)$ is singular. Then there exists a vector $v \neq 0$ such that

$$v^{\mathrm{T}} P(N \mid N)\, v = 0. \tag{7.178}$$

Let

$$S(k) \triangleq \Psi(N, k)\, P(k \mid k)\, \Psi^{\mathrm{T}}(N, k), \tag{7.179}$$

where

$$\Psi(k, k-1) = [I - K(k)\, M(k)]\, \Phi(k, k-1)$$

is the filter state transition matrix [see (7.167)]. It is easy to compute

$$\begin{aligned} S(k) - S(k-1) = \Psi(N, k)[[I - K(k)\, M(k)]\, \Gamma(k-1)\, Q_k \Gamma^{\mathrm{T}}(k-1) \\ \times [I - K(k)\, M(k)]^{\mathrm{T}} + K(k)\, R_k K^{\mathrm{T}}(k)]\, \Psi^{\mathrm{T}}(N, k), \end{aligned} \tag{7.180}$$

using Theorem 7.2 and (7.16). Now $S(0) \geqslant 0$, $S(k) - S(k-1) \geqslant 0$, and, in view of (7.178), $v^{\mathrm{T}} S(N) v = 0$. Therefore, $v^{\mathrm{T}} S(k) v = 0$ for all $k \leqslant N$. That is,

$$v^{\mathrm{T}} \Psi(N, k)\, P(k \mid k)\, \Psi^{\mathrm{T}}(N, k)\, v = 0, \qquad \text{all} \quad k \leqslant N, \tag{7.181}$$

or, equivalently,

$$P(k \mid k)\, \Psi^{\mathrm{T}}(N, k)\, v = 0, \qquad \text{all} \quad k \leqslant N. \tag{7.182}$$

Now let

$$w(k) \triangleq \Psi^{\mathrm{T}}(N, k)\, v, \qquad w(N) = v. \tag{7.183}$$

We easily compute

$$w(k-1) = \Psi^{\mathrm{T}}(k, k-1)\, w(k) \tag{7.184}$$

and

$$\Phi^{\mathrm{T}}(k-1, k)\, w(k-1) = (I - K(k)\, M(k))^{\mathrm{T}}\, \Psi^{\mathrm{T}}(N, k)\, v. \tag{7.185}$$

Since $v^{\mathrm{T}}[S(k) - S(k-1)]v = 0$, $1 \leqslant k \leqslant N$, using (7.185) in (7.180), we have that

$$\begin{aligned} w^{\mathrm{T}}(k-1)\, \Phi(k-1, k)\, \Gamma(k-1)\, Q_k \Gamma^{\mathrm{T}}(k-1)\, \Phi^{\mathrm{T}}(k-1, k)\, w(k-1) = 0, \\ 1 \leqslant k \leqslant N. \end{aligned} \tag{7.186}$$

Now (7.184) can be written as

$$w(k-1) = \Phi^{\mathrm{T}}(k, k-1)\, w(k) - \Phi^{\mathrm{T}}(k, k-1)\, M^{\mathrm{T}}(k)\, K^{\mathrm{T}}(k)\, w(k). \quad (7.187)$$

But

$$K^{\mathrm{T}}(k)\, w(k) = [M(k)\, P(k \mid k-1)\, M^{\mathrm{T}}(k) + R_k]^{-1}\, M(k)\, P(k \mid k-1)\, \Psi^{\mathrm{T}}(N, k)\, v,$$

and, using (7.29) and (7.182), it is easy to show that

$$P(k \mid k-1)\, \Psi^{\mathrm{T}}(N, k)\, v = 0, \qquad \text{all} \quad k \leqslant N.$$

Therefore,

$$w(k-1) = \Phi^{\mathrm{T}}(k, k-1)\, w(k),$$

which has the solution

$$w(k) = \Phi^{\mathrm{T}}(N, k)\, v. \quad (7.188)$$

As a result, (7.186) becomes

$$v^{\mathrm{T}}\Phi(N, k)\, \Gamma(k-1)\, Q_k \Gamma^{\mathrm{T}}(k-1)\, \Phi^{\mathrm{T}}(N, k)\, v = 0, \qquad 1 \leqslant k \leqslant N,$$

and

$$v^{\mathrm{T}}\left[\sum_{k=1}^{N} \Phi(N, k)\, \Gamma(k-1)\, Q_k \Gamma^{\mathrm{T}}(k-1)\, \Phi^{\mathrm{T}}(N, k)\right] v = 0,$$

which is

$$v^{\mathrm{T}}\mathscr{C}(N, 0)\, v = 0.$$

But this contradicts the hypothesis of uniform complete controllability, and the lemma is proved. ■

We are now ready to consider the stability of the filter. For the definitions of various types of stability, and stability theory in general, the reader is referred to Kalman and Bertram [30, 31] and Hahn [23]. We are concerned here only with the linear discrete system (7.165) or (7.167).

The linear system

$$z_{k+1} = \Psi(t_{k+1}, t_k)\, z_k + u_{k+1}, \qquad t_k \geqslant t_0, \quad (7.189)$$

is said to be *stable*[9] if

$$\| \Psi(t_k, t_0) \| \leqslant c_1 \qquad \text{for all} \quad t_k \geqslant t_0. \quad (7.190)$$

[9] When speaking of the stability of (7.189), we mean the stability of the homogeneous part of (7.189).

The system is *asymptotically stable* if, in addition,

$$\| \Psi(t_k , t_0)\| \to 0 \qquad (t_k \to \infty). \tag{7.191}$$

It is *uniformly asymptotically stable* if

$$\| \Psi(t_k , t_0)\| \leqslant c_2 \exp(-c_3(t_k - t_0)) \qquad \text{for all} \quad t_k \geqslant t_0 . \tag{7.192}$$

(The c_i are positive constants.) We note in passing that, in uniformly asymptotically stable systems, bounded inputs u_k produce bounded outputs z_k .

Kalman and Bertram [30, 31] show that (7.189) is uniformly asymptotically stable if there exist scalar functions $V(z_k , t_k)$, $\gamma_1(\| z_k \|)$, $\gamma_2(\| z_k \|)$, and $\gamma_3(\| z_k \|)$, such that, for some integers N, $M > 0$,

$$0 < \gamma_1(\| z_k \|) \leqslant V(z_k , t_k) \leqslant \gamma_2(\| z_k \|), \qquad z_k \neq 0, \qquad \text{all} \quad k \geqslant M, \tag{7.193}$$

where

$$\gamma_1(0) = \gamma_2(0) = 0, \qquad \gamma_1(\| z \|) \to \infty \qquad (\| z \| \to \infty);$$

and

$$V(z_k , t_k) - V(z_{k-N} , t_{k-N}) \leqslant \gamma_3(\| z_k \|) < 0, \quad z_k \neq 0, \quad \text{all} \quad k \geqslant M, \tag{7.194}$$

where

$$\gamma_3(0) = 0;$$

and $V(0, t_k) \equiv 0$; γ_1 , γ_2 , and γ_3 are continuous, and γ_1 and γ_2 are nondecreasing. The function V is called a *Lyapunov* function for the system (7.189).

We shall prove the uniform asymptotic stability of the linear filter by producing a Lyapunov function (and γ_1 , γ_2 , γ_3) for it.

Theorem 7.4. *If the dynamical system* (7.27), (7.2) *is uniformly completely observable and uniformly completely controllable, and if* $P_0 \geqslant 0$, *then the discrete linear filter of Theorem* 7.2 *is uniformly asymptotically stable.*

PROOF: We shall prove that

$$V(z_k , k) = z_k^{\mathrm{T}} P^{-1}(k \mid k)\, z_k \tag{7.195}$$

is a Lyapunov function for (7.165), or for

$$z_k = P(k \mid k)\, P^{-1}(k \mid k - 1)\, \Phi(k, k - 1)\, z_{k-1} . \tag{7.196}$$

In view of Lemma 7.3, $P(k \mid k) > 0$ for $k \geqslant N$. Then, by Lemmas 7.1 and 7.2,

$$\left(\frac{\alpha}{1+\alpha\beta}\right) I \leqslant P^{-1}(k \mid k) \leqslant \left(\frac{1+\alpha\beta}{\alpha}\right) I, \qquad k \geqslant 2N,$$

so that

$$\left(\frac{\alpha}{1+\alpha\beta}\right) \| z_k \|^2 \leqslant V(z_k, k) \leqslant \left(\frac{1+\alpha\beta}{\alpha}\right) \| z_k \|^2, \qquad k \geqslant 2N.$$

This establishes property (7.193). It remains to establish (7.194).

Following Deyst and Price [15], we define

$$\begin{aligned} u_k &\triangleq [P(k \mid k) P^{-1}(k \mid k-1) - I] \Phi(k, k-1) z_{k-1}, \\ \bar{z}_k &\triangleq \Phi(k, k-1) z_{k-1}. \end{aligned} \tag{7.197}$$

Then the system (7.196) can be written as

$$z_k = \bar{z}_k + [P(k \mid k) P^{-1}(k \mid k-1) - I] \bar{z}_k = \bar{z}_k + u_k. \tag{7.198}$$

Using (7.166), we have

$$\begin{aligned} & z_k^{\mathrm{T}} P^{-1}(k \mid k) z_k \\ &\quad = z_k^{\mathrm{T}} [P^{-1}(k \mid k-1) + M^{\mathrm{T}}(k) R_k^{-1} M(k)] z_k \\ &\quad = z_k^{\mathrm{T}} P^{-1}(k \mid k-1) z_k + 2 z_k^{\mathrm{T}} [P^{-1}(k \mid k) - P^{-1}(k \mid k-1)] z_k \\ &\qquad - z_k^{\mathrm{T}} M^{\mathrm{T}}(k) R_k^{-1} M(k) z_k + \bar{z}_k^{\mathrm{T}} P^{-1}(k \mid k-1) \bar{z}_k \\ &\qquad - \bar{z}_k^{\mathrm{T}} P^{-1}(k \mid k-1) \bar{z}_k \\ &\quad = \bar{z}_k^{\mathrm{T}} P^{-1}(k \mid k-1) \bar{z}_k - z_k^{\mathrm{T}} M^{\mathrm{T}}(k) R_k^{-1} M(k) z_k - u_k^{\mathrm{T}} P^{-1}(k \mid k-1) u_k \\ &\quad = z_{k-1}^{\mathrm{T}} [P(k-1 \mid k-1) \\ &\qquad + \Phi(k-1, k) \Gamma(k-1) Q_k \Gamma^{\mathrm{T}}(k-1) \Phi^{\mathrm{T}}(k-1, k)]^{-1} z_{k-1} \\ &\qquad - z_k^{\mathrm{T}} M^{\mathrm{T}}(k) R_k^{-1} M(k) z_k - u_k^{\mathrm{T}} P^{-1}(k \mid k-1) u_k \\ &\quad \leqslant z_{k-1}^{\mathrm{T}} P(k-1 \mid k-1) z_{k-1} \\ &\qquad - z_k^{\mathrm{T}} M^{\mathrm{T}}(k) R_k^{-1} M(k) z_k - u_k^{\mathrm{T}} P^{-1}(k \mid k-1) u_k. \end{aligned}$$

Therefore,

$$V(x_k, t_k) - V(x_{k-1}, t_{k-1}) \leqslant -z_k^{\mathrm{T}} M^{\mathrm{T}}(k) R_k^{-1} M(k) z_k - u_k^{\mathrm{T}} P^{-1}(k \mid k-1) u_k,$$

and

$$V(z_k, t_k) - V(z_{k-N-1}, t_{k-N-1}) \leqslant -\sum_{i=k-N}^{k} [z_i^{\mathrm{T}} M^{\mathrm{T}}(i) R_i^{-1} M(i) z_i + u_i^{\mathrm{T}} P^{-1}(i \mid i-1) u_i] \triangleq -J, \quad k \geqslant 2N+1. \tag{7.199}$$

Let J^0 be the minimum value of J. Then we have

$$V(z_k, t_k) - V(z_{k-N-1}, t_{k-N-1}) \leqslant -J \leqslant -J^0, \qquad k \geqslant 2N+1. \tag{7.200}$$

We are thus led to the problem of minimizing J with respect to $\{u_k\}$ for the system (7.198):

$$z_i = \Phi(i, i-1) z_{i-1} + u_i\,; \qquad i = k-N, \ldots, k; \qquad z_{k-N-1} \text{ given.} \tag{7.201}$$

This is an ordinary minimum problem once the z_i, $i \geqslant k-N$, are eliminated from J in favor of z_{k-N-1} via (7.201).

This problem is readily solved for $\{u_k^0\}$, the minimizing $\{u_k\}$ as a function of z_{k-N-1}, which results in J^0 as a function of z_{k-N-1}. Then, using the hypothesis of uniform complete observability, it can be shown that

$$J^0 \geqslant c_4 \| z_{k-N-1} \|^2, \qquad c_4 > 0.^{10}$$

Then, in view of (7.196),

$$J^0 \geqslant c_5 \| z_k \|^2, \qquad c_5 > 0,$$

and (7.200) becomes

$$V(z_k, t_k) - V(z_{k-N-1}, t_{k-N-1}) \leqslant -c_5 \| z_k \|^2, \qquad k \geqslant 2N+1,$$

which establishes property (7.194). ■

An immediate consequence of Theorem 7.4 is

Theorem 7.5. *Let the dynamical system* (7.27), (7.2) *be uniformly completely observable and uniformly completely controllable. Suppose* $P^1(k \mid k)$ *and* $P^2(k \mid k)$ *are any two solutions of* (7.166) *with respective initial conditions* P_0^1 *and* P_0^2; $P_0^1, P_0^2 \geqslant 0$. *Let* $\delta P(t_k) = P^1(k \mid k) - P^2(k \mid k)$. *Then*

$$\| \delta P(t_k) \| \leqslant c_2^2 e^{-2c_3(t_k - t_0)} \| P_0^1 - P_0^2 \| \to 0 \qquad (k \to \infty)$$

($c_2, c_3 > 0$).

[10] We assume, of course, that all the matrices in the system description are bounded.

PROOF: It is easy to verify that $\delta P(t_k)$ satisfies the difference equation

$$\delta P(t_k) = [I - K^1(k)\, M(k)]\, \Phi(k, k-1)\, \delta P(t_{k-1})\, \Phi^T(k, k-1)[I - K^2(k)\, M(k)]^T, \tag{7.202}$$

where K^1 and K^2 are the Kalman gains corresponding to $P^1(k \mid k)$ and $P^2(k \mid k)$, respectively. But

$$[I - K(k)\, M(k)]\, \Phi(k, k-1) = \Psi(k, k-1)$$

is the filter state transition matrix [see (7.167)]. Therefore, the solution of (7.202) is

$$\delta P(t_k) = \Psi_1(t_k, t_0)(P_0^1 - P_0^2)\, \Psi_2^T(t_k, t_0). \tag{7.203}$$

According to Theorem 7.4, the filter is uniformly asymptotically stable, and, by definition, its state transition matrix has property (7.192). Taking norms in (7.203) and using (7.192) proves the theorem. ■

Theorem 7.5 has two important applications. It essentially states that the effect of the initial statistic P_0 is forgotten as more and more data are processed. This is important, since P_0 is often poorly known and sometimes rather arbitrarily set. Furthermore, this means that the computation of $P(k \mid k)$ is stable. Numerical errors in $P(k \mid k)$ are also forgotten.

7. BOUNDS AND STABILITY—CONTINUOUS FILTER

The qualitative properties of the continuous filter (Theorem 7.3) are identical to those already developed for the discrete filter in the preceding section. That is, Lemmas 7.1–7.3 and Theorems 7.4 and 7.5 hold if the dynamical system (7.95), (7.96) is uniformly completely observable and uniformly completely controllable, and $P(t \mid t)$ is the solution of the variance equation (7.99). The homogeneous part of the filter equation is, of course [see (7.98)],

$$d\hat{x}_t^t/dt = [F(t) - P(t \mid t)\, M^T(t)\, R^{-1}(t)\, M(t)]\, \hat{x}_t^t.$$

The proofs of these lemmas and theorems are given by Kalman [36]. They are by and large analogous to those already given for the discrete case. Because of this, and because they are available in the book already cited, we shall not present these proofs here. In fact, the reader himself can adapt the proofs in the preceding section to the continuous case.

In the case of the continuous filter, one can also speak of the *steady-*

state solution of the variance equation. The existence of the steady-state solution and its properties for constant (stationary) systems are treated in detail by Kalman and Bucy [33] and Kalman [36].

8. ERROR SENSITIVITY—DISCRETE FILTER

In applying the linear filter (Theorem 7.2) to a specific system, the dynamical model parameters (Φ, Γ, M), noise statistics (Q, R), and *a priori* data $[\hat{x}_0, P(0)]$ must be specified. Since the system model is usually an approximation to a physical situation, the model parameters and noise statistics are seldom exact. That is, the system model used in constructing the filter differs from the (real) system that generates the observations. Sometimes such an approximation is intentional. For example, it may be desirable to use a system model of lower dimension than the dimension of the real system in order to gain computational speed and simplicity. It has already been remarked in several places that the prior data are seldom precisely known and are generally rather arbitrarily specified to start the filtering process. We have seen (Theorem 7.5) that the prior data are eventually forgotten, after sufficient observations have been processed.

It is clear that an inexact filter model will degrade the filter performance. In fact, such an inexact model may cause the filter to diverge (see Chapter 8). In designing a filter model, it is therefore important to evaluate the effect on performance of the various approximations made. This section is a study of the qualitative effects of such approximations.

Suppose that the *real* or *actual* system is described by

$$\begin{aligned} x_{k+1} &= \Phi(k+1, k)\, x_k + \varphi(k) + \Gamma(k)\, w_{k+1}, \\ y_k &= M(k)\, x_k + v_k, \\ w_k &\sim N[0, Q(k)], \qquad v_k \sim N[0, R(k)], \qquad x_0 \sim N[\hat{x}(0), P(0)], \end{aligned} \tag{7.204}$$

whereas we *model* this system by

$$\begin{aligned} \bar{x}_{k+1} &= \Phi_c(k+1, k)\, \bar{x}_k + \varphi_c(k) + \Gamma_c(k)\, \bar{w}_{k+1}, \\ \bar{y}_k &= M_c(k)\, \bar{x}_k + \bar{v}_k, \\ \bar{w}_k &\sim N[0, Q_c(k)], \qquad \bar{v}_k \sim N[0, R_c(k)], \qquad \bar{x}_0 \sim N[\hat{x}_c(0), P_c(0)]. \end{aligned} \tag{7.205}$$

We include the fixed bias $\varphi(k)$ in the dynamics to account in an approximate, yet convenient, way for errors due to nonlinearities, reduction in system dimension, and so on. In what follows, $\{\varphi(k)\}$ could readily be

treated as a random sequence. We assume that $\Phi(k+1, k)$, $\Phi_c(k+1, k)$, $\Gamma(k)$, $\Gamma_c(k)$, $M(k)$, $M_c(k)$, $Q(k)$, $Q_c(k)$, $R(k)$, $R_c(k)$ are uniformly bounded; that Φ and Φ_c are nonsingular; that $R(k)$, $R_c(k) > 0$; and that $P(0)$, $P_c(0) \geqslant 0$.

Using the model (7.205), our filter design is

$$\begin{aligned} \hat{x}_{k+1}^{k} &= \Phi_c(k+1, k)\, \hat{x}_k{}^{k} + \varphi_c(k), \\ \hat{x}_k{}^{k} &= [I - K_c(k)\, M_c(k)]\, \hat{x}_k^{k-1} + K_c(k)\, y_k, \end{aligned} \tag{7.206}$$

where

$$K_c(k) = P_c(k \mid k-1)\, M_c{}^{\mathrm{T}}(k)[M_c(k)\, P_c(k \mid k-1)\, M_c{}^{\mathrm{T}}(k) + R_c(k)]^{-1}, \tag{7.207}$$

and

$$\begin{aligned} P_c(k+1 \mid k) &= \Phi_c(k+1, k)\, P_c(k \mid k)\, \Phi_c{}^{\mathrm{T}}(k+1, k) + \Gamma_c(k)\, Q_c(k+1)\, \Gamma_c{}^{\mathrm{T}}(k), \\ P_c(k \mid k) &= [I - K_c(k)\, M_c(k)]\, P_c(k \mid k-1) \\ &= [I - K_c(k)\, M_c(k)]\, P_c(k \mid k-1)[I - K_c(k)\, M_c(k)]^{\mathrm{T}} \\ &\quad + K_c(k)\, R_c(k)\, K_c{}^{\mathrm{T}}(k), \\ P_c(0 \mid 0) &= P_c(0). \end{aligned} \tag{7.208}$$

Note that the filter operates on the *real* data $\{y_k\}$.

Now the *computed* matrix $P_c(k \mid k)$ *is not* the estimation error covariance matrix, since the filter model differs from the real model. Neither is this filter the minimum variance filter for the actual system (7.204). The *actual* estimation errors are

$$\begin{aligned} \tilde{x}_{k+1}^{k} &\triangleq x_{k+1} - \hat{x}_{k+1}^{k} = \Phi_c(k+1, k)\, \tilde{x}_k{}^{k} + \Delta\Phi(k+1, k)\, x_k + \Delta\varphi(k) + \Gamma(k) w_{k+1}, \\ \tilde{x}_k{}^{k} &\triangleq x_k - \hat{x}_k{}^{k} = [I - K_c(k)\, M_c(k)]\, \tilde{x}_k^{k-1} - K_c(k)\, \Delta M(k)\, x_k - K_c(k)\, v_k, \end{aligned} \tag{7.209}$$

where

$$\begin{aligned} \Delta\Phi(k+1, k) &\triangleq \Phi(k+1, k) - \Phi_c(k+1, k), \\ \Delta\varphi(k) &\triangleq \varphi(k) - \varphi_c(k), \\ \Delta M(k) &\triangleq M(k) - M_c(k). \end{aligned} \tag{7.210}$$

A measure of the filter performance is provided by the *actual estimation error covariance matrix*

$$P(k \mid k) \triangleq \mathscr{E}\{\tilde{x}_k{}^{k} \tilde{x}_k^{k\mathrm{T}}\}, \qquad P(k+1 \mid k) \triangleq \mathscr{E}\{\tilde{x}_{k+1}^{k} \tilde{x}_{k+1}^{k\mathrm{T}}\}. \tag{7.211}$$

We next develop a recursion for this matrix.

Using (7.209), we get

$$\begin{aligned} P(k+1 \mid k) = {} & \Phi_c(k+1,k)\, P(k \mid k)\, \Phi_c^{\mathrm{T}}(k+1,k) + \Gamma(k)\, Q(k+1)\, \Gamma^{\mathrm{T}}(k) \\ & + \Delta\Phi(k+1,k)\, X(k)\, \Delta\Phi^{\mathrm{T}}(k+1,k) \\ & + \Delta\Phi(k+1,k)\, C(k \mid k)\, \Phi_c^{\mathrm{T}}(k+1,k) \\ & + \Phi_c(k+1,k)\, C^{\mathrm{T}}(k \mid k)\, \Delta\Phi^{\mathrm{T}}(k+1,k) \\ & + \Delta\Phi(k+1,k)\, m(k)\, \Delta\varphi^{\mathrm{T}}(k) \\ & + \Delta\varphi(k)\, m^{\mathrm{T}}(k)\, \Delta\Phi^{\mathrm{T}}(k+1,k) + \Delta\varphi(k)\, \Delta\varphi^{\mathrm{T}}(k) \\ & + \Phi_c(k+1,k)\, \Delta m(k \mid k)\, \Delta\varphi^{\mathrm{T}}(k) \\ & + \Delta\varphi(k)\, \Delta m^{\mathrm{T}}(k \mid k)\, \Phi_c^{\mathrm{T}}(k+1,k), \end{aligned} \tag{7.212}$$

$$\begin{aligned} P(k \mid k) = {} & [I - K_c(k)\, M_c(k)]\, P(k \mid k-1)[I - K_c(k)\, M_c(k)]^{\mathrm{T}} \\ & + K_c(k)\, R(k)\, K_c^{\mathrm{T}}(k) \\ & - K_c(k)\, \Delta M(k)\, C(k \mid k-1)[I - K_c(k)\, M_c(k)]^{\mathrm{T}} \\ & - [I - K_c(k)\, M_c(k)]\, C^{\mathrm{T}}(k \mid k-1)\, \Delta M^{\mathrm{T}}(k) K_c^{\mathrm{T}}(k) \\ & + K_c(k)\, \Delta M(k)\, X(k)\, \Delta M^{\mathrm{T}}(k)\, K_c^{\mathrm{T}}(k), \end{aligned} \tag{7.213}$$

where

$$\begin{aligned} X(k) &\triangleq \mathscr{E}\{x_k x_k^{\mathrm{T}}\}, \\ C(k \mid k) &\triangleq \mathscr{E}\{x_k \tilde{x}_k^{k\mathrm{T}}\}, \qquad C(k+1 \mid k) \triangleq \mathscr{E}\{x_{k+1} \tilde{x}_{k+1}^{k\mathrm{T}}\}, \\ m(k) &\triangleq \mathscr{E}\{x_k\}, \qquad \Delta m(k \mid k) \triangleq \mathscr{E}\{\tilde{x}_k^{k}\}. \end{aligned} \tag{7.214}$$

Difference equations for these latter quantities are similarly computed to be

$$\begin{aligned} X(k+1) = {} & \Phi(k+1,k)\, X(k)\, \Phi^{\mathrm{T}}(k+1,k) + \varphi(k)\, \varphi^{\mathrm{T}}(k) \\ & + \Gamma(k)\, Q(k+1)\, \Gamma^{\mathrm{T}}(k) + \Phi(k+1,k)\, m(k)\, \varphi^{\mathrm{T}}(k) \\ & + \varphi(k)\, m^{\mathrm{T}}(k)\, \Phi^{\mathrm{T}}(k+1,k), \end{aligned} \tag{7.215}$$

$$\begin{aligned} C(k+1 \mid k) = {} & \Phi(k+1,k)\, C(k \mid k)\, \Phi_c^{\mathrm{T}}(k+1,k) \\ & + \Phi(k+1,k)\, X(k)\, \Delta\Phi^{\mathrm{T}}(k+1,k) \\ & + \Gamma(k)\, Q(k+1)\, \Gamma^{\mathrm{T}}(k) \\ & + \Phi(k+1,k)\, m(k)\, \Delta\varphi^{\mathrm{T}}(k) \\ & + \varphi(k)\, m^{\mathrm{T}}(k)\, \Delta\Phi^{\mathrm{T}}(k+1,k) \\ & + \varphi(k)\, \Delta m^{\mathrm{T}}(k \mid k)\, \Phi_c^{\mathrm{T}}(k+1,k) + \varphi(k)\, \Delta\varphi^{\mathrm{T}}(k), \end{aligned} \tag{7.216}$$

$$\begin{aligned} C(k \mid k) &= C(k \mid k-1)[I - K_c(k)\, M_c(k)]^{\mathrm{T}} - X(k)\, \Delta M^{\mathrm{T}}(k)\, K_c^{\mathrm{T}}(k), \\ m(k+1) &= \Phi(k+1,k)\, m(k) + \varphi(k), \end{aligned} \tag{7.217}$$

$$\begin{aligned}\Delta m(k+1 \mid k+1) = {} & [I - K_c(k+1)\, M_c(k+1)][\Phi_c(k+1, k)\, \Delta m(k \mid k) \\ & + \Delta\Phi(k+1, k)\, m(k) + \Delta\varphi(k)] \\ & - K_c(k+1)\, \Delta M(k+1)\, m(k+1). \end{aligned} \tag{7.218}$$

Initial conditions for (7.212)–(7.218) are

$$\begin{aligned} P(0 \mid 0) &= P(0), \\ X(0) &= \mathscr{E}\{x_0 x_0^{\mathrm{T}}\} = P(0) + \hat{x}(0)\, \hat{x}^{\mathrm{T}}(0), \\ C(0 \mid 0) &= \mathscr{E}\{x_0[x_0 - \hat{x}(0)]^{\mathrm{T}}\} = P(0), \\ m(0) &= \mathscr{E}\{x_0\} = \hat{x}(0), \\ \Delta m(0 \mid 0) &= \mathscr{E}\{x_0 - \hat{x}(0)\} = 0. \end{aligned} \tag{7.219}$$

Various special cases of the actual estimation error covariance matrix equations have been developed by several authors. Battin [3, p. 334] considered the effect of incorrect measurement error statistics. Fagin [16] was concerned with errors in Φ and Q. Heffes [24] considered errors in $P(0)$, Q, and R. Heffes also presents some numerical simulations that show that the actual estimation error variance is greater than or equal to the minimum variance (errors degrade the filter performance). His simulations also indicate that, under appropriate conditions, the actual error variance is less than or equal to the computed variance [$P(k \mid k) \leqslant P_c(k \mid k)$]. Nishimura [41, 42] considers initial errors only [errors in $P(0)$] and develops an upper bound on $P(k \mid k)$. His result is a special case of Theorem 7.6. Nishimura also presents applications and numerical simulations. Price [44] treats errors in $\varphi(k)$, Q, and R and proves, under certain conditions, that $P(k \mid k)$ is bounded. This result is contained in our Theorem 7.7. The most complete development is given by Griffin and Sage [22]. They consider all the errors except $\Delta\varphi$. They do not, however, prove any qualitative results but merely develop equations. Griffin and Sage also define and develop equations for local error sensitivities,

$$[\partial P(k \mid k)/\partial \xi_i]_{\bar{\xi}},$$

where ξ is a vector of parameters in the actual system, and $\bar{\xi}$ is its value used in the filter model. Such equations[11] are developed trivially by differentiating (7.212) and (7.213). Griffin and Sage also present some numerical results. Of some interest are their simulations of a model dimension error (via $\Delta\Phi$).

[11] These equations, as given by Griffin and Sage, appear to be in error in that they neglect to differentiate X and C.

The equations just developed enable the *a priori* determination of the effect of various approximations on filter performance. As a result, they are a useful tool for filter design. Analogous equations for the linear smoother are presented in Griffin and Sage [22].

Consider first the special case of errors in $P(0)$, $Q(k)$, and $R(k)$ only. In this case, Eqs. (7.212) and (7.213) reduce to

$$\begin{aligned} P(k+1 \mid k) &= \Phi(k+1, k)\, P(k \mid k)\, \Phi^{\mathrm{T}}(k+1, k) + \Gamma(k)\, Q(k+1)\, \Gamma^{\mathrm{T}}(k), \\ P(k \mid k) &= [I - K_c(k)\, M(k)]\, P(k \mid k-1)[I - K_c(k)\, M(k)]^{\mathrm{T}} \\ &\quad + K_c(k)\, R(k)\, K_c^{\mathrm{T}}(k), \end{aligned} \tag{7.220}$$

whereas

$$\begin{aligned} P_c(k+1 \mid k) &= \Phi(k+1, k)\, P_c(k \mid k)\, \Phi^{\mathrm{T}}(k+1, k) + \Gamma(k)\, Q_c(k+1)\, \Gamma^{\mathrm{T}}(k), \\ P_c(k \mid k) &= [I - K_c(k)\, M(k)]\, P_c(k \mid k-1)[I - K_c(k)\, M(k)]^{\mathrm{T}} \\ &\quad + K_c(k)\, R_c(k)\, K_c^{\mathrm{T}}(k). \end{aligned} \tag{7.221}$$

Define

$$\begin{aligned} E(k \mid k) &\triangleq P(k \mid k) - P_c(k \mid k), \\ E(k+1 \mid k) &= P(k+1 \mid k) - P_c(k+1 \mid k). \end{aligned} \tag{7.222}$$

Then

$$\begin{aligned} E(k+1 \mid k) &= \Phi(k+1, k)\, E(k \mid k)\, \Phi^{\mathrm{T}}(k+1, k) \\ &\quad + \Gamma(k)[Q(k+1) - Q_c(k+1)]\, \Gamma^{\mathrm{T}}(k), \\ E(k \mid k) &= [I - K_c(k)\, M(k)]\, E(k \mid k-1)[I - K_c(k)\, M(k)]^{\mathrm{T}} \\ &\quad + K_c(k)[R(k) - R_c(k)]\, K_c^{\mathrm{T}}(k). \end{aligned} \tag{7.223}$$

Suppose $Q(k) \leqslant Q_c(k)$, $R(k) \leqslant R_c(k)$, all k. Then, if $E(k \mid k-1) \leqslant 0$, it is evident from (7.223) that $E(k \mid k) \leqslant 0$ and $E(k+1 \mid k) \leqslant 0$. Therefore, by induction, we have

Theorem 7.6. *If* $P(0) \leqslant P_c(0)$ *and* $Q(k) \leqslant Q_c(k)$, $R(k) \leqslant R_c(k)$, *all* k, *then* $P(k \mid k) \leqslant P_c(k \mid k)$ *and* $P(k+1 \mid k) \leqslant P_c(k+1 \mid k)$ *for all* k.

The implications of this result are the following. Although the actual values $P(0)$, $Q(k)$, and $R(k)$ are not generally available, conservative estimates of these quantities can often be made. In that case, the *actual* estimation error variances are bounded by the *computed* "variances," which are generated as part of the filter solution. In fact, $P_c(k \mid k)$ may be precomputed to determine whether the conservative estimates of

$P(0)$, $Q(k)$, and $R(k)$ give satisfactory filter performance, or whether somewhat less conservative estimates are desirable. Theorem 7.6 has the following corollary.

Corollary. *Hypothesis of Theorem* 7.6. *Further suppose that the system model* (7.205) *is uniformly completely observable and uniformly completely controllable. Then there is an integer* $N > 0$ *such that* $P(k \mid k)$ *is uniformly bounded (from above) for all* $k \geqslant N$.

PROOF: By Lemma 7.1, we have

$$P_c(k \mid k) \leqslant \gamma I \qquad (\gamma > 0), \qquad k \geqslant N.$$

Combining this with Theorem 7.6 proves the corollary. ■

Consider now the case of errors in $P(0)$, $Q(k)$, $R(k)$, and $\varphi(k)$ only. As was mentioned earlier, $\varphi(k)$ accounts, in an approximate way, for possible neglected nonlinearities, reduction in system dimension, and so on. In this case, Eqs. (7.212) and (7.213) become

$$\begin{aligned} P(k+1 \mid k) &= \Phi(k+1, k)\, P(k \mid k)\, \Phi^{\mathrm{T}}(k+1, k) + \Gamma(k)\, Q(k+1)\, \Gamma^{\mathrm{T}}(k) \\ &\quad + \Delta\varphi(k)\, \Delta\varphi^{\mathrm{T}}(k) + \Phi(k+1, k)\, \Delta m(k \mid k)\, \Delta\varphi^{\mathrm{T}}(k) \\ &\quad + \Delta\varphi(k)\, \Delta m^{\mathrm{T}}(k \mid k)\, \Phi^{\mathrm{T}}(k+1, k), \\ P(k \mid k) &= [I - K_c(k)\, M(k)]\, P(k \mid k-1)[I - K_c(k)\, M(k)]^{\mathrm{T}} \\ &\quad + K_c(k)\, R(k)\, K_c^{\mathrm{T}}(k), \end{aligned}$$

and, combining these equations, we have

$$\begin{aligned} P(k+1 \mid k+1) &= [I - K_c(k+1)\, M(k+1)]\, \Phi(k+1, k)\, P(k \mid k)\, \Phi^{\mathrm{T}}(k+1, k) \\ &\quad \times [I - K_c(k+1)\, M(k+1)]^{\mathrm{T}} + F(k), \end{aligned} \tag{7.224}$$

where

$$\begin{aligned} F(k) &= [I - K_c(k+1)\, M(k+1)][\Gamma(k)\, Q(k+1)\, \Gamma^{\mathrm{T}}(k) + \Delta\varphi(k)\, \Delta\varphi^{\mathrm{T}}(k) \\ &\quad + \Phi(k+1, k)\, \Delta m(k \mid k)\, \Delta\varphi^{\mathrm{T}}(k) + \Delta\varphi(k)\, \Delta m^{\mathrm{T}}(k \mid k)\, \Phi^{\mathrm{T}}(k+1, k)] \\ &\quad \times [I - K_c(k+1)\, M(k+1)]^{\mathrm{T}} + K_c(k+1)\, R(k+1)\, K_c^{\mathrm{T}}(k+1), \end{aligned} \tag{7.225}$$

and where [from (7.218)]

$$\Delta m(k+1 \mid k+1) = [I - K_c(k+1)\, M(k+1)]\, \Phi(k+1, k)\, \Delta m(k \mid k) + f(k), \tag{7.226}$$

where

$$f(k) = [I - K_c(k+1)\,M(k+1)]\,\Delta\varphi(k). \tag{7.227}$$

Note that

$$[I - K_c(k+1)\,M(k+1)]\,\Phi(k+1, k) = \Psi_c(k+1, k) \tag{7.228}$$

is the state transition matrix of the filter [see (7.167)]. We have the following theorem due to Price [44].

Theorem 7.7. *If the system model* (7.205) *is uniformly completely observable and uniformly completely controllable, and if* $F(k)$ *is uniformly bounded and* $P(0)$ *is bounded, then* $P(k \mid k)$ *is uniformly bounded for all* k.

PROOF: We can write the solution of (7.224) as

$$P(k \mid k) = \Psi_c(k, 0)\,P(0)\,\Psi_c^{\mathrm{T}}(k, 0) + \sum_{i=0}^{k-1} \Psi_c(k, i+1)\,F(i)\,\Psi_c^{\mathrm{T}}(k, i+1). \tag{7.229}$$

Now in view of Theorem 7.4, the filter (7.206) is uniformly asymptotically stable; that is,

$$\| \Psi_c(k, j)\| \leqslant c_1 e^{-c_2(k-j)} \qquad (c_1\,, c_2 > 0), \qquad \text{all} \quad k \geqslant j. \tag{7.230}$$

Using this and the uniform bound on $F(k)$ in (7.229), it is not difficult to show that ($c_3\,, c_4 > 0$)

$$\| P(k \mid k)\| \leqslant c_3 \sum_{i=0}^{k} e^{-c_4 i} \leqslant c_3 \sum_{i=0}^{\infty} e^{-c_4 i} < \infty.$$

(This is an application of the well-known theorem that, in uniformly asymptotically stable systems, a uniformly bounded input produces a uniformly bounded response.) ■

Consider the problem of bounding $F(k)$, which is a hypothesis of the theorem. In view of the assumed uniform complete observability and controllability, and Lemma 7.1, $P_c(k \mid k)$ and $P_c(k+1 \mid k)$ are uniformly bounded, at least for $k \geqslant N$. It has been assumed that all the matrices in the description of system (7.205) are bounded (except φ and φ_c). It therefore remains to bound $\Delta\varphi(k)$ and $\Delta m(k \mid k)$ in $F(k)$. Suppose $\Delta\varphi(k)$ is uniformly bounded. Then $f(k)$ in (7.226) is uniformly bounded. Then Theorem 7.7, applied to $\Delta m(k \mid k)$, proves that it is uniformly bounded.

As a consequence, we have the

Corollary. *If the system model* (7.205) *is uniformly completely observable and uniformly completely controllable, and if* $\Delta\varphi(k)$ *is uniformly bounded and* $P(0)$ *is bounded, then* $P(k \mid k)$ *is uniformly bounded.*

Now if $\varphi(k)$ is known to be bounded, then it is easy to bound $\Delta\varphi(k)$ [set $\varphi_c(k) \equiv 0$, for example]. The difficult problem occurs if $\varphi(k)$ is unbounded. An unbounded $\varphi(k)$ can result, for example, if $\varphi(k)$ stands for neglected states (due, for example, to a reduction of system dimension), and these states are unstable. In that case, it appears that our infinite memory filter may diverge [unbounded $P(k \mid k)$], and we may have to resort to the limited memory filter techniques described in Section 10. Suppose a bound on $\Delta\varphi(k)$ is available. Since all the hypotheses of Theorem 7.7 involve bounds on the parameters of the system model (7.205), it is possible, at least in principle, to design the filter in such a way as to produce the desired bound on $P(k \mid k)$. In particular, $R_c(k)$ and $Q_c(k)$ can be so chosen. The hypothesis of uniform complete controllability points out the requirement of having *some* system noise (Q_c) in the filter.

Consider now the possibility of bounding $P(k \mid k)$ in the general situation described by Eqs. (7.212)–(7.218). Assume that *all* the matrices, including φ and φ_c, in systems (7.204) and (7.205) are uniformly bounded. If Eqs. (7.212) and (7.213) are combined, we see that the forcing term in the difference equation for $P(k \mid k)$ depends on $X(k)$, $C(k \mid k)$, and $m(k)$, as well as on $\Delta m(k \mid k)$. An inspection of the equations for X, C, and m reveals that these matrices can be bounded *if* the *actual system* (7.204) is uniformly asymptotically stable, that is, if $\Phi(k+1, k)$ has property (7.192). Thus, in order to bound $P(k \mid k)$ in the general case, we need the uniform asymptotic stability of the actual system in addition to the hypotheses of Theorem 7.7. This is a very stringent requirement. We can expect the filter to diverge [unbounded $P(k \mid k)$] in cases where the actual system is not uniformly asymptotically stable.

9. ERROR SENSITIVITY—CONTINUOUS FILTER

As in the discrete case, assume that the *actual* system is described by

$$\begin{aligned} dx_t/dt &= F(t)\,x_t + f(t) + G(t)\,w_t\,, \\ y_t &= M(t)\,x_t + v_t\,, \\ w_t \sim N[0, Q(t)], &\qquad v_t \sim N[0, R(t)], \qquad x_{t_0} \sim N[\hat{x}(t_0), P(t_0)], \end{aligned} \tag{7.231}$$

whereas its *model* is

$$\begin{aligned} d\bar{x}_t/dt &= F_c(t)\,\bar{x}_t + f_c(t) + G_c(t)\,\bar{w}_t\,, \\ \bar{y}_t &= M_c(t)\,\bar{x}_t + \bar{v}_t\,, \\ \bar{w}_t \sim N[0, Q_c(t)], \qquad \bar{v}_t &\sim N[0, R_c(t)], \qquad \bar{x}_{t_0} \sim N[\hat{x}_c(t_0), P_c(t_0)]. \end{aligned} \tag{7.232}$$

Boundedness requirements, analogous to those of Sect. 8, are imposed on the matrices in the description of systems (7.231) and (7.232). Our filter design is

$$\begin{aligned} d\hat{x}_t^t/dt &= F_c(t)\,\hat{x}_t^t + K_c(t)(y_t - M_c(t)\,\hat{x}_t^t), \\ dP_c(t \mid t)/dt &= F_c(t)\,P_c(t \mid t) + P_c(t \mid t)\,F_c^{\mathrm{T}}(t) \\ &\quad + G_c(t)\,Q_c(t)\,G_c^{\mathrm{T}}(t) - K_c(t)\,R_c(t)\,K_c^{\mathrm{T}}(t), \\ K_c(t) &= P_c(t \mid t)\,M_c^{\mathrm{T}}(t)\,R_c^{-1}(t), \\ P_c(t_0 \mid t_0) &= P_c(t_0). \end{aligned} \tag{7.233}$$

Define

$$\begin{aligned} \Delta F(t) &\triangleq F(t) - F_c(t), \\ \Delta f(t) &\triangleq f(t) - f_c(t), \\ \Delta M(t) &\triangleq M(t) - M_c(t), \end{aligned} \tag{7.234}$$

and

$$\begin{aligned} P(t \mid t) &\triangleq \mathscr{E}\{(x_t - \hat{x}_t^t)(x_t - \hat{x}_t^t)^{\mathrm{T}}\}, \\ X(t) &\triangleq \mathscr{E}\{x_t x_t^{\mathrm{T}}\}, \\ C(t \mid t) &\triangleq \mathscr{E}\{x_t(x_t - \hat{x}_t^t)^{\mathrm{T}}\}, \\ m(t) &\triangleq \mathscr{E}\{x_t\}, \\ \delta m(t \mid t) &\triangleq \mathscr{E}\{x_t - \hat{x}_t^t\}. \end{aligned} \tag{7.235}$$

Then, applying the limiting procedure of Example 7.10 to Eqs. (7.212)–(7.218), we (tediously) get the following differential equations for the *actual* estimation error covariance matrix $P(t \mid t)$, and for $X(t)$, $C(t \mid t)$, $m(t)$, and $\delta m(t \mid t)$:

$$\begin{aligned} dP(t \mid t)/dt &= [F_c(t) - K_c(t)\,M_c(t)]\,P(t \mid t) \\ &\quad + P(t \mid t)[F_c(t) - K_c(t)\,M_c(t)]^{\mathrm{T}} \\ &\quad + G(t)\,Q(t)\,G^{\mathrm{T}}(t) + K_c(t)\,R(t)\,K_c^{\mathrm{T}}(t) \\ &\quad + \delta m(t \mid t)\,\Delta f^{\mathrm{T}}(t) + \Delta f(t)\,\delta m^{\mathrm{T}}(t \mid t) \\ &\quad + [\Delta F(t) - K_c(t)\,\Delta M(t)]\,C(t \mid t) \\ &\quad + C^{\mathrm{T}}(t \mid t)[\Delta F(t) - K_c(t)\,\Delta M(t)]^{\mathrm{T}}, \end{aligned} \tag{7.236}$$

$$dX(t)/dt = F(t)\,X(t) + X(t)F^{\mathrm{T}}(t) + G(t)Q(t)\,G^{\mathrm{T}}(t) + m(t)f^{\mathrm{T}}(t) + f(t)\,m^{\mathrm{T}}(t), \tag{7.237}$$

$$dC(t \mid t)/dt = F(t)\,C(t \mid t) + C(t \mid t)[F_c(t) - K_c(t)\,M_c(t)]^{\mathrm{T}} + G(t)Q(t)\,G^{\mathrm{T}}(t) + f(t)\,\delta m^{\mathrm{T}}(t \mid t) + m(t)\,\Delta f^{\mathrm{T}}(t) + X(t)[\Delta F(t) - K_c(t)\,\Delta M(t)]^{\mathrm{T}}, \tag{7.238}$$

$$dm(t)/dt = F(t)\,m(t) + f(t), \tag{7.239}$$

$$d\,\delta m(t \mid t)/dt = [F_c(t) - K_c(t)\,M_c(t)]\,\delta m(t \mid t) + \Delta f(t) + [\Delta F(t) - K_c(t)\,\Delta M(t)]\,m(t). \tag{7.240}$$

Initial conditions are those given in (7.219).

Several authors have developed special cases of the actual estimation error covariance equation (7.236). Nishimura [43], Friedland [18], and Mehra [38] considered the case of errors in $P(t_0)$, $Q(t)$, and $R(t)$. Fitzgerald [17] and Griffin and Sage [21] consider all the errors except $\Delta f(t)$. Nishimura, Fitzgerald, and Griffin and Sage present examples and numerical simulations. Analogous equations for the linear smoother are developed by Griffin and Sage and by Mehra. Fitzgerald discusses the case of constant coefficient systems in detail and determines conditions under which divergence will occur.

The analogs of Theorems 7.6 and 7.7 hold in the continuous case. We shall develop these theorems below, but our treatment will be sketchy, since it essentially duplicates the discrete results. The reader should refer to Section 8 for discussion, since everything carries over from the discrete to the continuous case.

In case of errors in $P(t_0)$, $Q(t)$, and $R(t)$ only, (7.236) reduces to

$$dP(t \mid t)/dt = [F(t) - K_c(t)\,M(t)]\,P(t \mid t) + P(t \mid t)[F(t) - K_c(t)\,M(t)]^{\mathrm{T}} + G(t)Q(t)\,G^{\mathrm{T}}(t) + K_c(t)\,R(t)\,K_c^{\mathrm{T}}(t), \tag{7.241}$$

whereas

$$dP_c(t \mid t)/dt = F(t)\,P_c(t \mid t) + P_c(t \mid t)F^{\mathrm{T}}(t) + G(t)Q_c(t)\,G^{\mathrm{T}}(t) - K_c(t)\,R_c(t)\,K_c^{\mathrm{T}}(t). \tag{7.242}$$

Defining

$$E(t) \triangleq P(t \mid t) - P_c(t \mid t), \tag{7.243}$$

we have

$$dE(t)/dt = [F(t) - K_c(t)\,M(t)]\,E(t) + E(t)[F(t) - K_c(t)\,M(t)]^{\mathrm{T}} + G(t)[Q(t) - Q_c(t)]\,G^{\mathrm{T}}(t) + K_c(t)[R(t) - R_c(t)]\,K_c^{\mathrm{T}}(t). \tag{7.244}$$

Equation (7.244) is linear and has the solution

$$E(t) = \Psi_c(t, t_0)\, E(t_0)\, \Psi_c^{\mathrm{T}}(t, t_0) + \int_{t_0}^{t} \Psi_c(t, \tau)\, G(\tau)[Q(\tau) - Q_c(\tau)]\, G^{\mathrm{T}}(\tau)\, \Psi_c^{\mathrm{T}}(t, \tau)\, d\tau$$

$$+ \int_{t_0}^{t} \Psi_c(t, \tau)\, K_c(\tau)[R(\tau) - R_c(\tau)]\, K_c^{\mathrm{T}}(\tau)\, \Psi_c^{\mathrm{T}}(t, \tau)\, d\tau, \tag{7.245}$$

where $\psi_c(t, \tau)$ is the state transition matrix of the filter (see Section 7); that is,

$$d\Psi_c(t, \tau)/dt = [F(t) - K_c(t)\, M(t)]\, \Psi_c(t, \tau), \qquad \Psi_c(\tau, \tau) = I. \tag{7.246}$$

Inspection of (7.245) produces the theorem due to Nishimura.

Theorem 7.8. *If $P(t_0) \leqslant P_c(t_0)$ and $Q(t) \leqslant Q_c(t)$, $R(t) \leqslant R_c(t)$, all t, then $P(t \mid t) \leqslant P_c(t \mid t)$ for all t.*

The following corollary is immediate if we apply the continuous analog of Lemma 7.1 (see Section 7).

Corollary. *Hypotheses of Theorem 7.8. Further suppose that the system model (7.232) is uniformly completely observable and uniformly completely controllable. Then there is a $\sigma > 0$ such that $P(t \mid t)$ is uniformly bounded for all $t \geqslant t_0 + \sigma$.*

Now consider the case of errors in $P(t_0)$, $Q(t)$, $R(t)$, and $f(t)$. Equation (7.236) becomes

$$dP(t \mid t)/dt = [F(t) - K_c(t)\, M(t)]\, P(t \mid t) + P(t \mid t)[F(t) - K_c(t)\, M(t)]^{\mathrm{T}} + D(t), \tag{7.247}$$

where

$$D(t) = G(t)\, Q(t)\, G^{\mathrm{T}}(t) + K_c(t)\, R(t)\, K_c^{\mathrm{T}}(t) + \delta m(t \mid t)\, \Delta f^{\mathrm{T}}(t) + \Delta f(t)\, \delta m^{\mathrm{T}}(t \mid t), \tag{7.248}$$

and where, from (7.240),

$$d\, \delta m(t \mid t)/dt = [F(t) - K_c(t)\, M(t)]\, \delta m(t \mid t) + \Delta f(t). \tag{7.249}$$

Theorem 7.9. *If the system model (7.232) is uniformly completely observable and uniformly completely controllable, and if $\Delta f(t)$ is uniformly bounded and $P(t_0)$ is bounded, then $P(t \mid t)$ is uniformly bounded.*

PROOF: Note that both (7.249) and (7.247) are linear, and their solutions may be written down using the filter state transition matrix defined in (7.246). By Theorem 7.4 for the continuous filter, (7.249) is uniformly asymptotically stable, and, since $\Delta f(t)$ is uniformly bounded, so is $\delta m(t \mid t)$. As a result, $D(t)$ is uniformly bounded. But then, by the same arguments, $P(t \mid t)$ is uniformly bounded. ■

10. LINEAR LIMITED MEMORY FILTER

The results and discussion of Sections 8 and 9 indicate that, if the dynamical model parameters are not sufficiently well known, the "optimal" filter may diverge. Even if the actual estimation error covariance matrix is bounded, the actual estimation errors may be too large and the estimate unusable. In that case, for all practical purposes, the filter is diverging. The onset of divergence is, of course, determined by statistical inconsistency of the actual estimation errors with the computed error covariance matrix.

The problem of divergence is discussed in detail in the next chapter, where various remedies for divergence are outlined. One of these is the idea of limiting the memory of the filter, thus curtailing the growth of the actual estimation error covariance matrix. This has already been discussed in Section 6.11, as well as in Section 8. The limited memory filter appears to be the only device for preventing divergence in the presence of unbounded $\Delta\varphi(k)$ or $\Delta f(t)$ (see Theorems 7.7 and 7.9).

Since the limited memory filter has a sound theoretical basis (most other remedies are *ad hoc*), we develop its theory here. We note that the idea of limiting the filter memory by various means has been advocated by a number of authors. Schweppe [49] develops a form of the limited memory filter. Similar ideas are contained in Cosaert and Gottzein [12]. Schmidt [48] advocates overweighting the most recent observations, and limiting information by other means. Our development here follows that of Jazwinski [28].

We assume that our discrete dynamical system is noise-free:

$$x_{k+1} = \Phi(k+1, k)\, x_k\,, \qquad x_0 \sim N(\hat{x}_0\,, P_0), \qquad P_0 > 0, \tag{7.250}$$

with observations

$$y_k = M(k)\, x_k + v_k\,, \qquad v_k \sim N[0, R(k)], \qquad R(k) > 0. \tag{7.251}$$

We propose to limit the filter memory sufficiently so that (7.250), (7.251) is an adequate approximation to the real system over the limited time

arcs. The absence of noise in (7.250) is fundamental to our theory. We now specialize the results of Section 6.11 to this linear case.

Using the notation (Y_m, Y_n, Y_N) defined in Section 6.11, let

$$\begin{aligned}
&\hat{x}_n{}^n \triangleq \mathscr{E}\{x_n \mid Y_n\},\\
&P_n{}^n \triangleq \mathscr{E}\{(x_n - \hat{x}_n{}^n)(x_n - \hat{x}_n{}^n)^{\mathrm{T}} \mid Y_n\},\\
&\hat{x}_n{}^m \triangleq \mathscr{E}\{x_n \mid Y_m\} = \Phi(n, m)\, \hat{x}_m{}^m,\\
&P_n{}^m \triangleq \mathscr{E}\{(x_n - \hat{x}_n{}^m)(x_n - \hat{x}_n{}^m)^{\mathrm{T}} \mid Y_m\} = \Phi(n, m)\, P_m{}^m \Phi^{\mathrm{T}}(n, m).
\end{aligned} \tag{7.252}$$

Theorem 7.10. *If x_n is completely observable with respect to Y_N, then $\hat{x}_n^{N|}$, $P_n^{N|}$ (given below) are, respectively, the conditional mean of x_n and the conditional covariance matrix, where the conditioning is on Y_N only:*

$$\begin{aligned}
\hat{x}_n^{N|} &= P_n^{N|}(P_n^{n^{-1}} \hat{x}_n{}^n - P_n^{m^{-1}} \hat{x}_n{}^m),\\
P_n^{N|^{-1}} &= P_n^{n^{-1}} - P_n^{m^{-1}}.
\end{aligned} \tag{7.253}$$

$\hat{x}_n^{N|}$ is also the maximum likelihood estimate of x_n based on Y_N.

PROOF: Since $P_0 > 0$, and in view of the discussion following Lemma 7.1, $P_n{}^n, P_n{}^m > 0$, so that the densities appearing in Lemma 6.5 are well-defined Gaussian densities. According to Example 7.2, the parameters of $p(x_n \mid Y_N)$ (that is, its mean and covariance matrix) are, respectively,

$$\begin{aligned}
\mathscr{E}\{x_n \mid Y_N, p(x_0)\} &= \operatorname{cov}\{x_n \mid Y_N, p(x_0)\}\\
&\quad \times \left[\sum_{i=m+1}^{n} \Phi^{\mathrm{T}}(i, n)\, M^{\mathrm{T}}(i)\, R^{-1}(i)\, y_i + \Phi^{\mathrm{T}}(0, n)\, P_0^{-1} \hat{x}_0\right],
\end{aligned} \tag{7.254}$$

$$\operatorname{cov}^{-1}\{x_n \mid Y_N, p(x_0)\} = \Phi^{\mathrm{T}}(0, n)\, P_0^{-1} \Phi(0, n) + \mathscr{J}(n, m+1),$$

where

$$\mathscr{J}(n, m+1) = \sum_{i=m+1}^{n} \Phi^{\mathrm{T}}(i, n)\, M^{\mathrm{T}}(i)\, R^{-1}(i)\, M(i)\, \Phi(i, n) > 0 \tag{7.255}$$

by hypothesis. Therefore, the $P_0^{-1} = 0$ limits in (7.254) exist, so that $p(x_n \mid Y_N |)$ is a well-defined Gaussian density [defined according to (6.133)].

We now compute the parameters of $p(x_n \mid Y_N |)$ using Theorem 6.8.

Now $p(x_n \mid Y_N)$ has parameters $\hat{x}_n^n$, P_n^n; and $p(x_n \mid Y_m)$ has parameters $\hat{x}_n^m$, P_n^m. Therefore, by Theorem 6.8,

$$p(x_n \mid Y_N |) = c_2 \exp\{-\tfrac{1}{2}[(x_n - \hat{x}_n^n)^T P_n^{n-1}(x_n - \hat{x}_n^n) - (x_n - \hat{x}_n^m)^T P_n^{m-1}(x_n - \hat{x}_n^m)]\}. \tag{7.256}$$

Since this density is Gaussian, its mean coincides with its mode. The mode minimizes the quantity in brackets in (7.256). Setting the gradient (with respect to x_n) of that quantity to zero, we get

$$P_n^{n-1}(x_n - \hat{x}_n^n) - P_n^{m-1}(x_n - \hat{x}_n^m) = 0. \tag{7.257}$$

$\hat{x}_n^{N|}$ is the solution of (7.257). $P_n^{N|-1}$ is the matrix of second derivatives

$$P_n^{n-1} - P_n^{m-1} \equiv P_n^{N|-1}.$$

Note that $P_n^{N|} = \mathscr{J}^{-1}(n, m+1)$, which is positive definite by hypothesis, so that $P_n^{N|}$ is nonsingular. This establishes (7.253) and proves the theorem. ■

Several remarks concerning Theorem 7.10 are in order. We note that $\hat{x}_n^n$, P_n^n are the outputs of the discrete Kalman filter of Theorem 7.2. $\hat{x}_n^m$, P_n^m are predicted values computed via (7.28). We may solve (7.253) for $\hat{x}_n^n$ and P_n^n and use the resulting equations to process batches of preprocessed data (preprocessed via a least-squares procedure) and characterized by $\hat{x}_n^{N|}$ and $P_n^{N|}$. If $m = 0$, Eqs. (7.253) can be used to remove the conditioning of a Kalman estimate on any initial data. In fact, we can remove $\hat{x}_0$, P_0 and *replace* it by $\hat{x}_0'$, P_0'. We may note that, with $n = m$ ($N = 0$), $P_n^{N|-1} = 0$ (no information), and $\hat{x}_n^{N|}$ is arbitrary. For N large, $P_n^{N|-1} \approx P_n^{n-1}$ and $\hat{x}_n^{N|} \approx \hat{x}_n^n$, which are the intuitively correct properties.

Equations (7.253) generate the minimum variance estimate based on a "moving window" of the most recent N observations. Two Kalman filters and a predictor are required in their implementation, and N observations have to be stored in memory. Three indicated matrix inversions are required. Now, aside from the large computational load, this scheme is clearly no solution to the problem of divergence. The two required Kalman filters would diverge!

Based on Theorem 7.10, we propose the following limited memory filter, which, at the same time, decreases the computational load and eliminates the storage requirement. The conditioning of the estimate on old data is discarded in batches of N. We filter N observations and also

predict over the same time arc. This produces $\hat{x}_N{}^N$, $P_N{}^N$, $\hat{x}_N{}^0$, $P_N{}^0$. Equations (7.253) are used to compute $\hat{x}_N^{N|}$, $P_N^{N|}$ (we have just eliminated the prior data). We next set $\hat{x}_N^{N|}$, $P_N^{N|}$ as initial conditions for both the filter and predictor, and repeat the procedure.

The limited memory filter just described produces limited memory estimates with memory varying between N and $2N$. Clearly, variations on this are possible, and N itself may be varied in the process. N should be chosen so that the model (7.250) represents an adequate approximation to the real system over time arcs of $2N$.

We next discuss some alternate computational procedures. The second of Eqs. (7.253) involves two matrix inversions and a subtraction, and could be ill-conditioned computationally. Recognizing that $P_n^{N|} = \mathscr{I}^{-1}(n, m+1)$ (see 7.255), we may compute this matrix by the recursion

$$\mathscr{I}(k, m+1) = \Phi^{\mathrm{T}}(k-1, k)\,\mathscr{I}(k-1, m+1)\,\Phi(k-1, k) + M^{\mathrm{T}}(k)\,R^{-1}(k)\,M(k), \qquad k = m+1,\ldots, n, \tag{7.258}$$

with

$$\mathscr{I}(m, m+1) = 0.$$

Then the first of Eqs. (7.253) can be written as

$$\hat{x}_n^{N|} = \hat{x}_n{}^n + \mathscr{I}^{-1}(n, m+1)\,P_n^{m-1}(\hat{x}_n{}^n - \hat{x}_n{}^m). \tag{7.259}$$

Equation (7.258) requires the inverse of the state transition matrix, but this is sometimes available in closed form.[12] In that case, P_n^{m-1} can also be computed recursively:

$$P_k^{m-1} = \Phi^{\mathrm{T}}(k-1, k)\,P_{k-1}^{m-1}\Phi(k-1, k), \qquad k = m+1,\ldots, n. \tag{7.260}$$

Recursions (7.258) and (7.260) can be carried along with the Kalman filter.

As was mentioned earlier, Schweppe [49] develops another form of the limited memory filter with a constant memory length. That algorithm, however, requires the storage of N observations together with $M(k)$ and $R(k)$, $k = m+1,\ldots, n$; the inverse of the state transition matrix; prediction over arcs of length N every time a new observation is acquired; and inversion of an information matrix every time a new observation is acquired. Although a constant memory length is desirable, the computational load associated with the filter just described is very large.

[12] In orbit determination, closed-form approximations to the state transition matrix and its inverse are available.

APPENDIX 7A

CLASSICAL PARAMETER ESTIMATION

In this Appendix, we point out some interesting connections between filtering theory and classical parameter estimation theory. We shall see that estimation and filtering are based on essentially the same theory.

Consider the problem of estimating the (fixed) parameter x_k from the observations

$$y_i = M(i)\,\Phi(i, k)\,x_k + v_i\,, \qquad i = 1,\ldots, k, \tag{7A.1}$$

where $\{v_i\}$ is a white, but *not necessarily* Gaussian, vector sequence with

$$\mathscr{E}\{v_i\} = 0, \qquad \mathscr{E}\{v_i v_j^{\mathrm{T}}\} = R_i\,\delta_{ij}\,, \qquad R_i > 0.$$

The parameter x_k may be viewed as the state, at time t_k, of the dynamical system

$$x_{i+1} = \Phi(i+1, i)\,x_i\,, \qquad y_i = M(i)\,x_i + v_i\,.$$

We assume that x_k is completely observable with respect to $\{y_1,\ldots, y_k\}$.

We wish to find the *linear, unbiased, minimum variance* estimator for x_k. That is, we seek that estimate $x_k{}^*$ in the class of estimates $\{\xi_k\}$,

$$\xi_k = \sum_{i=1}^{k} A_i y_i\,, \tag{7A.2}$$

which is unbiased,

$$\mathscr{E}\{x_k{}^*\} = \mathscr{E}\{x_k\} = x_k\,, \tag{7A.3}$$

and for which

$$\operatorname{tr} \mathscr{E}\{(x_k{}^* - x_k)(x_k{}^* - x_k)^{\mathrm{T}}\}$$

is a minimum. The answer is provided in the classical

Theorem 7A.1 (Gauss–Markov). *Let x_k be completely observable with respect to $\{y_1,\ldots, y_k\}$. Let $\{v_i\}$ be a zero-mean, white, not necessarily Gaussian vector sequence, $R_i > 0$. Then the linear, unbiased, minimum variance estimate of x_k is*

$$x_k{}^* = \mathscr{I}_{k,1}^{-1} \sum_{i=1}^{k} \Phi^{\mathrm{T}}(i, k)\,M^{\mathrm{T}}(i)\,R_i^{-1} y_i\,,$$

where

$$\mathscr{I}_{k,1} = \sum_{i=1}^{k} \Phi^{\mathrm{T}}(i, k)\,M^{\mathrm{T}}(i)\,R_i^{-1} M(i)\,\Phi(i, k)$$

is the information matrix. The covariance matrix of $x_k{}^*$ is $\mathscr{I}_{k,1}^{-1}$.

PROOF: Consider any linear, unbiased estimate

$$\xi_k = \sum_{i=1}^{k} A_i y_i \,.$$

We have

$$\mathscr{E}\{\xi_k\} = \mathscr{E}\left\{\sum_{i=1}^{k} A_i[M(i)\,\Phi(i,k)\,x_k + v_i]\right\} = \sum_{i=1}^{k} A_i M(i)\,\Phi(i,k)\,x_k$$

and

$$\xi_k - x_k = \xi_k - \mathscr{E}\{\xi_k\}.$$

Therefore,

$$\begin{aligned}\mathscr{E}\{(\xi_k - x_k)(\xi_k - x_k)^{\mathrm{T}}\} &= \sum_{i=1}^{k} A_i R_i A_i^{\mathrm{T}} \\ &= \sum_{i=1}^{k} [\mathscr{I}_{k,1}^{-1} L_i + (A_i - \mathscr{I}_{k,1}^{-1} L_i)]\, R_i [\mathscr{I}_{k,1}^{-1} L_i + (A_i - \mathscr{I}_{k,1}^{-1} L_i)]^{\mathrm{T}}, \quad \text{(7A 4)}\end{aligned}$$

where

$$L_i \triangleq \Phi^{\mathrm{T}}(i,k)\, M^{\mathrm{T}}(i)\, R_i^{-1}.$$

Now

$$\sum_{i=1}^{k} (A_i - \mathscr{I}_{k,1}^{-1} L_i)\, R_i L_i^{\mathrm{T}} \mathscr{I}_{k,1}^{-1} = \left[\sum_{i=1}^{k} A_i M(i)\,\Phi(i,k) - I\right] \mathscr{I}_{k,1}^{-1} = 0,$$

since ξ_k is unbiased. [From (7A.3),

$$0 = \mathscr{E}\{\xi_k\} - x_k = \left[\sum_{i=1}^{k} A_i M(i)\,\Phi(i,k) - I\right] x_k$$

for all x_k]. As a result, (7A.4) becomes

$$\begin{aligned}\mathscr{E}\{(\xi_k - x_k)(\xi_k - x_k)^{\mathrm{T}}\} &= \sum_{i=1}^{k} [\mathscr{I}_{k,1}^{-1} L_i R_i L_i^{\mathrm{T}} \mathscr{I}_{k,1}^{-1} + (A_i - \mathscr{I}_{k,1}^{-1} L_i)\, R_i (A_i - \mathscr{I}_{k,1}^{-1} L_i)^{\mathrm{T}}]. \quad \text{(7A.5)}\end{aligned}$$

Since $R_i > 0$, the trace of this matrix is minimum when $A_i = \mathscr{I}_{k,1}^{-1} L_i$, that is, when $\xi_k = x_k{}^*$. From (7A.5), it then follows that

$$\mathscr{E}\{(x_k{}^* - x_k)(x_k{}^* - x_k)^{\mathrm{T}}\} = \mathscr{I}_{k,1}^{-1}. \quad \blacksquare$$

Compare the linear, unbiased, minimum variance estimate (Theorem 7A.1) with the estimate (7.41) of Example 7.2, which derives the discrete Kalman–Bucy filter. We see that with $P_0^{-1} = 0$ they are identical. Thus we see again that the absence of *a priori* information about x_k is expressed formally by setting $P_0^{-1} = 0$. We also have

Theorem 7A.2. *The minimum variance filter in the Gaussian case is also the linear minimum variance filter in the general case, without any assumption of Gaussianness.*

This conclusion can also be understood in the following way. If we seek the best *linear* estimate, then only the first two moments of the $\{y_k\}$ process need be known. These moments define a unique Gaussian process. Therefore, the linear minimum variance filter in the Gaussian case must be the same as in the general case. But in the Gaussian case, the variance is minimized by the conditional mean which *is* a linear functional of the $\{y_k\}$ process. Therefore, in the Gaussian case, the minimum variance and linear minimum variance filters are the same. Thus the conclusion follows.

Recall that, in the derivation of Example 7.2, no statistical significance was attached to the R_i, that is, the errors v_i were not considered random variables with a given distribution. Now if one is willing to stipulate that the v_i are *zero-mean*, one can conclude that the estimate (of Theorem 7A.1) is *unbiased*. If one is further willing to specify the *second moments* of v_i, that is, R_i, one can conclude by the Gauss–Markov theorem that the estimate is a *linear*, *minimum variance*, *unbiased* estimate. If the v_i are furthermore *Gaussian*, then Theorem 7A.2 asserts that the estimate is the *minimum variance* estimate. Thus there is a direct relationship between the assumptions made about the noise and the conclusions that can be drawn about the estimate.

APPENDIX 7B

SOME MATRIX EQUALITIES

Let P, R, and M be $n \times n$, $m \times m$, and $m \times n$ matrices, respectively. Suppose $P \geqslant 0$ and $R > 0$. Then the following equalities hold:

$$(I + PM^{\mathrm{T}}R^{-1}M)^{-1} = I - PM^{\mathrm{T}}(MPM^{\mathrm{T}} + R)^{-1}M, \tag{7B.1}$$

$$(I + PM^{\mathrm{T}}R^{-1}M)^{-1}P = P - PM^{\mathrm{T}}(MPM^{\mathrm{T}} + R)^{-1}MP, \tag{7B.2}$$

$$(I + PM^{\mathrm{T}}R^{-1}M)^{-1}PM^{\mathrm{T}}R^{-1} = PM^{\mathrm{T}}(MPM^{\mathrm{T}} + R)^{-1}. \tag{7B.3}$$

If, in addition, $P > 0$, then

$$(I + PM^TR^{-1}M)^{-1}P = (P^{-1} + M^TR^{-1}M)^{-1}. \tag{7B.4}$$

Using (7B.4) in (7B.2) and (7B.3), we get (for P, $R > 0$)

$$(P^{-1} + M^TR^{-1}M)^{-1} = P - PM^T(MPM^T + R)^{-1}MP, \tag{7B.5}$$

$$(P^{-1} + M^TR^{-1}M)^{-1}M^TR^{-1} = PM^T(MPM^T + R)^{-1}. \tag{7B.6}$$

These matrix equalities are relatively well known in numerical analysis (e.g., Bodewig [6, p. 30]). Equation (7B.5) was introduced to the filtering field by Ho [26].

Equation (7B.1) is easily proved by direct verification:

$$\begin{aligned}(I + PM^TR^{-1}M)&(I - PM^T(MPM^T + R)^{-1}M)\\ &= I + PM^TR^{-1}M - PM^TR^{-1}R(MPM^T + R)^{-1}M\\ &\quad - PM^TR^{-1}MPM^T(MPM^T + R)^{-1}M\\ &= I.\end{aligned}$$

Equation (7B.2) follows trivially. To prove (7B.3), multiply (7B.2) on the right by M^TR^{-1}:

$$\begin{aligned}&(I + PM^TR^{-1}M)^{-1}PM^TR^{-1}\\ &\quad = PM^TR^{-1} - PM^T(MPM^T + R)^{-1}MPM^TR^{-1}\\ &\qquad + [PM^T(MPM^T + R)^{-1} - PM^T(MPM^T + R)^{-1}RR^{-1}]\\ &\quad = PM^T(MPM^T + R)^{-1}.\end{aligned}$$

Equation (7B.4) follows from

$$[(I + PM^TR^{-1}M)^{-1}P]^{-1} = P^{-1}(I + PM^TR^{-1}M) = P^{-1} + M^TR^{-1}M.$$

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